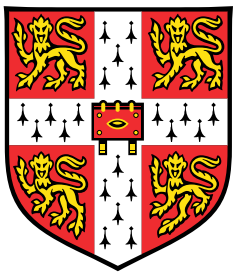
Ghost Fluid Method Comparison



Jiayi Wu

Department of Physics
University of Cambridge

This dissertation is submitted for the degree of
Master of Philosophy

Girton College August 2025

I would like to dedicate this thesis to my loving parents ...

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements. This dissertation contains fewer than 65,000 words including appendices, bibliography, footnotes, tables and equations and has fewer than 150 figures.

Jiayi Wu
August 2025

MPhil SC Research Project – First Report: signed declaration

Title of Research Project:

Ghost Fluid Method Comparison

I declare that this project report is work done as part of the MPhil in Scientific Computing Examination. I have read and understood the Statement on Plagiarism for this course in the Student Handbook and have abided by it. This project report is the result of my own work, and except where explicitly stated otherwise, only includes material undertaken since October 2024, and includes nothing which was performed in collaboration. No part of this project report has been submitted, or is concurrently being submitted, for any degree, diploma or similar qualification at any university or similar institution.

Signed: 吴佳怡

Name in CAPITALS: JIAYI WU

Date: 29/5/2025 College: Girton

Title of Research Project:

Ghost Fluid Methods Comparison

I declare that this project report is work done as part of the MPhil in Scientific Computing Examination. I have read and understood the Statement on Plagiarism for this course in the Student Handbook and have abided by it. This project report is the result of my own work, and except where explicitly stated otherwise, only includes material undertaken since October 2024, and includes nothing which was performed in collaboration. No part of this project report has been submitted, or is concurrently being submitted, for any degree, diploma or similar qualification at any university or similar institution.

Signed: 吴佳屹

Name in CAPITALS: JIA YI WU

Date: 14/8/2025 College: Girton

Acknowledgements

As I write this acknowledgment, my mind drifts through the quiet corridors of time—over two decades of learning, faltering, and becoming. From the early days when numbers danced beyond my grasp, to the quiet joy of graduating with first-class honors, and finally to the solitude and strength of completing my Scientific Computing project, this journey has not been mine alone. The unwavering support and companionship of those around me have guided me through challenges and brightened even the most difficult moments. Their presence remains a source of strength and encouragement, shaping the person I am today.

First and foremost, I wish to express my heartfelt gratitude to my supervisor, Dr. Riccardo Dematte. Your patience, understanding, insightful guidance, and unwavering trust have been nothing short of a lifeline. In moments of doubt and difficulty, your steady encouragement lit the path forward and gave me the strength to persist. Thank you for believing in me and for always challenging me to grow.

My deepest thanks belong to my mother, Yanhua Wu. No matter the challenges I faced, you stood by me silently and steadfastly, with unconditional love, boundless sacrifice, and unshakeable faith. You lifted me up when I fell and celebrated every victory, no matter how small, with a smile only a mother can give. Thank you for teaching me what it means to be resilient and hopeful, and for embracing every part of who I am, even the imperfect ones.

To my sister, Qianyu Yao, thank you for being a radiant light in my life. Your endless optimism and joyful spirit never fail to inspire me. Even in my busiest, most stressful days, your words reminded me to pause and rediscover the beauty in small moments.

Special thanks go to my senior colleague, Zefeng Chen. Your technical expertise, thoughtful advice, and willingness to help during critical moments gave me clarity and confidence when I needed it most. To my friends Yaozu Li and Yuechen Wang, thank you for being beside me through late-night brainstorming sessions and shared frustrations. Those long hours of problem-solving and laughter became some of my fondest memories.

To all the family members and friends who have quietly supported me along the way, your encouragement, kindness, and faith carried me forward, even when I questioned myself. Your love made the difference.

To my grandfather, who, I believe, would be deeply proud to see the completion of this project.

Cambridge winters are long and bitterly cold, but thanks to my friends, this unfamiliar place became a second home. Your warmth and companionship softened the harshest days, and in your company, I found comfort, belonging, and joy.

My last tribute is to those who know I am not perfect but still love me. Thank you for accepting my flaws, embracing my imperfections, and helping me believe I am worthy of love and happiness.

As I close this chapter, I stand at the threshold of the unknown, unsure of what lies ahead, yet ready to face it with steady steps. The path may be challenging, but I carry with me the lessons, support, and encouragement I've received along the way. Whatever comes next, I'll meet it with patience, effort, and a clear heart. And so, I walk on. Ever forward, in every sense of the phrase.

Abstract

This report presents a comprehensive development, implementation, and evaluation of six Ghost Fluid Method (GFM) variants for sharp-interface simulation in compressible multiphase flow. Building on a two dimensional finite volume solver with MUSCL–Hancock reconstruction and an HLLC Riemann solver, we implement Original GFM, Riemann GFM, Real GFM, Modified GFM, Practical GFM, and Interface GFM. Each scheme is subjected to eight benchmark problems: four one dimensional tests (Sod shock tube, Karni–Quirk mixed-EOS, water air shock tube, and underwater gas water interaction), two two dimensional problems (cylindrical explosion and shock bubble interaction), and two advanced 1D scenarios (Toro Test 2 double-rarefaction and Toro Test 5 extreme shock/contact). We assess performance across five practical dimensions: interface accuracy, shock stability, implementation ease, two-dimensional robustness, and computational cost—using quantitative error metrics and qualitative diagnostics. Results show that all six GFMs capture basic wave features in single-phase and mild multiphase flows; however, under strong shocks, high density ratios, or vacuum conditions, Original GFM fails, and Riemann/Real GFM sacrifice sharpness via smoothing or oscillations. In contrast, Modified, Practical, and Interface GFM deliver superior fidelity and robustness, with Interface GFM achieving the highest accuracy at the expense of increased complexity. We summarize each method's strengths and weaknesses in Table 10.4 and provide targeted recommendations for selecting or hybridizing GFMs based on local flow features. Finally, we outline future extensions to three dimensional grids, non-ideal equations of state, interface topological changes, and adaptive method switching to further enhance the accuracy, stability, and efficiency of multiphase flow simulations.

Table of contents

List of figures **xvii**

List of tables **xxiii**

1 Introduction and background **1**
1.1 Motivation for Multi-Fluid Simulations 1
1.2 Sharp Interface Method 2
1.3 Development of Ghost Fluid Methods 3
1.4 Objectives of this project 4

2 Numerical Methods **5**
2.1 Euler Equations 5
2.1.1 One-Dimensional Euler Equations 6
2.1.2 Two-Dimensional Euler Equations 6
2.2 Second-order Finite Volume Scheme 8
2.2.1 Time Step Calculation 8
2.2.2 Data Reconstruction 9
2.2.3 Slope Limiting 10
2.2.4 Half-time Update 11
2.2.5 HLLC Riemann Solver 11
2.2.6 The MUSCL-Hancock Scheme 13
2.3 Level Set Method 13
2.4 Fast Sweeping Method 14
2.5 Equations of State 17
2.5.1 Ideal Gas 17
2.5.2 Stiffened Gas 17

3 The Ghost Fluid Methods **19**
3.1 The Original Ghost Fluid Method 20

3.2 The Riemann Ghost Fluid Method 23
3.3 The Real Ghost Fluid Method 28
3.4 The Modified Ghost Fluid Method 29
3.5 The Practical Ghost Fluid Method 30
3.6 The Interface Ghost Fluid Method 32

4 Tests Results and Analysis **37**
4.1 The Shock Tube Test 37
4.2 The Karni and Quirk's Test 38
4.3 The Water Air Shock Tube Test 40
4.4 Underwater shock impacting on a gas-water interface 41
4.5 The Cylindrical Explosion Test 43
4.6 The 2D Shock Bubble Test 45

5 Midterm Conclusion **49**

6 Introduction to Report 2 **51**

7 1D Validation Tests **53**
7.1 Single-material Sod Shock Tube Test 53
7.2 The Multi-material Karni and Quirk's Test 54
7.3 The Water Air Shock Tube Test 55
7.4 Underwater shock impacting on a gas-water interface 57

8 2D Validation Tests **59**
8.1 The Cylindrical Explosion Test 59
8.2 The 2D Shock Bubble Test 61

9 Supplementary Tests for GFM Comparison **65**
9.1 Toro's Test 2 65
9.2 Toro's Test 5 67

10 Comparative Analysis and Discussion **71**
10.1 Overview of GFMs 71
10.2 Evaluation Strategy: Criteria and Benchmark Suite 72
10.3 Results and Cross-Test Comparison 72
10.4 Final Conclusion 76

References **79**

Appendix A OpenMP Parallelization 81

List of figures

2.1 Schematic of the control volume associated with grid node (i, j) in a two-dimensional finite-volume discretization. The rectangle spans the interval $\left(x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}\right) \times \left(y_{j-\frac{1}{2}}, y_{j+\frac{1}{2}}\right)$, with the gray dot marking the cell center. Arrows denote the numerical fluxes: $F_{i\pm\frac{1}{2},j}$ across the left/right faces and $G_{i,j\pm\frac{1}{2}}$ across the bottom/top faces, where the arrow direction defines the positive sign convention. The spacings Δx and Δy indicate the grid widths in the x and y directions, respectively, and the dotted lines show the coordinate lines through the cell center. 7

2.2 Wave structure in the HLLC Riemann solver. Within the star region, the solution is composed of two uniform states that are separated by a middle wave traveling at speed S^* 12

3.1 Scheme of ghost cells indentification 20

3.2 Schematic of the Original Ghost Fluid Method at a 1D interface. The black circles are real fluid cells; the red squares are ghost cells. From the interface cell i, we copy its density ρ_I, tangential velocity $\mathbf{v}_{t,I}$, and pressure p_I. From the nearest real neighbor in the opposite material (cell $i+1$), we copy its pressure p_{i+1} and normal velocity $V_{n,i+1}$. The ghost-cell density is then computed via $\rho_G = \rho_I \left(p_{i+1}/p_I\right)^{1/\gamma_i}$, the ghost-cell velocity by $\mathbf{v}_G = V_{n,i+1}\,\mathbf{n} + \mathbf{v}_{t,I}$, and the ghost-cell pressure is set to $p_G = p_{i+1}$. No Riemann solver is required. 21

3.3 Illustration of neighbor selection in the Original GFM: the interface cell I discards neighbours in the same material (grey) and picks the remaining cell whose center best aligns with the normal $\mathbf{n}$ (pink). 22

3.4 Geometric illustration of the closest point. The cell center $\mathbf{x}_I$ lies inside the Material 2 (gray region), whose closest point $\mathbf{x}_P$ is computed along the normal direction $\mathbf{n}$. 24

3.5 Geometric illustration of interface-normal interpolation points used in the Riemann Ghost Fluid Method. The interface cell centered at $\mathbf{x}_I$ contains the interface, whose closest point $\mathbf{x}_P$ is projected along the local normal direction. Two states are defined at points $\mathbf{x}_{P1}$ and $\mathbf{x}_{P2}$, which are symmetrically located on either side of the interface. Material 1 ($\phi > 0$) and Material 2 ($\phi < 0$) are indicated by the labeled regions. 25

3.6 Geometric illustration of the bilinear interpolation process for estimating the value at point $P(x,y)$. The four known corner values are $Q_{11}, Q_{21}, Q_{12}, Q_{22}$, located at $(x_1,y_1), (x_2,y_1), (x_1,y_2), (x_2,y_2)$, respectively. Intermediate values R_1 and R_2 are obtained by linear interpolation in the x-direction, and the final interpolated value at P is obtained by linear interpolation in the y-direction using R_1 and R_2. 26

3.7 Illustration of the Practical GFM stencil. Real fluid cells (black circles) on the left of the interface supply p and ρ directly to their corresponding ghost fluid counterparts (red circles), while the normal velocity in each ghost cell is set ny mirror-reflection about the interface velocity u_I. i.e. $u^* = 2u_I - u_{real}$ 31

3.8 Schematic of two adjacent grid cells have two different materials. The left cell at index i carries state $W_l = (\rho_l, u_l, p_l)$ and the right cell at index $i+1$ carries state $W_r = (\rho_r, u_r, p_r)$ before interaction. 32

3.9 Schematic of the two adjacent half–cells immediately after interface interaction. The left half–cell at index i carries the interface state $W_{I,\ell} = (\rho_{I,\ell}, u_I, p_I)$, and the right half–cell at index $i+1$ carries $W_{I,r} = (\rho_{I,r}, u_I, p_I)$. 33

3.10 Ghost cell construction for Material 1 before interaction. The real fluid cell at index i holds state $W_l = (\rho_l, u_l, p_l)$. The ghost fluid cell holds state $W_{gl} = (\rho_{gl}, u_{gl}, p_{gl})$. 34

3.11 Ghost cell construction for Material 1 after interaction. The real fluid cell at index i holds state $W_{g,l} = (\rho_{g,l}, u_g, p_g)$. The ghost fluid cell holds state $W_{g,gl} = (\rho_{g,gl}, u_g, p_g)$. 34

4.1 One-dimensional Sod shock-tube solutions at t = 0.25 computed with the Riemann GFM, Real GFM and Original GFM. Panels show the spatial profiles of (a) density ρ, (b) velocity u, (c) pressure p and (d) specific internal energy e. A CFL number of 0.8 was used. 38

4.2 Karni–Quirk test at $t = 0.3$ s (400 cells, CFL = 0.8). Red step-line = reference; blue dots = Riemann GFM, green square = Real GFM, purple triangle = Original GFM. 39

4.3 Comparison of numerical and reference solutions for the 1D water–air shock tube test at final time $t = 0.000229$. Shown are the profiles of density ρ, velocity u, pressure p, and total energy E. The numerical results were obtained using the Riemann GFM (blue) and the Real GFM (green), and are compared against the exact reference solution (red). The simulation was performed on a uniform grid with 400 cells and a CFL number of 0.8. . . . 41
4.4 1D underwater shock on a gas–water interface with CFL=0.9, run for 150 time steps, mesh resolution: 200 cells. 42
4.5 Comparison along $y = 1$ at $t = 0.25\,\mathrm{s}$ for the 2D cylindrical-explosion test (mesh 300×300, CFL=0.8). Panels show density ρ, velocity u, pressure p, and internal energy e. 43
4.6 Surface plots of density (left) and pressure (right) at $t = 0.25\,\mathrm{s}$ for the 2D cylindrical-explosion test on a 300×300 grid (CFL=0.8). Shown are results from (a) the Riemann GFM, (b) the Real GFM, and (c) the Original GFM. . 44
4.7 Evolution of the two-phase interface under a planar shock computed with the Riemann GFM on a 1000×394 grid (CFL = 0.8). Shown are density contours (left column) and corresponding shock–schlieren images (right column) at four successive time instants. 46
4.8 Evolution of the two-phase interface under a planar shock computed with the Real GFM on a 1000×394 grid (CFL = 0.8). Shown are density contours (left column) and corresponding shock–schlieren images (right column) at four successive time instants. 47
4.9 Evolution of the two-phase interface under a planar shock computed with the Original GFM on a 1000×394 grid (CFL = 0.8). Shown are density contours (left column) and corresponding shock–schlieren images (right column) at four successive time instants. 48

7.1 One-dimensional Sod shock-tube solutions at t = 0.25 computed with the Interface GFM, Practical GFM and Modified GFM. Panels show the spatial profiles of (a) density ρ, (b) velocity u, (c) pressure p and (d) specific internal energy e. A CFL number of 0.8 was used. 54
7.2 Karni–Quirk test at $t = 0.3\,\mathrm{s}$ (400 cells, CFL $= 0.8$). Red step-line = reference; blue dots = Interface GFM, green square = Practical GFM, purple triangle = Modified GFM. 55

7.3 Comparison of numerical and reference solutions for the 1D water–air shock tube test at final time $t = 0.000229$. Shown are the profiles of density ρ, velocity u, pressure p, and total energy E. The numerical results were obtained using the Interface GFM (blue), the Practical GFM (green), the Modified GFM (purple) and are compared against the exact reference solution (red). The simulation was performed on a uniform grid with 400 cells and a CFL number of 0.8. 56
7.4 1D underwater shock on a gas–water interface with CFL=0.9, run for 150 time steps, mesh resolution: 200 cells. 57

8.1 Comparison along $y = 1$ at $t = 0.25\,\mathrm{s}$ for the 2D cylindrical-explosion test (mesh 300×300, CFL=0.8). Panels show density ρ, velocity u, pressure p, and internal energy e. 60
8.2 Surface plots of density (left) and pressure (right) at $t = 0.25\,\mathrm{s}$ for the 2D cylindrical-explosion test on a 300×300 grid (CFL=0.8). Shown are results from (a) the Interface GFM, (b) the Practical GFM, and (c) the Modified GFM. 61
8.3 Evolution of the two-phase interface under a planar shock computed with the Interface GFM on a 1000×394 grid (CFL = 0.8). Shown are density contours (left column) and corresponding shock–schlieren images (right column) at four successive time instants. 62
8.4 Evolution of the two-phase interface under a planar shock computed with the Practical GFM on a 1000×394 grid (CFL = 0.8). Shown are density contours (left column) and corresponding shock–schlieren images (right column) at four successive time instants. 63
8.5 Evolution of the two-phase interface under a planar shock computed with the Modified GFM on a 1000×394 grid (CFL = 0.8). Shown are density contours (left column) and corresponding shock–schlieren images (right column) at four successive time instants. 64

9.1 Comparison of numerical results for Toro Test 2 using Original GFM, Real GFM, and Riemann GFM. The four panels show the profiles of density, velocity, pressure, and internal energy at $t = 0.15$ with mesh resolution 300×1 and CFL $= 0.9$. The reference solution (red line) corresponds to the exact Riemann solution. 66

9.2 Comparison of numerical results for Toro Test 2 using Interface GFM, Practical GFM, and Modified GFM. The four panels show the profiles of density, velocity, pressure, and internal energy at $t = 0.15$ with mesh resolution 300×1 and CFL $= 0.9$. The reference solution (red line) corresponds to the exact Riemann solution. 67
9.3 Comparison of numerical results for Toro Test 5 using Original GFM, Real GFM, and Riemann GFM. The four panels show the profiles of density, velocity, pressure, and internal energy at $t = 0.012$ with mesh resolution 300×1 and CFL $= 0.9$. The reference solution (red line) corresponds to the exact Riemann solution. 68
9.4 Comparison of numerical results for Toro Test 5 using Interface GFM, Practical GFM, and Modified GFM. The four panels show the profiles of density, velocity, pressure, and internal energy at $t = 0.012$ with mesh resolution 300×1 and CFL $= 0.9$. The reference solution (red line) corresponds to the exact Riemann solution. 69

List of tables

2.1 Physical interpretation of the level set function $\phi(x,t)$ 14

3.1 Assignment of ghost cell values based on level set sign and Riemann solver results. 26

4.1 Setup for the Sod shock-tube benchmarks in 2D 37
4.2 initial condition of the Karni and Quirk's test 39
4.3 Initial condition and numerical setup for liquid-gas shock interaction 40
4.4 Underwater shock impacting on a gas–water interface – initial conditions and numerical settings 42
4.5 Initial conditions and numerical setup for the 2-D cylindrical-explosion test. 43
4.6 Initial conditions and numerical setup for the shock–bubble interaction test . 45

9.1 Initial conditions for Test 2 (Double Rarefaction Case). Mesh resolution: 300×1, CFL = 0.9, $t_{\text{end}} = 0.15$. 65

9.2 Initial conditions and numerical setup for Test 5 68
10.1 Algorithmic kernels of the six GFMs. 71

10.2 Five practical evaluation dimensions used in the comparative study. 72
10.3 Benchmark suite for GFM comparison. 73
10.4 Summary of observed advantages, weaknesses, and practical recommendations for the six GFMs. 78

Chapter 1

Introduction and background

1.1 Motivation for Multi-Fluid Simulations

Multi-fluid simulations are widely used in engineering and scientific research, especially for handling complex systems that involve the interaction of different fluids [1]. These simulations are essential for understanding and predicting the behavior of multiphase flows, such as in chemical, energy, environmental, and biomedical fields.

In chemical processes, multi-fluid simulations are used to model gas-liquid, liquid-liquid, or solid-liquid systems. Examples include particle behavior in fluidized bed reactors, gas-liquid exchange in bubble columns, and droplet formation and breakup in emulsions. These simulations help optimize reactor design, improve mass transfer efficiency, and reduce energy consumption[2].

In the energy sector, especially in the oil and gas industry, multi-fluid simulations are used to study three-phase flows of oil, water, and gas. They help engineers design better pipelines and separation devices [3]. In nuclear and renewable energy systems, these simulations support the analysis of coolant flow, phase change, and heat transfer processes, improving system safety and efficiency.

In environmental engineering, multi-fluid simulations are used to model pollutant transport, deposition, and resuspension in water bodies. They support water quality management and pollution control [4]. In atmospheric science, simulations help study aerosol formation, transport, and deposition, which are important for air quality prediction and climate research.

In biomedical engineering, multi-fluid simulations help study blood flow, drug delivery, and tissue engineering problems. For example, simulating blood flow in artificial heart valves helps optimize valve design and reduce the risk of blood clots [5]. Simulations are also used to analyze how aerosol drugs deposit in the respiratory tract, guiding the development of inhaled therapies [6].

1.2 Sharp Interface Method

In multi-fluid simulations, we often need to describe clearly the interface between different fluids. The sharp-interface method helps track or capture exactly where the interface is. This method is important because it clearly shows the big differences in properties like density, pressure, and speed on each side of the interface [7]. It is very useful for simulations where the interface has clear, sudden changes.

Sharp-interface methods can be put into two main groups:

1. **Interface Tracking:** This method explicitly tracks the location of the interface. For example, the Front Tracking method uses points or small surfaces to represent the interface [8]. The main advantage is accuracy, especially when the interface shape does not change much. However, when the interface breaks or merges, this method becomes complicated and hard to manage [9].

2. **Interface Capturing:** This method captures the interface implicitly. Common techniques are Volume of Fluid (VOF) and Level-Set methods [10]. These methods naturally handle complex changes at the interface, like breaking and merging. The VOF method records how much fluid is in each grid cell. The Level-Set method represents the interface using a special function called the signed-distance function.

Among these capturing methods, the **Level-Set method** is widely used in multi-fluid simulations. It enables a combination of the two approaches mentioned above [11]. The main idea of the Level-Set method is to define a function $\phi(x,t)$ to represent the interface implicitly:

$$\phi(x,t) = 0 \tag{1.1}$$

shows the interface location, while

$$\phi(x,t) > 0 \quad \text{and} \quad \phi(x,t) < 0 \tag{1.2}$$

show the regions of different fluids. Usually, ϕ is chosen as the shortest distance from any point to the interface. The sign (positive or negative) indicates which side of the interface the point is on.

The Level-Set method is very good at handling complex interface changes automatically, without special treatment. It also easily calculates the normal vectors and curvature at the interface, useful in complex simulations like those involving surface tension. However, the method has some drawbacks, such as numerical diffusion, which means the interface might

become less sharp and accurate over time [12]. More details of the level set method will be discussed in Section 2.3.

1.3 Development of Ghost Fluid Methods

As noted in [11], the idea of Ghost Fluid Method(GFM) is initially introduced by Glimm et al.[13], which has emerged as an efficient and versatile approach for numerical treatment of multi-fluid flows.

Later, numerous variants and improvements based on GFM have been proposed and applied in different problems. By defining ghost fluid states at the interfaces, this approach allows multi-fluid problems to be solved using a single-phase numerical scheme. In this section, we will discuss the development of GFM after 2000.

Fedkiw et al. initially proposed the Original Ghost Fluid Method(OGFM) [14], where the ghost fluid state is constructed by using the local fluid velocity and pressure from the real fluid region. Although the OGFM is easy to implement and is convenient to extend to 2D, it still cannot accurately capture the complex interface dynamics when the interface is subjected to strong shock impacts, often resulting in non-physical oscillations and numerical errors near the interface.

To more accurately account for the complex interactions of waves and diverse material properties at interfaces, Liu et al. proposed the Modified Ghost Fluid Method (MGFM)[15]. This approach integrates characteristic analysis of waves as they reach the interface, combined with the local Riemann problem solution.

Inspired by this Riemann-probem-based strategy, more advancements include the Interface-interaction Ghost Fluid Method (IGFM)[11] and the Real Ghost Fluid Method (RGFM) [16], both of which aim to enhance the fidelity and robustness of the interface capturing.

Moreover, previous studies have shown that MGFM achieves a third-order accurate error estimate near the interface when applied to multi-medium Riemann problems. To simplify the implementation while retaining the accuracy, the Practical Ghost Fluid Method (PGFM) [17] was proposed. Similar to the MGFM, the PGFM maintains the third-order accuracy for the interface errors regardless of the solution type. However, the PGFM significantly simplifies the definition of ghost fluid states. Unlike MGFM, which requires specifying all interface quantities: pressure, velocity, and density, the PGFM only requires specifying the interfacial velocity.

1.4 Objectives of this project

This report (Report 1) focuses on the sharp-interface problem in multiphase flows. Using a 2D finite-volume scheme with MUSCL–Hancock reconstruction and an HLLC Riemann solver, we implement and compare three ghost fluid methods, Original GFM, Riemann GFM, and Real GFM on four 1D tests (the classic Sod shock tube, the Karni–Quirk oscillation problem, the water–air shock tube, and an underwater shock interacting with a gas–water interface) and two 2D tests (a cylindrical explosion and a shock-bubble deformation). We compare each method's shock-location accuracy, oscillation suppression at the contact, interface stability, and ability to capture complex structure evolution, then summarize their strengths and limitations in different flow scenarios.

In the next report (Report 2), we will implement the Modified GFM, Practical GFM, and Interface GFM and evaluate their performance.

The report is structured as follows: Chapter 1 introduces the motivation for multiphase flow simulation, the sharp-interface concept, the development of GFMs, and the project objectives. Chapter 2 describes the 2D Euler equations, the second-order finite-volume scheme (time stepping, MUSCL–Hancock reconstruction, HLLC solver), the level-set and fast-sweeping methods, and the equations of state. Chapter 3 presents the algorithms and implementation details of Original GFM, Riemann GFM, and Real GFM, Chapter 4 shows the results of the six tests (Sections 4.1–4.6) with both qualitative and quantitative analysis. Chapter 5 summarizes our main conclusions, discusses limitations, and outlines the work planned for the next report. and Appendix A details the OpenMP parallelization.

Chapter 2

Numerical Methods

2.1 Euler Equations

In this project, we use the Euler equations to model how fluids move. These equations are used for fluids that have no viscosity and can be compressed. They are based on three basic physical laws: conservation of mass, conservation of momentum, and conservation of energy. Because the Euler equations are written in a conservative form, they are good for finite-volume methods, which we use in our simulation. Below, we show the Euler equations in their standard form.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{2.1}$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + p\mathbf{I}) = 0 \tag{2.2}$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\mathbf{v}] = 0 \tag{2.3}$$

where ρ is fluid density, $\mathbf{v}$ is velocity, p is pressure, and E is total energy per unit volume. We can calculate the total energy by summing the internal energy and the kinetic energy:

$$E = \rho e + \frac{1}{2}\rho v^2 \tag{2.4}$$

where e is the specific internal energy given by the Equation of State (EOS).

$$e = e(\rho, p) \tag{2.5}$$

There are many different EOS depend on different materials we simulate, more details about different EOS are given in Section 2.5.

We can also write the Euler equations in two dimensions, where the fluid can move in both the x- and y-directions. This will be explained later in this section.

2.1.1 One-Dimensional Euler Equations

In one-dimensional problems, the Euler equations describe how fluid properties change in space x and time t. They can be written in the conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} = 0 \tag{2.6}$$

where $\mathbf{U}$ is the vector of conserved variables, and $\mathbf{F}(\mathbf{U})$ is the flux vector. For compressible flow, these are defined as:

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho v \\ E \end{bmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho v \\ \rho v^2 + p \\ v(E+p) \end{bmatrix}$$

2.1.2 Two-Dimensional Euler Equations

In two-dimensional problems, the fluid can move in both the x- and y-directions. The Euler equations in 2D describe the conservation of mass, momentum, and energy in space and time. They can be written in the conservative form as:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = 0 \tag{2.7}$$

Here, $\mathbf{U}$ is the vector of conserved variables, and $\mathbf{F}$ and $\mathbf{G}$ are the fluxes in the x- and y-directions, respectively. These are defined as:

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ E \end{bmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(E+p) \end{bmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(E+p) \end{bmatrix}$$

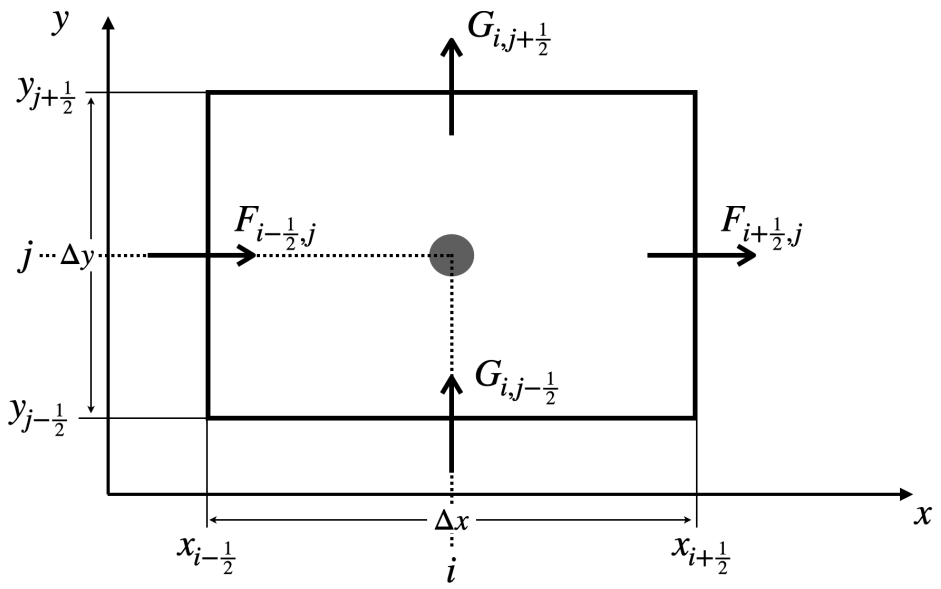


Fig. 2.1 Schematic of the control volume associated with grid node (i, j) in a two-dimensional finite-volume discretization. The rectangle spans the interval $\left(x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}\right) \times \left(y_{j-\frac{1}{2}}, y_{j+\frac{1}{2}}\right)$, with the gray dot marking the cell center. Arrows denote the numerical fluxes: $F_{i\pm\frac{1}{2},j}$ across the left/right faces and $G_{i,j\pm\frac{1}{2}}$ across the bottom/top faces, where the arrow direction defines the positive sign convention. The spacings Δx and Δy indicate the grid widths in the x and y directions, respectively, and the dotted lines show the coordinate lines through the cell center.

Figure 2.1 shows the placement of fluxes around a 2D grid. $\mathbf{F}(\mathbf{U})$ is evaluated on the horizontal cell faces, which accounts for transport in the $x-$direction, whereas $\mathbf{G}(\mathbf{U})$ is evaluated on the vertical faces, which represents transport in the $y-$ direction.

The face-centred numerical fluxes that enter the cell $F_{i+\frac{1}{2},j}$ and $G_{i,j+\frac{1}{2}}$ are computed with the HLLC approximate Riemann solver, the formulas are given in Section 2.2.5.

In order to simplify the numerical treatment of the two-dimensional Euler equations, especially when dealing with fluxes in both directions, operator splitting techniques are often employed. This approach allows the 2D problem to be decomposed into a sequence of 1D problems, typically solving fluxes in the x-direction and y-direction separately at each time step. In the following, we adopt a Strang splitting strategy to alternate between x- and y-direction updates, enabling efficient and accurate integration of the 2D system.

Firstly we rewrite the governing equation in Eqn.(2.7) as Eqn.(2.8):

$$\mathbf{U}_t + \mathbf{F}(\mathbf{U})_x + \mathbf{G}(\mathbf{U})_y = 0 \tag{2.8}$$

This splitting strategy can be expressed in operator form as:

$$\mathbf{U}^{n+1} = \mathcal{Y}^{(\Delta t)}\left(\mathcal{X}^{(\Delta t)}\left(\mathbf{U}^{n}\right)\right) \tag{2.9}$$

where $\mathscr{X}^{(\Delta t)}$ and $\mathscr{Y}^{(\Delta t)}$ are flux update operators that operate over one full time step Δt. In practice, this means we first update x-direction:

$$\mathbf{U}_t + \mathbf{F}(\mathbf{U})_x = 0, \tag{2.10}$$

and then update y-direction:

$$\mathbf{U}_t + \mathbf{G}(\mathbf{U})_y = 0, \tag{2.11}$$

The sequential application in Eqn.2.9-2.11 is first order accurate in time. In this project, we achieve a second order accuracy in time by using the following equation:

$$\mathbf{U}^{n+1} = \frac{1}{2}\left[\mathscr{X}^{(\Delta t)}\mathscr{Y}^{(\Delta t)} + \mathscr{Y}^{(\Delta t)}\mathscr{X}^{(\Delta t)}\right]\left(\mathbf{U}^{n}\right), \tag{2.12}$$

2.2 Second-order Finite Volume Scheme

To solve the Euler equations numerically, we use the finite volume method. This method works by dividing the domain into small cells and computing the average value of each variable in each cell. Then we update these values using the fluxes across the cell boundaries. The finite volume method is especially useful for solving conservation laws, because it ensures that mass, momentum, and energy are conserved.

A simple finite volume method gives only first-order accuracy in space, which means it can produce large numerical diffusion, especially near sharp changes like shock waves. To improve the accuracy, we use a second-order scheme. This involves reconstructing the values at the cell edges using linear functions and applying a slope limiter to avoid oscillations. We also use a better time-stepping method to improve accuracy in time. These techniques together form a second-order accurate scheme in both space and time.

In this project, we use the MUSCL-Hancock scheme to achieve second-order accuracy, and apply the HLLC Riemann solver to compute the numerical fluxes at cell interfaces.

2.2.1 Time Step Calculation

To maintain numerical stability, the time step Δt is selected based on the *Courant–Friedrichs–Lewy (CFL)* condition, which ensures that information does not travel more than one cell per time

step. In two-dimensional simulations, the time step is typically computed as:

$$\Delta t = C\frac{\min(\Delta x, \Delta y)}{a_{\max}}, \tag{2.13}$$

where C is the CFL number, typically chosen such that $C \leq 1$. The grid spacings in the x- and y-directions are denoted by Δx and Δy, respectively. The quantity $a_{\max}$ represents the maximum wave propagation speed and is given by:

$$a_{\max} = \max_{i,j} \left(|\mathbf{v}_{i,j}| + c_{s,i,j} \right), \tag{2.14}$$

where $|\mathbf{v}_{i,j}|$ is the local velocity magnitude and $c_{s,i,j}$ is the corresponding local sound speed. Since this definition accounts for the fastest signal speeds in both directions, a single uniform time step can be applied consistently throughout the computational domain.

2.2.2 Data Reconstruction

To update the solution in each cell using the finite volume method, we must compute the numerical flux at the interface between cells. This flux is obtained by solving a local Riemann problem at each interface. The Riemann problem describes the evolution of a fluid system that starts with two different constant states separated by a discontinuity. The initial condition is given by:

$$\mathbf{U}(x,0) = \begin{cases} \mathbf{U}_L, & x \leq 0 \\ \mathbf{U}_R, & x > 0 \end{cases} \tag{2.15}$$

where $\mathbf{U}_L$ and $\mathbf{U}_R$ are the fluid states on the left and right sides of the interface.

To provide the inputs for the Riemann solver, we first reconstruct these left and right states by using a linear approximation inside each cell. This step increases the accuracy of the scheme by estimating how the solution varies across each cell. Specifically, we assume that the fluid state changes linearly within each cell. For any cell i, the fluid state U at any position x within the cell can be expressed as:

$$U_i(x) = U_i^n + (x - x_i)\frac{\Delta_i}{\Delta x} \tag{2.16}$$

where U_i^n is the state at the center of cell i at time step n, x_i is the location of the cell center, Δx is the width of the cell, and Δ_i is the slope of the fluid state, representing how quickly the fluid state changes within the cell.

This slope is calculated using the states of neighboring cells to better capture the flow features:

$$\Delta_i = \frac{1}{2}(1+\omega)(U_i - U_{i-1}) + \frac{1}{2}(1-\omega)(U_{i+1} - U_i) \tag{2.17}$$

where the parameter $\omega \in [-1,1]$ controls the smoothness of reconstruction, with $\omega = 0$ typically used to give a balanced approximation.

Once the slope is determined, we can define the left and right interface states in cell i, which serve as inputs for the Riemann solver:

$$U^i_{L,i} = U^n_i - \frac{1}{2}\Delta_i, \quad U^i_{R,i} = U^n_i + \frac{1}{2}\Delta_i \tag{2.18}$$

However, such a straightforward linear reconstruction can introduce numerical oscillations near sharp changes in the fluid state, such as shock waves. To prevent these unrealistic fluctuations, we introduce a slope limiter, discussed in the next section.

2.2.3 Slope Limiting

The slope limiter is essential for producing reliable and oscillation-free solutions by modifying the raw slope Δ_i. The first step is to compute the slope ratio by using the formula below:

$$r_i = \frac{\Delta_{i-1/2}}{\Delta_{i+1/2}}, \tag{2.19}$$

The next step is to apply the slope limiter function $\zeta(r_i)$ to modify the slope in each cell. This function adjusts the slope based on the smoothness of the solution, measured by the ratio r_i. A good limiter helps reduce oscillations near discontinuities without reducing accuracy in smooth regions. In this project, we use the Minbee limiter, which is defined as:

$$\zeta(r) = \begin{cases} 0, & r \leq 0 \\ r, & 0 < r \leq 1 \\ \min\left(1, \frac{2}{1+r}\right), & r > 1 \end{cases} \tag{2.20}$$

After applying the limiter, we redefine the left and right states using the limited slope $\zeta(r_i)\Delta_i$ as below:

$$U_{L,i} = U^n_i - \frac{1}{2}\zeta(r_i)\Delta_i, \quad U_{R,i} = U^n_i + \frac{1}{2}\zeta(r_i)\Delta_i \tag{2.21}$$

This step ensures the reconstruction is more stable and accurate, especially near sharp gradients.

2.2.4 Half-time Update

After applying the slope limiter, the reconstructed values become more accurate and stable in space. However, to achieve full second-order accuracy, we must also improve the time integration. If we update the solution directly using the current states, time discretization errors will still exist.

To reduce these errors, we apply the *Half-Time Update* method. This approach estimates the state at the midpoint of the time step ($t^{n+\frac{1}{2}}$) by evolving the left and right interface values forward in time by half a step. The update formulas are:

$$\bar{\mathbf{u}}_i^{L,n+\frac{1}{2}} = \bar{\mathbf{u}}_i^{L,n} - \frac{1}{2}\frac{\Delta t}{\Delta x}\left(f(\bar{\mathbf{u}}_i^{R,n}) - f(\bar{\mathbf{u}}_i^{L,n})\right) \tag{2.22}$$

$$\bar{\mathbf{u}}_i^{R,n+\frac{1}{2}} = \bar{\mathbf{u}}_i^{R,n} - \frac{1}{2}\frac{\Delta t}{\Delta x}\left(f(\bar{\mathbf{u}}_i^{R,n}) - f(\bar{\mathbf{u}}_i^{L,n})\right) \tag{2.23}$$

Once the half-step values are computed, we use them to update the final cell average at the next time level:

$$\mathbf{u}_i^{n+1} = \mathbf{u}_i^{n} - \frac{\Delta t}{\Delta x}\left(f(\mathbf{u}_{i+\frac{1}{2}}) - f(\mathbf{u}_{i-\frac{1}{2}})\right) \tag{2.24}$$

This ensures that the numerical scheme is second-order accurate in both space and time. This technique, together with the reconstruction and limiter steps, forms the basis of the *MUSCL-Hancock scheme*.

2.2.5 HLLC Riemann Solver

After obtaining the left and right states through reconstruction and slope limiting, the next step is to compute the numerical flux across each interface. This is done by solving the Riemann problem at the cell boundaries.

In this project, we use the HLLC (Harten–Lax–van Leer–Contact) Riemann solver. The HLLC Riemann solver is based on an approximate solution of the Riemann problem that includes three waves: the left-going wave S_L, the right-going wave S_R, and the contact wave S_* in the middle. This structure allows the solver to capture contact discontinuities, which are not resolved by the simpler HLL method. Figure 2.2 shows this arrangement.

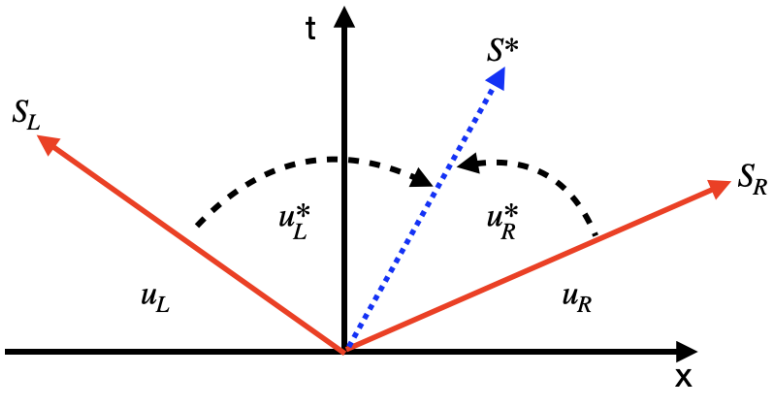


Fig. 2.2 Wave structure in the HLLC Riemann solver. Within the star region, the solution is composed of two uniform states that are separated by a middle wave traveling at speed S^*

The HLLC method works the same way in both the x and y directions. So in this section, we only explain how it works in the x direction. The same steps can be used for the y direction.

According to this wave structure displayed in Figure 2.2, we can get the following equation:

$$\mathbf{u}(x,t) = \begin{cases} u_L & \text{if } S_L \geq 0 \\ u_L^* & \text{if } S_L < 0 \leq S^* \\ u_R^* & \text{if } S_* < 0 \leq S_R \\ u_R & \text{if } S_R < 0 \end{cases} \tag{2.25}$$

$$\mathbf{f}^*_{i+\frac{1}{2}} = \begin{cases} \mathbf{f}_L & \text{if } S_L \geq 0 \\ \mathbf{f}_L + S_L\left(\mathbf{u}_L^* - \mathbf{u}_L\right) & \text{if } S_L < 0 \leq S^* \\ \mathbf{f}_R + S_R\left(\mathbf{u}_R^* - \mathbf{u}_R\right) & \text{if } S^* < 0 \leq S_R \\ \mathbf{f}_R & \text{if } S_R < 0 \end{cases} \tag{2.26}$$

In Eqn.(2.25), $\mathbf{u}(x,t)$ denotes the approximate solution, where S_L, S^*, and S_R are wave speeds that separate the left, middle, and right states. The variables u_L and u_R represent the original left and right states, while u_L^* and u_R^* are the star-region states in the HLLC Riemann solver, and Eqn.(2.26) describes the corresponding HLLC flux.

Then, by applying the Rankine–Hugoniot Conditions, we can derive the following:

$$\begin{aligned} \mathbf{f}_L^* - \mathbf{f}_L &= S_L\left(\mathbf{u}_L^* - \mathbf{u}_L\right) \\ \mathbf{f}_R^* - \mathbf{f}_L^* &= S^*\left(\mathbf{u}_R^* - \mathbf{u}_L^*\right) \\ \mathbf{f}_R - \mathbf{f}_R^* &= S_R\left(\mathbf{u}_R - \mathbf{u}_R^*\right) \end{aligned} \tag{2.27}$$

Solving Eqn.(2.27) we can get the wave speed of the contact discontinuity:

$$S^* = \frac{p_R - p_L + \rho_L v_L(S_L - v_L) - \rho_R v_R(S_R - v_R)}{\rho_L(S_L - v_L) - \rho_R(S_R - v_R)} \tag{2.28}$$

where p_L and p_R are the pressures in the left and right states, and ρ_L and ρ_R are the corresponding densities. we can get the speed of two shock waves S_L and S_R by:

$$S^+ = \max\left(|v_L| + c_{s,L},\ |v_R| + c_{s,R}\right), \quad S_L = -S^+, \quad S_R = S^+ \tag{2.29}$$

Then we can get the intermediate state by using:

$$\mathbf{u}_K^* = \rho_K \frac{S_K - v_K}{S_K - S^*} \begin{pmatrix} 1 \\ S^* \\ \frac{E_K}{\rho_K} + (S^* - v_K)\left[S^* + \frac{\rho_K}{\rho_K(S_K - v_K)}\right] \end{pmatrix} \tag{2.30}$$

2.2.6 The MUSCL-Hancock Scheme

The MUSCL–Hancock method begins with data reconstruction and slope limiting (see Sections(2.2.2-2.2.3) to compute left and right interface values within each cell. These reconstructed values are then advanced by a half time step using the procedure described in Eq.(2.22), Eq.(2.23) (see Section2.2.4), producing the half-time states $\bar{u}_i^{L,n+\frac{1}{2}}$ and $\bar{u}_i^{R,n+\frac{1}{2}}$.

These intermediate states are passed to the HLLC Riemann solver (Section2.2.5) to calculate the numerical fluxes at each interface. With these fluxes, the solution is updated over a full time step Δt. In 1D, we use the update Eq.(2.24) (see Section2.2.4). This completes one MUSCL–Hancock update from time level t^n to t^{n+1}. For 2D problems, the same approach is applied sequentially in each coordinate direction.

2.3 Level Set Method

The Level Set Method is a numerical approach for tracking interfaces in multiphase flow problems. Instead of following the interface directly with a moving mesh, this method implicitly captures its motion using a scalar function $\phi(x,y)$ defined across the entire domain. This function represents the signed distance from any point to the interface.

Table 2.1 illustates different meaning of the ϕ function.

This signed distance representation is particularly useful because many subsequent numerical operations such as interpolating values between neighboring cells or computing interface normals depend on knowing the exact distance from each grid cell to the interface.

Value of $\phi(x,t)$	Physical Meaning
$\phi(x,t) > 0$	Inside fluid1
$\phi(x,t) = 0$	On the interface
$\phi(x,t) < 0$	Inside fluid2

Table 2.1 Physical interpretation of the level set function $\phi(x,t)$

Keeping ϕ as a signed distance function ensures geometric accuracy and consistency in these calculations.

The time evolution of the level-set function is governed by a Hamilton-Jacob equation of advection form:

$$\frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi = 0 \tag{2.31}$$

where $\mathbf{u}(x,y,t)$ is the local fluid velocity. Physically, Eqn.(2.31) states the zero-level contour, that is, the material interface.

While Eqn.(2.31) advances the position of the interface, many subsequent operations, such as constant extrapolation which introduced in Section 3.1, require its orientation. This orientation can be expressed through the unit normal vector, which can be obtained from the level-set function:

$$\mathbf{n} = \frac{\nabla \phi}{\|\nabla \phi\|}, \qquad \|\nabla \phi\| = \sqrt{\left(\partial \phi / \partial x\right)^2 + \left(\partial \phi / \partial y\right)^2 \left(+\left(\partial \phi / \partial z\right)^2\right)} \tag{2.32}$$

For the numerical solution of Eqn.(2.31), a first-order upwind scheme is employed, that is, choosing the differencing stencil according to the sign of the corresponding velocity.

$$\left(\frac{\partial \phi}{\partial x}\right)_i^n = \begin{cases} \dfrac{\phi_{i,j,k} - \phi_{i-1,j,k}}{\Delta x}, & u_{x,i}^n > 0, \\ \dfrac{\phi_{i+1,j,k} - \phi_{i,j,k}}{\Delta x}, & u_{x,i}^n < 0, \\ 0, & u_{x,i}^n = 0. \end{cases} \tag{2.33}$$

Eqn.(2.33) shows the upwind scheme used in solving the advection equation of the level set function ϕ.

2.4 Fast Sweeping Method

During long-time advection, the level-set function ϕ gradually loses its signed-distance property. Numerical diffusion may smear the zero contour. Even a small perturbation in ϕ

can shift the interface by severl grid cells. To restore accuracy, a reinitialisation step of ϕ is employed. i.e. forces it back to a signed-distance function while keeping the location of the interface unchanged.

Mathematically, reinitialisation seeks a steady solution of the Eikonal equation:

$$|\nabla\phi| = 1, \tag{2.34}$$

subject to the constraint $\phi = 0$ on the interface. In this project, we employ a Fast Sweeping Method (FSM)[18] to carry out the reinitialisation process.

Within the computational domain, each grid point is marked as either "known" (interface points) or "unkown" (non-interface points). FSM updates unknown points in a seires of alternating sweeps: first along $+x$, then $-x$, following by $+y$ and $-y$, until convergence. During a sweep, $\phi_{i,j}$ is computed from its up-to-date neighbours ϕ_x, ϕ_y by enforcing Eqn.(2.34) locally:

$$\left(\frac{\widehat{\phi}_{i,j}-\phi_x}{\Delta x}\right)^2 + \left(\frac{\widehat{\phi}_{i,j}-\phi_y}{\Delta y}\right)^2 = 1, \tag{2.35}$$

If the resulting $\widehat{\phi}_{i,j}$ is smaller in magnitude than the current value, it is accepted; otherwise the original value is retained. The complete procedure is summarised in Algorithm 1.

Algorithm 1 Fast Sweeping Method

for *iterate through all cells* **do**
  if *cell* $i, j \neq$ *interface cell* **then**
    $\phi_{i,j} = \phi_{\text{unknown}} = 999$ (any sufficiently large value is suitable.)
  end
end
for *x and y,* $+$*ve or* $-$*ve sweeps* **do**
  if *cell* $i, j \neq$ *interface cell* **then**
    if $\phi_{i,j} > 0$ **then**
      define mininum x and y-neighbour:
      $\phi_x = \min(\phi_{i+1,j}, \phi_{i-1,j})$
      $\phi_y = \min(\phi_{i,j+1}, \phi_{i,j-1})$
      plug ϕ_x, ϕ_y into Eqn.(2.35) to find the solution $\widehat{\phi}$.
      if *square root term of the solution* $\widehat{\phi} < 0$ *(i.e. ill-defined)* **then**
        remove $\max(\phi_x, \phi_y)$
      else
        if $\widehat{\phi}_{i,j} < \phi_{i,j}$ **then**
          $\phi_{i,j} = \widehat{\phi}_{i,j}$
        else
          $\phi_{i,j}$ remains
        end
      end
    else
      define maximum x and y-neighbour:
      $\phi_x = \max(\phi_{i+1,j}, \phi_{i-1,j})$
      $\phi_y = \max(\phi_{i,j+1}, \phi_{i,j-1})$
      plug ϕ_x, ϕ_y into Eqn.(2.35) to find the solution $\widehat{\phi}$.
      if *square root term of the solution* $\widehat{\phi} < 0$ **then**
        remove $\max(\phi_x, \phi_y)$
      else
        if $\widehat{\phi}_{i,j} > \phi_{i,j}$ **then**
          $\phi_{i,j} = \widehat{\phi}_{i,j}$
        else
          $\phi_{i,j}$ remains
        end
      end
    end
  else
    ϕ stays constant
  end
end

Algorithm(1) outlines the whole procedure used in FSM. In essence, reinitialisation is done for $\phi > 0$ and $\phi < 0$, only non-interface cells are updated, while values adjacent to the interface remain fixed.

2.5 Equations of State

We will use the following equations of state in the numerical tests.

2.5.1 Ideal Gas

For an ideal gas, the pressure p is related to the density ρ and specific internal energy e by:

$$p \;=\; \rho\, e\,(\gamma - 1) \tag{2.36}$$

where the adiabatic index is set to: $\gamma = 1.4$ for air, and $\gamma = 1.67$ for helium (which we may use in the numerical tests later).

The Energy E is the sum of kinetic energy and internal energy:

$$E = \frac{1}{2}\rho v^2 + \rho e \tag{2.37}$$

In 2D, the velocity $v = \sqrt{(v_x^2 + v_y^2)}$. The corresponding sound speed is:

$$c \;=\; \sqrt{\frac{\partial p}{\partial \rho}\Big|_s} \;=\; \sqrt{\frac{\gamma p}{\rho}} \tag{2.38}$$

2.5.2 Stiffened Gas

For liquid water we adopt the stiffened gas equation of state:

$$p \;=\; (\gamma - 1)\,\rho\, e - \gamma p_\infty \tag{2.39}$$

where p_∞ is a constant quantity which can be considered to represent the attraction between water molecules. For water, we take $p_\infty = 1 \times 10^9 Pa$ and $\gamma = 2.35$ from the literature.

The Energy is therefore become:

$$E = \frac{1}{2}\rho v^2 + \frac{p + \gamma p_\infty}{\gamma - 1} \tag{2.40}$$

and the sound speed is:

$$c = \sqrt{\frac{\partial p}{\partial \rho}\Big|_s} = \sqrt{\frac{\gamma(p+p_\infty)}{\rho}} \tag{2.41}$$

in this case.

Chapter 3

The Ghost Fluid Methods

Before presenting the individual GFMs, we first outline the interface detection and ghost-cell labelling procedure common to all GFMs.

As discussed in the level set method, the interface location is defined by the zero level set, i.e. $\phi = 0$. In this project, $\phi < 0$ represents Material 1, while $\phi > 0$ represents Material 2.

To locate the ghost cells, we check the level set values of adjacent cells. Specifically, if the product of the level set values of two neighbouring cells is negative, it indicates that the interface passes between them. Mathematically, the condition for identifying an interface cell at position (i, j) is:

$$\phi_{i,j}\phi_{i+1,j} < 0, or \quad \phi_{i,j}\phi_{i-1,j} < 0, or \quad \phi_{i,j}\phi_{i,j+1} < 0, or \quad \phi_{i,j}\phi_{i,j-1} < 0 \tag{3.1}$$

Once an interface cell is identified, it is labeled according to the sign of its ϕ value:

$$\text{The cell is labeled as } \begin{cases} \text{Material 1 ghost cell}, & \text{if } \phi_{i,j} > 0 \\ \text{Material 2 ghost cell}, & \text{if } \phi_{i,j} < 0 \end{cases} \tag{3.2}$$

A schematic diagram illustrating this procedure will be provided for further clarity.

As shown in Figure 3.1, the blue cells are labeled as ghost cells of Material 1 because their ϕ values are negative and adjacent to interface cells. Similarly, the red cells are labeled as Ghost cells of Material 2 because their ϕ values are positive and the interface also passes through them.

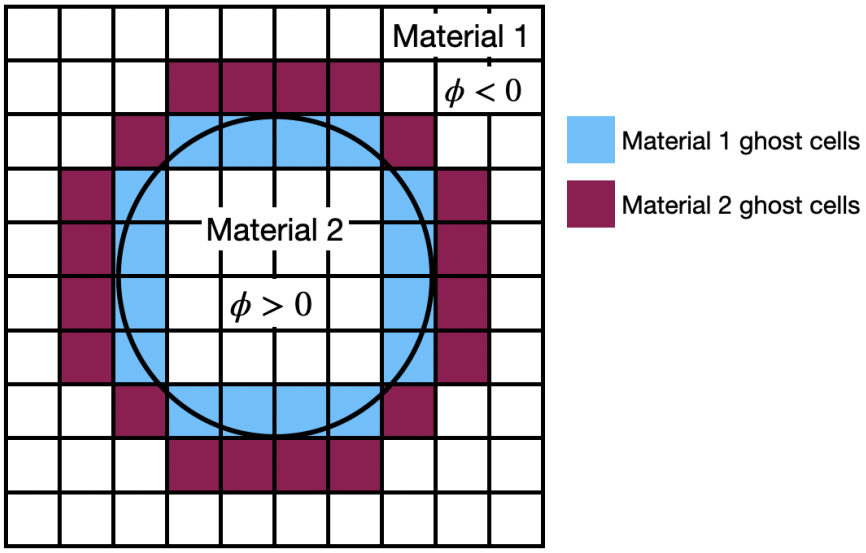


Fig. 3.1 Scheme of ghost cells indentification

3.1 The Original Ghost Fluid Method

In the family of GFMs the Original GFM is the earliest and most straightforward. Unlike the Riemann GFM and the Real GFM where a Riemann problem must be solved at every interface cell, the Original GFM constructs each ghost cell by a "copy and mix" of primitive variables from its neighbouring real cells. Specifically, it extracts density, tangential velocity and pressure from the real cell on its own side of the interface, copies pressure and normal velocity directly from the nearest real cell on the opposite side, computes the ghost cell density and reconstructs the full ghost cell velocity by combining the normal and the tangential components.

By completely avoiding ang Riemann solver, the Original GFM is extremely easy to implement and highly efficient, and it therefore provides a solid foundation for subsequent variants. In the sections that follow, we will first detail the algorithm and implementation of the Original GFM, and then introduce the Riemann GFM and the Real GFM in turn.

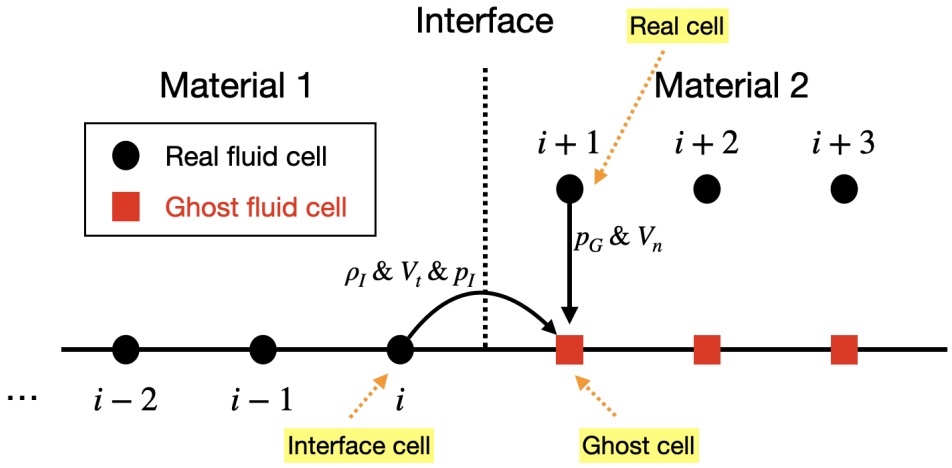


Fig. 3.2 Schematic of the Original Ghost Fluid Method at a 1D interface. The black circles are real fluid cells; the red squares are ghost cells. From the interface cell i, we copy its density ρ_I, tangential velocity $\mathbf{v}_{t,I}$, and pressure p_I. From the nearest real neighbor in the opposite material (cell $i+1$), we copy its pressure p_{i+1} and normal velocity $V_{n,i+1}$. The ghost-cell density is then computed via $\rho_G = \rho_I \left(p_{i+1}/p_I\right)^{1/\gamma_i}$, the ghost-cell velocity by $\mathbf{v}_G = V_{n,i+1}\,\mathbf{n} + \mathbf{v}_{t,I}$, and the ghost-cell pressure is set to $p_G = p_{i+1}$. No Riemann solver is required.

Figure 3.2 shows the scheme of the Original GFM. Hence the first step is to find the ghost cell by using the method described at the start of the chapter, then find its corresponding interface cell and extract its ρ, v_t, p.

To find the adjacent interface cell in the Original GFM, we examine all of the neighboring cells and aim to pick just one cell. (There are 8 cells in 2D, as illustrated in Figure 3.3.). We discard any neighbours that lie in the same material as the interface cell, and select one "most suitable" neighbour from the remaining.

That "suitability" is measured by how closely aligned the candidate cell is with the local interface normal $\mathbf{n}$.

Concretely, we compute the normal vector at both current interface cell and the candidate cell, $\mathbf{A}$ and $\mathbf{B}$ respectively, say. Then the cosine similarity can be calculated by:

$$\cos\theta = \frac{\mathbf{A}\cdot\mathbf{B}}{\|\mathbf{A}\|\,\|\mathbf{B}\|} = \mathbf{A}\cdot\mathbf{B}. \tag{3.3}$$

Since $\mathbf{A}$ and $\mathbf{B}$ are the unit normal vector ($\|\mathbf{A}\| = \|\mathbf{B}\| = 1$).

The neighbour with the largest value of cosine similarity (i.e. the one whose relative position aligns most directly with $\mathbf{n}$) is chosen as the adjacent cell.

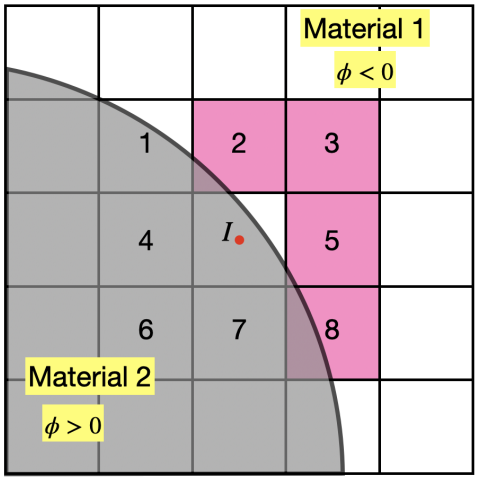


Fig. 3.3 Illustration of neighbor selection in the Original GFM: the interface cell I discards neighbours in the same material (grey) and picks the remaining cell whose center best aligns with the normal $\mathbf{n}$ (pink).

After we finding the interface cell using the method described above, we extract interface-side quantities: Density ρ_I, pressure p_I and tangential velocity $V_{t,I}$. and also extract quantities from the real-side: the pressure p and the normal velocity V_n. Since these two quantities are directly copied to the ghost cell, we write these two quantities as: p_G and $V_{n,G}$.

We then compute the density of the ghost cell by:

$$\rho_G = \rho_I \left(\frac{p_G}{p_I} \right)^{\frac{1}{\gamma_I}}, \tag{3.4}$$

and then reconstruct the full ghost cell velocity by combining:

$$\mathbf{v}_G = V_{n,G}\,\mathbf{n} + \mathbf{v}_{t,I}. \tag{3.5}$$

coupled with p_G from the real cell, we complete this procedure.

Therefore, each ghost cell inherits its thermodynamics (pressure and normal motion) from the real cell, and its material properties (density and tangential velocity) from the interface cell. No Riemann solver is needed, making the Original GFM simple to implement.

After finishing the procedure described above, we only set the values in cells adjacent to the interface. We have to extrapolate these values.

Consider an arbitrary variable Q, which could be any of the quantities from either type of GFMs. In this case, Q consists of the interface quantities $(\rho_I, \mathbf{v}_{t,I}, p_I)$. A constant

extrapolation aim to solve:

$$\mathbf{n} \cdot \nabla Q = 0 \tag{3.6}$$

where $\mathbf{n} = \frac{\nabla \phi}{|\nabla \phi|}$ is the normal computed from the level-set function. The equation states that the directional derivative of Q along $\mathbf{n}$ is zero. i.e. Q is constant along normal direction. This form form is closely analogous to the reinitialisation (Eikonal) equation (see Eqn(2.34)) which can be written in the similar form as above: $\mathbf{n} \cdot \nabla \phi = 1$. Hence, the numerical strategy we used earlier to reinitialise ϕ can be reused here.

We use the same fast-sweeping procedure presented in Algorithm1, the only change is that we now solve a linear equation:

$$n_x \frac{Q_{i,j} - Q_x}{\Delta x} + n_y \frac{Q_{i,j} - Q_y}{\Delta y} = 0 \tag{3.7}$$

instead of the quadratic equation that appeared in the level-set method.

Specifically, the procedure is:

1. Create a 4-component array Q whose entries at each interface cell (i, j): $Q_{i,j} = (\rho_I, v_{t,I,x}, v_{t,I,y}, p_I)$
2. Solve $\mathbf{n} \cdot \nabla Q = 0$ using the exact same algorithm (Algorithm1), but applied only to this 4-component array.
3. After sweeping, each ghost cell holds ectrapolated interface values $(\rho_I, v_{t,I,x}, v_{t,I,y}, p_I)$. At that point we copy in the opposide-side real pressure and normal velocity $(p_G, V_{n,G})$ and combine them to form the ghost-cell state exactly exactly as described above.

However, because the Original GFM relies solely on simple "copy and mix" of neighbouring states, it lacks a rigorous treatment of strong shocks and discontinuities and can produce oscillations or smearing in the presence of strong shocks. To overcome these limitations, the Riemann GFM, where a local Riemann problem is solved at each interface cell was subsequently proposed, and will be discussed in the next section.

3.2 The Riemann Ghost Fluid Method

The Riemann GFM augments the common ghost-cell framework by solving a one-dimensional Riemann problem at the interface. After locating an interface cell I as described at the atsrt of the chapter, we

1. Compute the unit normal vector $\mathbf{n}$ via Eqn.(2.32).

2. Obtain two points p_1 and p_2 .

3. Sample primitive variables on p_1 and p_2 to form the left and right states.

These two states define the Riemann problem at the interface.

To accurately place the left and right states, we have to locate the position on the interface closest to the center of the identified interface cell. Since ϕ is a signed distance function, the closet point x_p can be computed as:

$$x_p = x_I - \phi n \tag{3.8}$$

where x_I is the coordinate of the center of the interface cell.

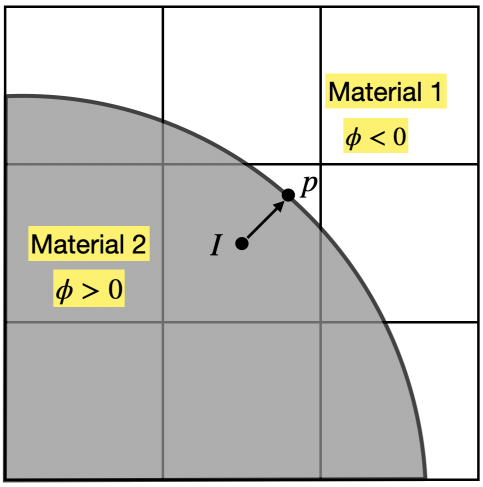


Fig. 3.4 Geometric illustration of the closest point. The cell center $\mathbf{x}_I$ lies inside the Material 2 (gray region), whose closest point $\mathbf{x}_P$ is computed along the normal direction $\mathbf{n}$.

Figure 3.4 shows a geometric construction of the closest point $\mathbf{x}_p$ on the interface from the center of an interface cell $\mathbf{x}_I$. The interface (indicated by the curved black line) separates Material 1 and Material 2.

Once the interface position $\mathbf{x}_p$ has been determined, we proceed to compute two points $\mathbf{x}_{p1}$ and $\mathbf{x}_{p2}$ that lie along the same normal direction n and are eqidistant from the interface. These points are defines as:

$$\begin{aligned} \mathbf{x}_{p1} &= \mathbf{x}_p + 1.5\Delta x \mathbf{n} \\ \mathbf{x}_{p2} &= \mathbf{x}_p - 1.5\Delta x \mathbf{n} \end{aligned} \tag{3.9}$$

Figure 3.5 illustrates this construction, showing how two interpolation points are symmetrically positioned around the interface.

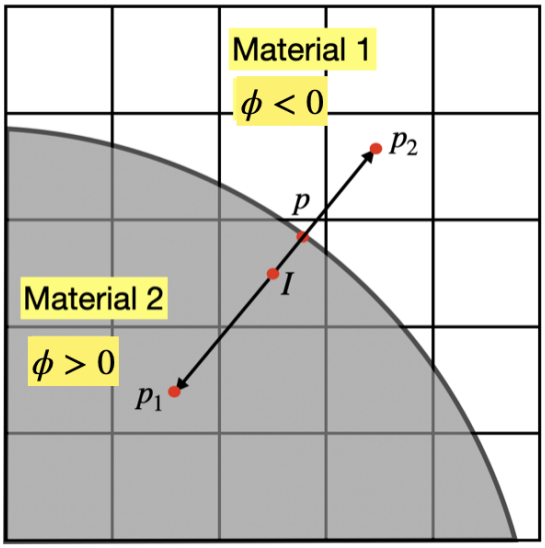


Fig. 3.5 Geometric illustration of interface-normal interpolation points used in the Riemann Ghost Fluid Method. The interface cell centered at $\mathbf{x}_I$ contains the interface, whose closest point $\mathbf{x}_P$ is projected along the local normal direction. Two states are defined at points $\mathbf{x}_{P1}$ and $\mathbf{x}_{P2}$, which are symmetrically located on either side of the interface. Material 1 ($\phi > 0$) and Material 2 ($\phi < 0$) are indicated by the labeled regions.

Once we found $\mathbf{x}_{p1}$ and $\mathbf{x}_{p2}$, we have to compute the flow variables there. We treat $\mathbf{x}_{p2}$ as the L-state and $\mathbf{x}_{p1}$ as the R-state. In order to interpolate these two points, we use a bilinear interpolation.

Consider the rectangular stencil shown in Figure 3.6: the corner coordinates are: (x_1,y_1),(x_2,y_1), (x_1,y_2),(x_2,y_2) with $x_1 \le x \le x_2$ and $y_1 \le y \le y_2$.

Let the cell-centered quantity be z. Bilinear interpolation proceeds in two one-dimentional steps:

Firstly, interpolate in x at the lower and upper rows:

$$\begin{aligned} z(x,y_1) &= \frac{x_2-x}{x_2-x_1} z(x_1,y_1) + \frac{x-x_1}{x_2-x_1} z(x_2,y_1), \\ z(x,y_2) &= \frac{x_2-x}{x_2-x_1} z(x_1,y_2) + \frac{x-x_1}{x_2-x_1} z(x_2,y_2). \end{aligned} \tag{3.10}$$

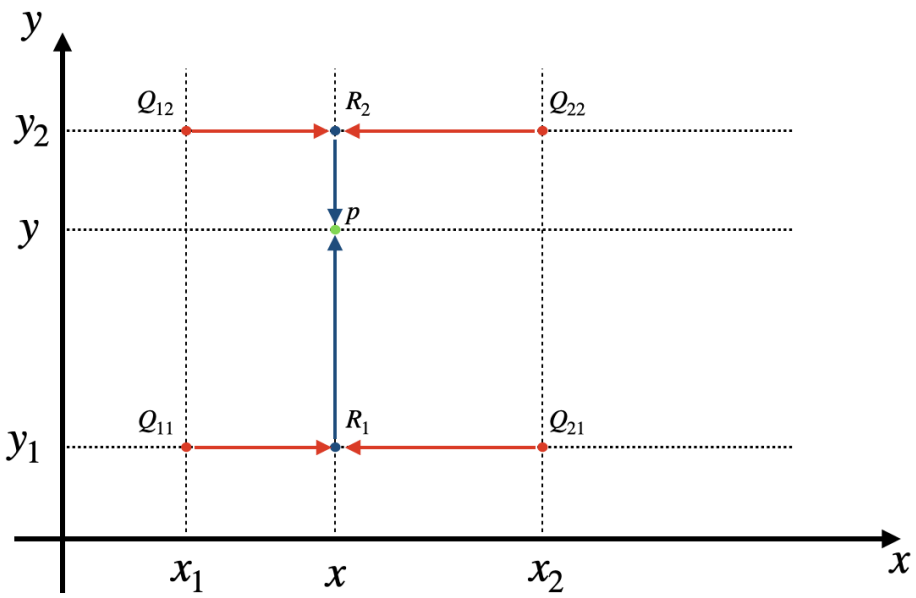


Fig. 3.6 Geometric illustration of the bilinear interpolation process for estimating the value at point $P(x,y)$. The four known corner values are $Q_{11}, Q_{21}, Q_{12}, Q_{22}$, located at $(x_1,y_1),(x_2,y_1),(x_1,y_2),(x_2,y_2)$, respectively. Intermediate values R_1 and R_2 are obtained by linear interpolation in the x-direction, and the final interpolated value at P is obtained by linear interpolation in the y-direction using R_1 and R_2.

Secondly, interpolate the two intermediate values in y:

$$z(x,y) = \frac{y_2 - y}{y_2 - y_1} z(x,y_1) + \frac{y - y_1}{y_2 - y_1} z(x,y_2). \tag{3.11}$$

After performing the bilinear interpolation at both $\mathbf{x}_{p1}$ and $\mathbf{x}_{p2}$, we obtain L-state and R-state as described above. These two states are then used as inputs to the Riemann solver i.e. the HLLC solver in this project, which returns two star states: a left star state and a right star state.

To assign values to the ghost cells, we apply the following logic listed in Table3.1:

Condition	Ghost Cell Type	Assigned State
$\phi > 0$	Material 1 ghost cell	Left star state
$\phi < 0$	Material 2 ghost cell	Right star state

Table 3.1 Assignment of ghost cell values based on level set sign and Riemann solver results.

We now get two states:

$$\text{L-state} = \big(\rho_L,\ v_{x,L},\ v_{y,L},\ p_L\big), \tag{3.12}$$

$$\text{R-state} = \big(\rho_R,\ v_{x,R},\ v_{y,R},\ p_R\big). \tag{3.13}$$

These two states are the two inputs of the HLLC Riemann Solver described in Section2.2.5.

For a multidimensional Riemann problem, the tangential components of the velocity remains unchanged. Hence we should decompose the velocity into the tangential velocity and the normal velocity by using:

$$\begin{aligned} V_n &= \mathbf{n}\cdot\mathbf{v}, \\ V_t &= \mathbf{v} - V_n\mathbf{n}. \end{aligned} \tag{3.14}$$

We therefore pass to the HLLC solver the reduced states:

$$\text{L-state} = \big(\rho_L,\ v_{n,L},\ p_L\big), \tag{3.15}$$

$$\text{R-state} = \big(\rho_R,\ v_{n,R},\ p_R\big). \tag{3.16}$$

Using the formulae introduced in Eqn.(2.28) and (2.30), we compute: the contact-discontinuity wave speed S^*, the intermediate density and pressure (ρ^*, p^*), and the intermediate normal velocity V_n^*.

Because V_t is unchanged through the procedure, the full velocity in the star region is reconstructed as:

$$\mathbf{v}^* = V_n^* \cdot \mathbf{n} + \mathbf{v}_{t,k} \tag{3.17}$$

where $k = L$ if $\phi > 0$ (Material 1) and $k = R$ if $\phi < 0$ (Material 2).

Then we assign the ghost cells using the variables that we obtained:

If $\phi > 0$ (inside Material 2): combine $(\rho^*, \mathbf{v}^*, p^*)$ with the equation of state to form conservative variables and store them in the Material 1 ghost cell.

If $\phi < 0$ (inside Material 1): repeat the same procedure using the R-state and write the result into the Material 2 ghost cell.

The whole procedure of assigning ghost cell states after the bilinear interpolation process is summarised in Algorithm 2.

Algorithm 2 Ghost-cell update for bilinear interpolation

for *iterate through all interface cell* **do**
  if $\phi > 0$ *(i.e. inside Material 2, need to compute ghost cells for Material 1)* **then**
    Decompose L-state and R-state into ($V = V_t + nV_n$):
    normal component V_n
    tangential component V_t
    Feed them into the HLLC solver to obtain L-star state.
    Assign L-star to the Material 1 ghost cell.
  else
    Decompose L-state and R-state into ($V = V_t + nV_n$):
    normal component V_n
    tangential component V_t
    Feed them into the HLLC solver to obtain R-star state.
    Assign R-star to the Material 2 ghost cell.
  end
end

After finishing the procedure described above, we do the constant extrapolation as described in the Original GFM.

3.3 The Real Ghost Fluid Method

Both Riemann GFM and the Real GFM are based on solving one-dimensional Riemann problems between pairs of adjacent cells to generate ghost cell states. The key difference lies in how they choose the adjacent cell.

In Riemann GFM we pick two points x_{p1} and x_{p2} on either side of the interface, interpolate into their four surrounding cells, and solve the Riemann problem there. In the Real GFM, by contrast, we examine all of the neighboring cells and aim to pick just one cell by using the same method described in the Original GFM(Sec.3.1).

The neighbour with the largest value of cosine similarity is chosen as the adjacent cell for the Riemann problem to solve.

Finally, we feed the interface cell state and this best-aligned neighbour into the HLLC Riemann slover exactly as before, and then proceed with the star-state assignment and constant extrapolation in the same way in the Riemann GFM.

The logic of implementing the method is summarized in Algorithm3.

Algorithm 3 Ghost-cell update for Real GFM

for *iterate through all cell* **do**
  if *cell* (i, j) *is interface &* $\phi > 0$ *(i.e. inside Material 2, the ghost cells of Material 1)* **then**
    find closest cell with $\phi < 0$ using maximum cosine similarity of normal vectors.
  end
  if *cell* (i, j) *is interface &* $\phi < 0$ *(i.e. inside Material 1, the ghost cells of Material 2)* **then**
    find closest cell with $\phi > 0$ using maximum cosine similarity of normal vectors.
  end
  if $\phi > 0$ *(i.e. aim to get the states of ghost cells of Material 1.)* **then**
    Use the cell we found by cosine similarity as the L-state and the current interface state as the R-state.
    Feed them into the HLLC Riemnn solver to get the intermediate state.
    The intermediate state is passed to the ghost cells of Material 1.
  else
    Use the cell we found by cosine similarity as the R-state and the current interface state as the L-state.
    Feed them into the HLLC Riemnn solver to get the intermediate state.
    The intermediate state is passed to the ghost cells of Material 2.
  end
end

3.4 The Modified Ghost Fluid Method

Although the original GFM cannot adequately handle strong shock interactions, it still offers a relatively simple and effective framework for simulating multi-material flows, especially when compared to more complex methods such as the Riemann GFM and the Real GFM, which require solving Riemann problems explicitly. The original GFM avoids this complexity by not relying on Riemann solvers, making it attractive for practical implementations.

To preserve the simplicity and general structure of the original GFM while overcoming its limitations under strong shock conditions, the Modified GFM has been proposed. The key improvement in the modified GFM is a more accurate way of estimating the pressure and velocity at the interface, based on characteristic lines and shock relations. This helps the method deal with strong discontinuities much more reliably than the original version.

As shown in Figure 3.2, the interface is located between i and $i+1$. The procedure is the same as that in the Original GFM, we first find the ghost cell, then we use the predicte pressure p and the normal velocity v_n instead of copying these values from the interface cell.

The predicted pressure p and normal velocity v_n is obtained by solving:

$$\frac{p_I - p_{i-1}^n}{W_L^n} + \left(v_I - v_{i-1}^n\right) = 0, \tag{3.18}$$

$$\frac{p_I - p_{i+2}^n}{W_R^n} - \left(v_I - v_{i+2}^n\right) = 0, \tag{3.19}$$

$$(W_L^n)^2 = \rho_I^L \rho_{i-1}^n \frac{p_I - p_{i-1}^n}{\rho_I^L - \rho_{i-1}^n}, \tag{3.20}$$

$$(W_R^n)^2 = \rho_I^R \rho_{i+2}^n \frac{p_I - p_{i+1}^n}{\rho_I^R - \rho_{i+2}^n}. \tag{3.21}$$

We therefore need the relationship of density and pressure to close the four equations:

$$\rho_I^L = \frac{\left[f_2\left(\rho_I^L\right)+1\right] p_I + g_2\left(\rho_I^L\right) - 0.5\left(p_I - p_{i-1}^n\right)}{\left[f_2\left(\rho_{i-1}^n\right)+1\right] p_{i-1}^n + g_2\left(\rho_{i-1}^n\right) + 0.5\left(p_I - p_{i-1}^n\right)} \rho_{i-1}^n, \tag{3.22}$$

$$\rho_I^R = \frac{\left[f_1\left(\rho_I^R\right)+1\right] p_I + g_1\left(\rho_I^R\right) - 0.5\left(p_I - p_{i+2}^n\right)}{\left[f_1\left(\rho_{i+2}^n\right)+1\right] p_{i+2}^n + g_1\left(\rho_{i+2}^n\right) + 0.5\left(p_I - p_{i+2}^n\right)} \rho_{i+2}^n. \tag{3.23}$$

where for a general EOS,

$$\rho e = f(\rho) \cdot p + g(\rho) \tag{3.24}$$

After obtaining p and v_n from the interface by solving the above equations, we then copy the tangential velocity v_t from the real cell, and reconstruct the velocity just same as described in the Original GFM.

3.5 The Practical Ghost Fluid Method

The Practical GFM can be viewed as a streamlined evolution of the Modified GFM. In the Modified GFM, we need three variables from the interface to construct the ghost fluid state. The Practical GFM, by contrust, reduces this to a single degree of freedom: the interfacial velocity u_I. Once u_I is determined, the ghost fluid pressure and density are simply mirrored from the real side, and only the velocity is reflected.

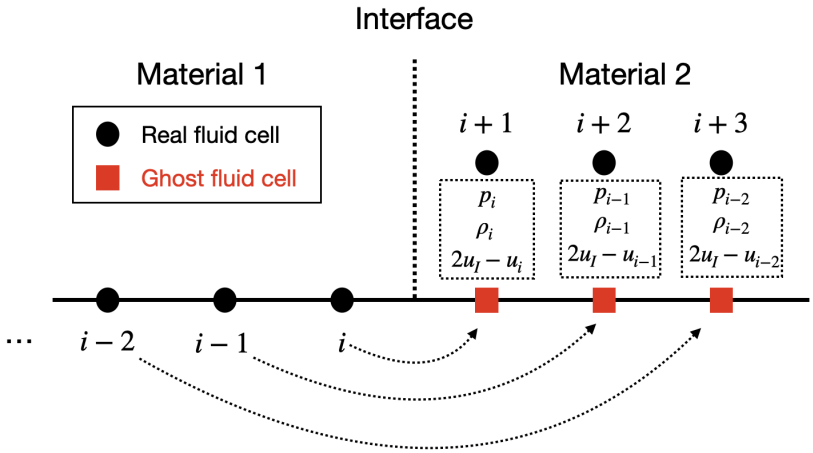


Fig. 3.7 Illustration of the Practical GFM stencil. Real fluid cells (black circles) on the left of the interface supply p and ρ directly to their corresponding ghost fluid counterparts (red circles), while the normal velocity in each ghost cell is set ny mirror-reflection about the interface velocity u_I. i.e. $u^* = 2u_I - u_{real}$

.

Figure 3.7 illustrates the scheme for the Practical GFM. Real fluid cells on the left of the interface (black circles) directly supply their pressure p_i and density ρ_i to the corresponding ghost fluid cells on the right (red squares). The normal velocity is set by reflecting about the interface velocity u_I. Specifically,

$$u^* = 2u_I - u_{real} \tag{3.25}$$

As a result, the Practical GFM uses only one degree of freedom (only u_I need to be computed) to define the ghost cells, which is much more simpler than the Modified GFM.

To compute the interfacial velocity u_I, we solve the following equations:

$$\begin{aligned}
u_L - u_I^a &= \sqrt{\left(p_I^a - p_L\right)\left(\tfrac{1}{\rho_L} - \tfrac{1}{\rho_{IL}^a}\right)}, \\
u_I^a - u_R &= \sqrt{\left(p_I^a - p_R\right)\left(\tfrac{1}{\rho_R} - \tfrac{1}{\rho_{IR}^a}\right)}, \\
e\left(p_L, \rho_L\right) - e\left(p_I^a, \rho_{IL}^a\right) &= \tfrac{1}{2}\left(p_I^a + p_L\right)\left(\tfrac{1}{\rho_{IL}^a} - \tfrac{1}{\rho_L}\right), \\
e\left(p_I^a, \rho_{IR}^a\right) - e\left(p_R, \rho_R\right) &= -\tfrac{1}{2}\left(p_I^a + p_R\right)\left(\tfrac{1}{\rho_{IR}^a} - \tfrac{1}{\rho_R}\right).
\end{aligned} \tag{3.26}$$

Eqn.(3.26) are known as an ARPS, which is based on a doubled-shock structure as in the Modified GFM. The superscript "a" denotes an approximation to the exact state. u_I is set to the velocity solution u_I^a of Eqn. (3.26). Here, e denotes the specific internal energy, which for each material is given by its EOS:

$$e = e(p, \rho) \tag{3.27}$$

In other words, whenever $e(p_L, \rho_L)$, $e(p_I^a, \rho_{IL}^a)$, etc. appear above, they are evaluated by plugging the corresponding pressure and density into the EOS of the material.

In 2D, Eqn.(3.25) yields the normal velocity, and the tangential velocity is simply extrapolated from the real cell just as that in the Modified GFM. By completing the procedure described above, we can therefore obtain the ghost fluid states.

After defining the ghost cells, we then apply the constant extrapolation just as before extrapolate these values to the whole computational grid.

3.6 The Interface Ghost Fluid Method

The Interface GFM departs from the Practical GFM by fully resolving the Riemann problem at the interface itself under two core assumptions: (a) The interaction take place at the interface. (b) There is no entropy exchange between the two fluids throughout the interaction.

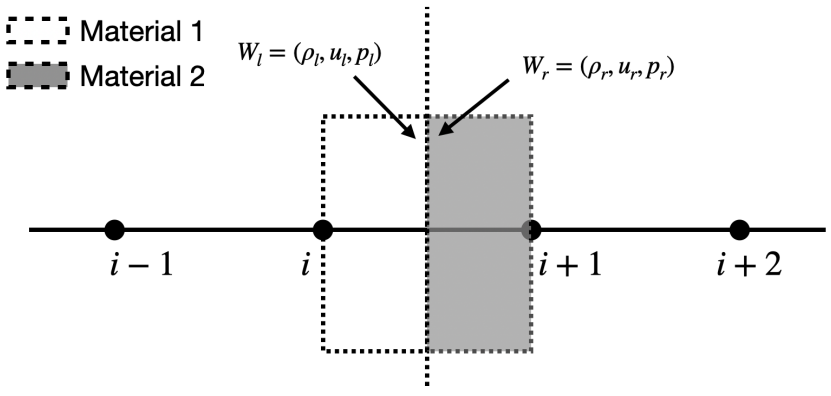


Fig. 3.8 Schematic of two adjacent grid cells have two different materials. The left cell at index i carries state $W_l = (\rho_l, u_l, p_l)$ and the right cell at index $i+1$ carries state $W_r = (\rho_r, u_r, p_r)$ before interaction.

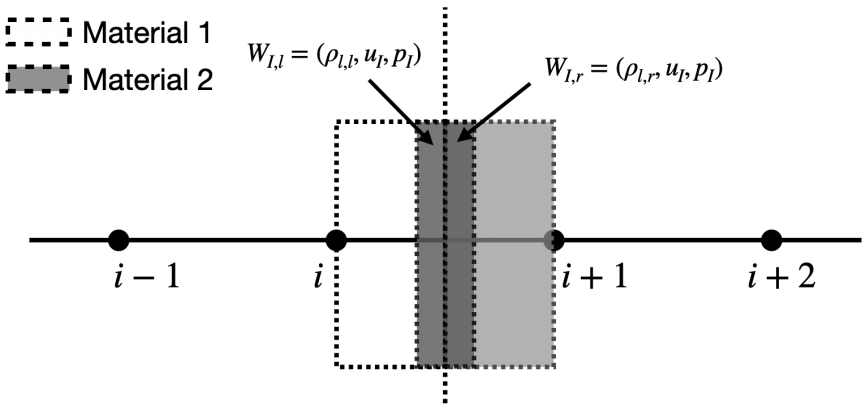


Fig. 3.9 Schematic of the two adjacent half–cells immediately after interface interaction. The left half–cell at index i carries the interface state $W_{I,\ell} = (\rho_{I,\ell}, u_I, p_I)$, and the right half–cell at index $i+1$ carries $W_{I,r} = (\rho_{I,r}, u_I, p_I)$.

.

As shown in Figure 3.8, assume that the adjacent grid cells have two different materials and their states are $W_l = (\rho_l, u_l, p_l)$ and $W_r = (\rho_r, u_r, p_r)$. After interaction, the interface takes interface velocity u_I and pressure p_I. The densities of the two materials near the interface have changed to $\rho_{I,l}$ and $\rho_{I,r}$ respectively as shown in Figure 3.9.

To obtain the values of u_I, p_I, we solve the following equations using Newton's method:

$$\begin{aligned} u_I &= u_\ell - \frac{2\,c_\ell}{\gamma_\ell - 1}\Big[\big(\frac{f(p_I)}{f(p_\ell)}\big)^{\frac{\gamma_\ell - 1}{2\gamma_\ell}} - 1\Big], \\ u_I &= u_r + \frac{2\,c_r}{\gamma_r - 1}\Big[\big(\frac{f(p_I)}{f(p_r)}\big)^{\frac{\gamma_r - 1}{2\gamma_r}} - 1\Big]. \end{aligned} \tag{3.28}$$

The term $f(p)$ represents the pressure function appearing in the chosen EOS. i.e. $f(p) = p$ for an ideal gas EOS and $f(p) = p + p_\infty$ for a stiffened gas EOS. The sound speed $c = \sqrt{\frac{\gamma f(p)}{\rho}}$.

After obtaing u_I and p_I, we can then calculate $\rho_{I,l}$ and $\rho_{I,r}$ using:

$$\rho_{I,\ell} \;=\; \rho_\ell \left(\frac{f(p_I)}{f(p_\ell)}\right)^{1/\gamma_\ell}, \qquad \rho_{I,r} \;=\; \rho_r \left(\frac{f(p_I)}{f(p_r)}\right)^{1/\gamma_r}. \tag{3.29}$$

After obtaining the interface velocity u_I, pressure p_I, and the interface densities $\rho_{I,l}$ and $\rho_{I,r}$ on both sides, we can proceed to the "ghost interaction" step to construct the ghost cell states.

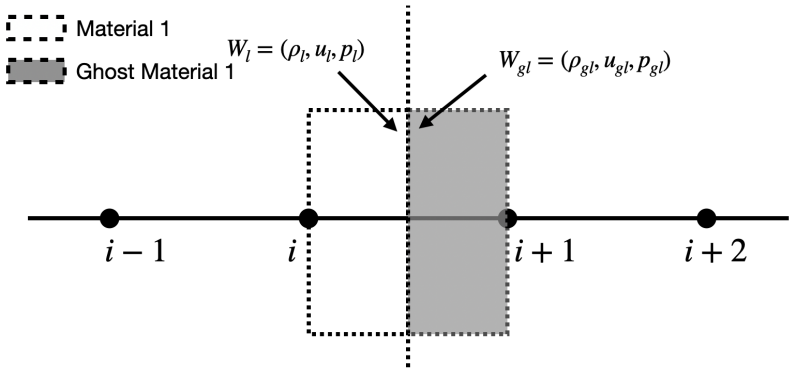


Fig. 3.10 Ghost cell construction for Material 1 before interaction. The real fluid cell at index i holds state $W_l = (\rho_l, u_l, p_l)$. The ghost fluid cell holds state $W_{gl} = (\rho_{gl}, u_{gl}, p_{gl})$.

.

Figure 3.10 shows the schematic of the ghost cell construction before interaction. Assume the real cell carries the state $W_l = (\rho_l, u_l, p_l)$, and the ghost cell carries the state $W_{gl} = (\rho_{gl}, u_{gl}, p_{gl})$. Both the real and the ghost cell are treated using the same EOS as for Material 1.

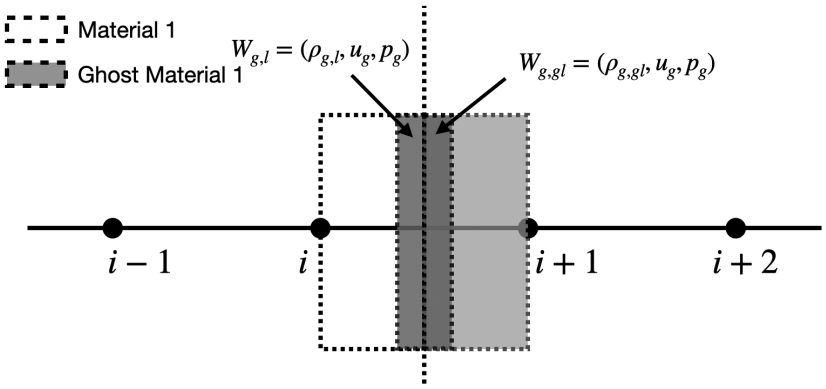


Fig. 3.11 Ghost cell construction for Material 1 after interaction. The real fluid cell at index i holds state $W_{g,l} = (\rho_{g,l}, u_g, p_g)$. The ghost fluid cell holds state $W_{g,gl} = (\rho_{g,gl}, u_g, p_g)$.

.

As shown in Figure 3.11, after interaction, the interface assumes the velocity u_g and pressure p_g. The densities of the two sides at the cell wall changed to $\rho_{g,l}$ and $\rho_{g,gl}$.

We now set $u_g = u_I$, $p_g = p_I$, and $\rho_{g,l} = \rho_{I,l}$ to ensure that both the real fluid interaction and the ghost interaction produce identical interface conditions for the real fluid. Therefore,

the ghost cell states can be obtained by solving:

$$u_I = u_{g,\ell} + \int_{p_{g,\ell}}^{p_I} \frac{\mathrm{d}p}{\rho_{g,\ell} c_{g,\ell,s}}, \tag{3.30}$$

$$\rho_{g,\ell} = \rho_\ell \left(\frac{f(p_{g,\ell})}{f(p_\ell)} \right)^{1/\gamma}. \tag{3.31}$$

To solve the above Eqn.(3.30) and (3.31), we define $p_{gl} = p_I$. Hence the integral in Eqn.(3.30) becomes zero and thus $u_{gl} = u_I$.

In 2D case, we compute the normal velocity v_n using the method described above, and the tangential velocity v_t is just a simple extrapolation from the real cell.

The summerized version of the Interface GFM is listed below:

1. Locate the interface cells via the level-set method described in the Original GFM.
2. Solve for the interface states using the coupled equations in Eqn.3.28 and 3.29 to get u_I, p_I, $\rho_{I,l}$ and $\rho_{I,r}$.
3. Identify the corresponding ghost cells for each material and solve ghost interaction equations (Eqn.(3.30) and (3.31)) to assign each ghost cell its pressure, normal velocity, and density based on the interface state.
4. Reconstruct the tangential velocity by coping the real cell's tangential velocity component into each ghost cell so that the full velocity is obtained.
5. Apply the constant extrapolation step as before.

Chapter 4

Tests Results and Analysis

In this chapter, we assess and compare the three GFMs: Original GFM, Riemann GFM and Real GFM. We will use a suite of 1D tests and several 2D tests to verify their correctness, measure their accuracy and evaluate the robustness.

The remaining GFMs will be tested in the next report.

4.1 The Shock Tube Test

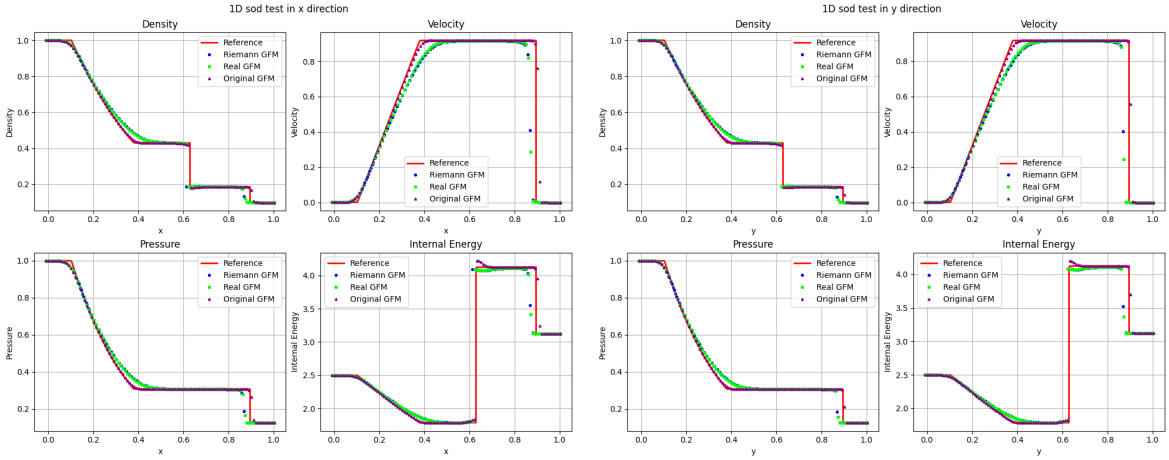
In this section, we aim to perform the classical sod shock-tube test to verify the validity of the three GFMs in an essentially 1D scenario within a 2D domain. In the first case, the initial discontinuity ia at $x = 0.5$, producing a shock in $x-$direction; In the second case, it is aligned at $y = 0.5$, producing a shock in the $y-$direction. Both runs use indentical left/right states, mesh resolution, CFL number and final time. The initial conditions are listed in the Table 4.1:

Table 4.1 Setup for the Sod shock-tube benchmarks in 2D

Parameter	**Shock in x–direction**	**Shock in y–direction**
Domain	$[0,1] \times [0,1]$	$[0,1] \times [0,1]$
Initial discontinuity	$x = 0.5$	$y = 0.5$
Left state $(\rho, v_x, v_y, p)_L$	$(1.0, 0, 0, 1.0)$	$(1.0, 0, 0, 1.0)$
Right state $(\rho, v_x, v_y, p)_R$	$(0.125, 0, 0, 0.1)$	$(0.125, 0, 0, 0.1)$
Mesh resolution	200×1	1×200
CFL number	0.8	0.8
Final time t	0.25	0.25

Figure 4.1 present the one-dimensional Sod shock-tube test carried out on a 200×1 grid and a 1×200 grid with a CFL number of 0.8 up to $t = 0.25$. In every variable: density, velocity, pressure and specific internal energy, the results produced by the three GFMs (Riemann GFM, Real GFM and Original GFM) aligns with the reference solution (solid red line). Shocks, contact discontinuities, and rarefaction waves are all captured at the correct locations and with the correct strength.

The fact that this overlap is observed for sweeps in both the x and $y$directions demonstrates that all three GFMs are able to reproduce single-material interaction in this basic test case.



(a) Sod shock tube in the x–direction (200×1 grid).

(b) Sod shock tube in the y–direction (1×200 grid).

Fig. 4.1 One-dimensional Sod shock-tube solutions at t = 0.25 computed with the Riemann GFM, Real GFM and Original GFM. Panels show the spatial profiles of (a) density ρ, (b) velocity u, (c) pressure p and (d) specific internal energy e. A CFL number of 0.8 was used.

4.2 The Karni and Quirk's Test

In this section, we present the test case that described in [19]. The initial conditions are listed in Table 4.2.

As shown in Figure 4.2, all three GFMs produce good results compared with the reference solution which is produced by the exact Riemann solver. Each material interface at 0.25, 0.4, 0.6 is correctly resolved, which demonstrates that all methods can correctly capture the material interface.

Table 4.2 initial condition of the Karni and Quirk's test

Mesh resolution = 400, **CFL** = 0.8, **Final time** $t = 0.3\,\mathrm{s}$

Interval x	Material	ρ	u	p	γ
$0 \le x \le 0.25$	air (post-shock)	1.3765	0.3948	1.57	1.40
$0.25 \le x < 0.40$	air (pre-shock)	1.0000	0.0000	1.00	1.40
$0.40 \le x < 0.60$	Helium	0.1380	0.0000	1.00	1.67
$0.60 \le x \le 1.0$	air (pre-shock)	1.0000	0.0000	1.00	1.40

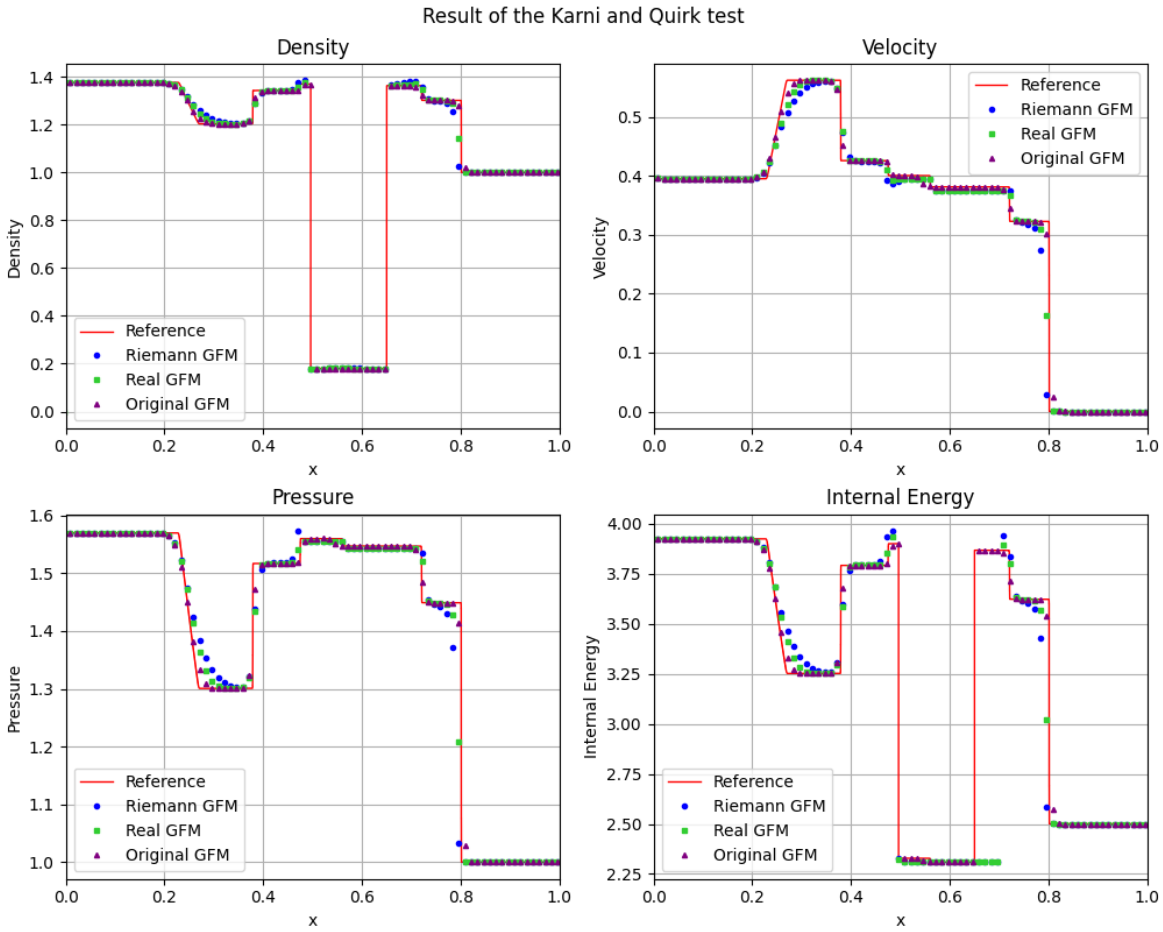


Fig. 4.2 Karni–Quirk test at $t = 0.3\,\mathrm{s}$ (400 cells, CFL $= 0.8$). Red step-line = reference; blue dots = Riemann GFM, green square = Real GFM, purple triangle = Original GFM.

4.3 The Water Air Shock Tube Test

To test the robustness of the Original GFM, the Riemann GFM and the Real GFM, we perform a 1D water air shock tube test described in saurel1999 [20]. The initial conditions are summerized in Table 4.3.

Table 4.3 Initial condition and numerical setup for liquid-gas shock interaction

Parameter	**Liquid (left)**	**Gas (right)**
Density ρ [kg/m^3]	1000	50
Pressure P [Pa]	10^9	10^5
Velocity u [m/s]	0	0
Gamma γ	4.4	1.4
p_∞ [Pa]	6×10^8	0
Region	if $x < 0.7$	otherwise
Discontinuity location	$x = 0.7$	
Final time t [s]	0.000229	
Mesh resolution	400×1	
CFL number	0.8	
Domain	$[0,1] \times [0,1]$	

Figure 4.3 shows the results of the 1D water air shock tube test at final time $t = 0.000229$. The numerical solutions obtained using the Riemann GFM and the Real GFM are in agreement with the reference solution, accurately capturing the profiles of density ρ, velocity u, pressure p, and total energy E. In contrast, the Original GFM failed to complete this test. The simulation crashed after only a few time steps due to numerical instability at the material interface.

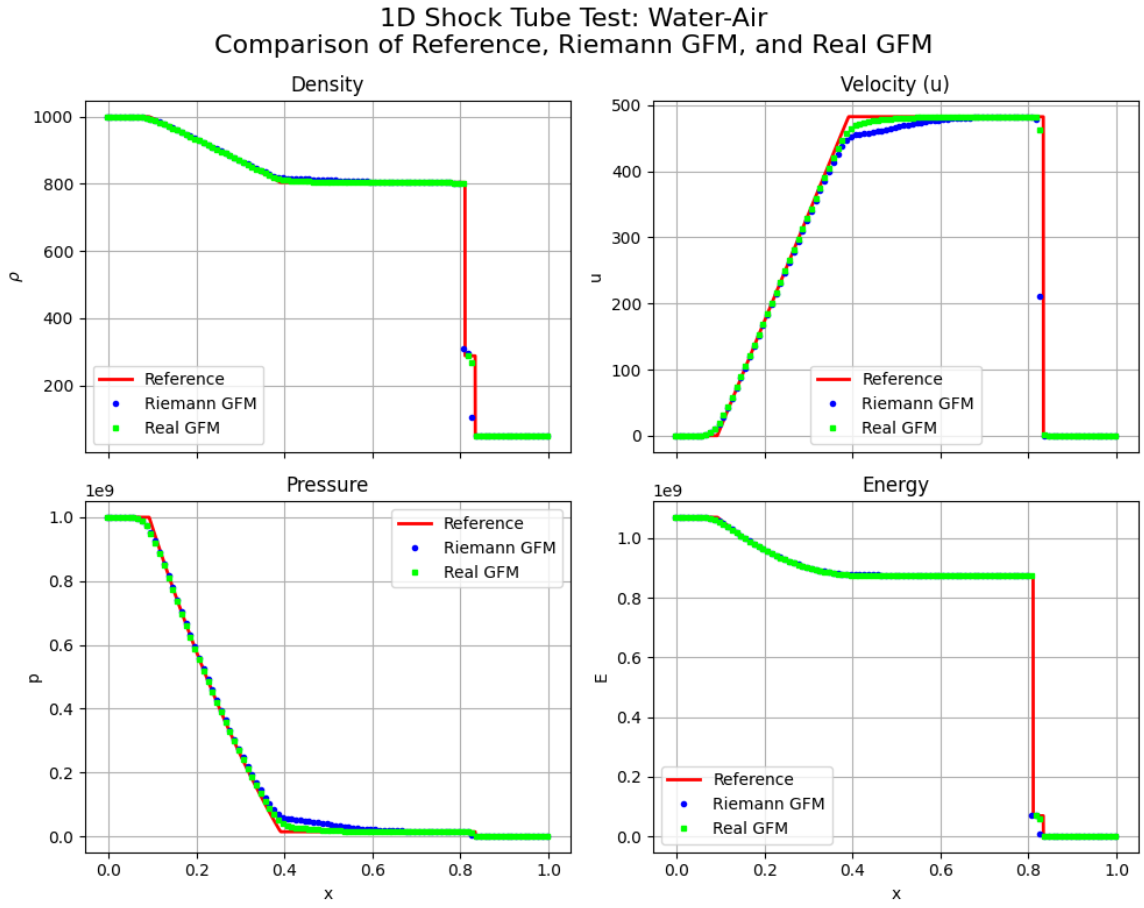


Fig. 4.3 Comparison of numerical and reference solutions for the 1D water–air shock tube test at final time $t = 0.000229$. Shown are the profiles of density ρ, velocity u, pressure p, and total energy E. The numerical results were obtained using the Riemann GFM (blue) and the Real GFM (green), and are compared against the exact reference solution (red). The simulation was performed on a uniform grid with 400 cells and a CFL number of 0.8.

4.4 Underwater shock impacting on a gas-water interface

In this section, we reproduce the test decribed in case 4, problem 8 of [15], which is an underwater shock impacting on a gas–water interface. The initial conditions are listed in Table 4.4.

Figure 4.4 shows the results obtained with the Riemann GFM and the Real GFM; the Original GFM still fails to run this test, demonstrating that it cannot stably handle such strong shock–interface interactions.

Table 4.4 Underwater shock impacting on a gas–water interface – initial conditions and numerical settings

Region	ρ	u	p	γ
$x \leq 0.7$ (water)	1000.0	0.0	1000.0	7.15
$x > 0.7$ (gas)	1.0	0.0	1.0	1.40
Mesh resolution	200 cells			
CFL number	0.9			
Time steps	150			

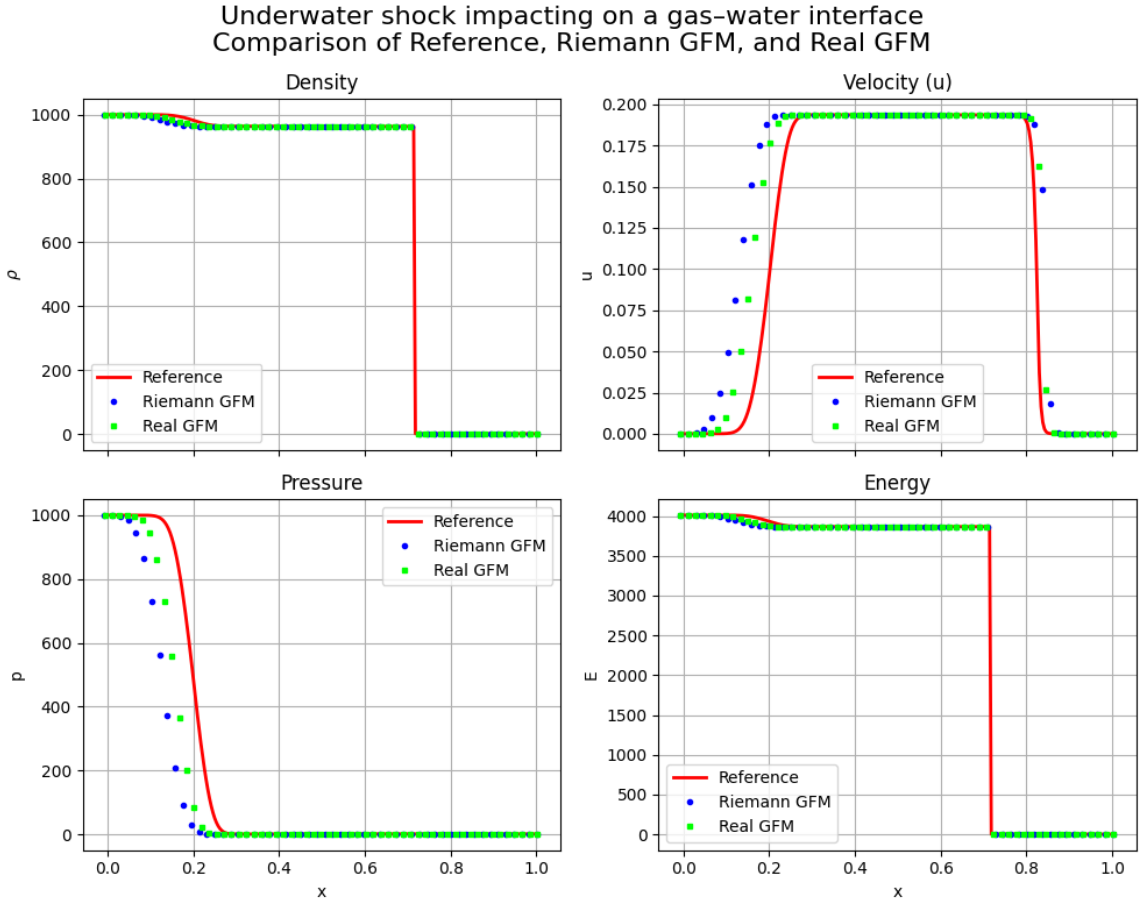


Fig. 4.4 1D underwater shock on a gas–water interface with CFL=0.9, run for 150 time steps, mesh resolution: 200 cells.

Table 4.5 Initial conditions and numerical setup for the 2-D cylindrical-explosion test.

Mesh resolution = 300×300, **CFL** = 0.8, **Final time** $t = 0.25\,\mathrm{s}$

Region	ρ	v_x	v_y	p
Inside the circle ($r < 0.4$)	1	0	0	1
Outside the circle ($r \geq 0.4$)	0.125	0	0	0.1

4.5 The Cylindrical Explosion Test

In this section, we perform the cylindrical explosion test described in Toro's book[21] of the same material to further verify the validity of the Riemann GFM, the Real GFM and the Original GFM in 2D case.

The initial conditions are listed in Table 4.5.

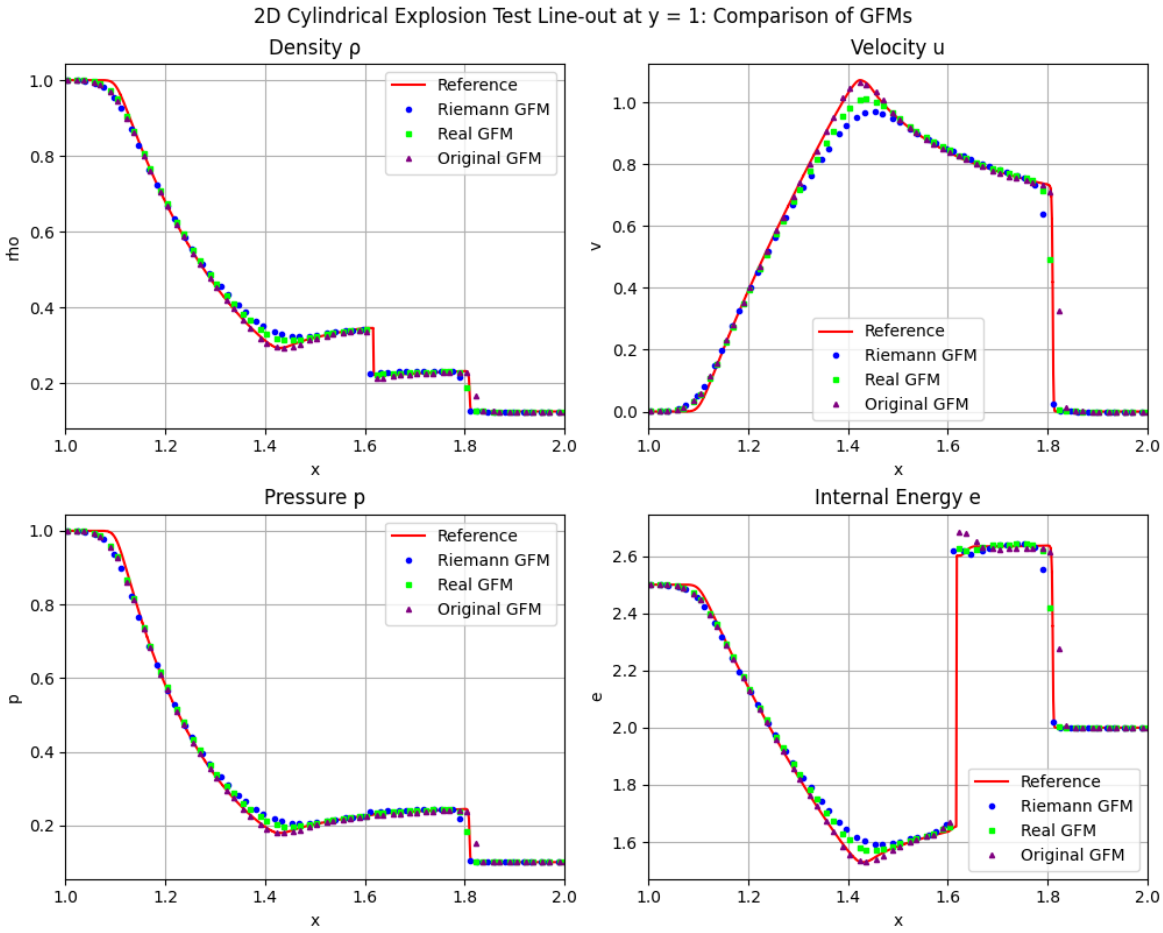
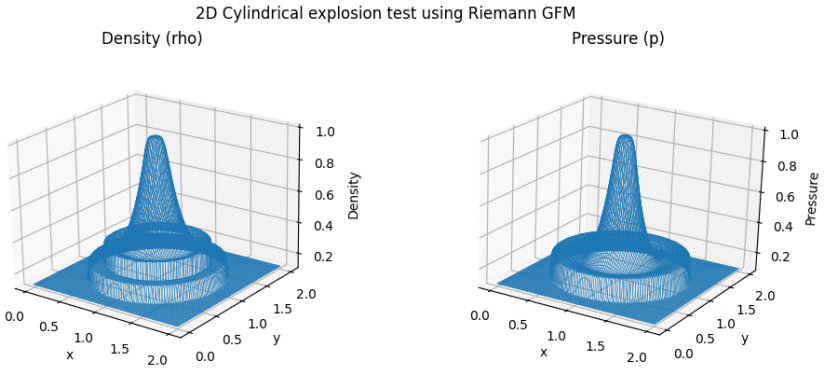
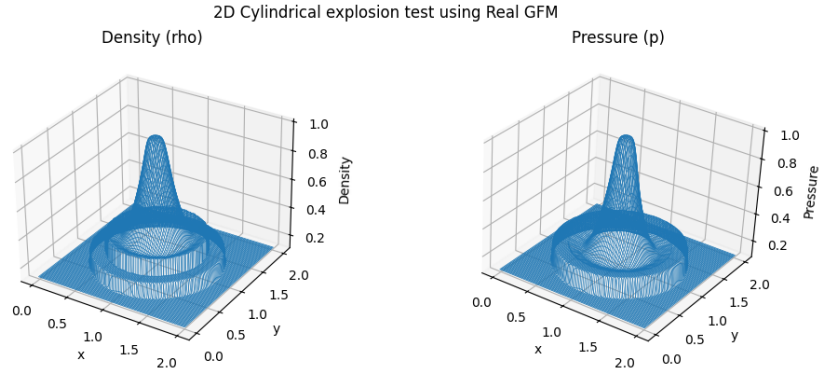


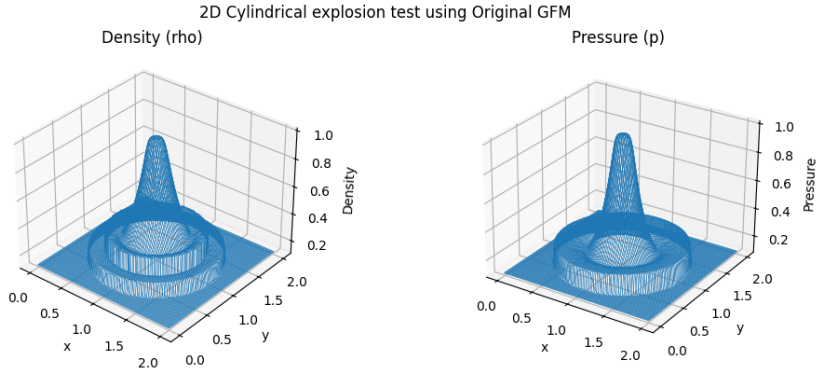
Fig. 4.5 Comparison along $y = 1$ at $t = 0.25\,\mathrm{s}$ for the 2D cylindrical-explosion test (mesh 300×300, CFL=0.8). Panels show density ρ, velocity u, pressure p, and internal energy e.



(a) Riemann GFM



(b) Real GFM



(c) Original GFM

Fig. 4.6 Surface plots of density (left) and pressure (right) at $t = 0.25\,\mathrm{s}$ for the 2D cylindrical-explosion test on a 300×300 grid (CFL=0.8). Shown are results from (a) the Riemann GFM, (b) the Real GFM, and (c) the Original GFM.

Figure 4.5 and 4.6 together demonstrates that all three GFM accurately capture the classical 2D cylindrical-explosion test using same material. In Figure 4.5, we extract a 1D lineout at $y = 1$ and compare density, velocity, pressure and internal energy at $t = 0.25$. Each GFM closely aligns with the reference solution. Figure 4.6 presents full surface plots of

density and pressure at the same time for the Riemann (a), Real (b) and Original (c) methods. All three GFMs produce virtually circular shock fronts and radial profiles, which confirms that each GFM is both stable and accurate for the same material 2D explosion test.

4.6 The 2D Shock Bubble Test

In this section, we perform a 2D test with two different materials (air and helium). The initial conditions are listed below in Table 4.6.

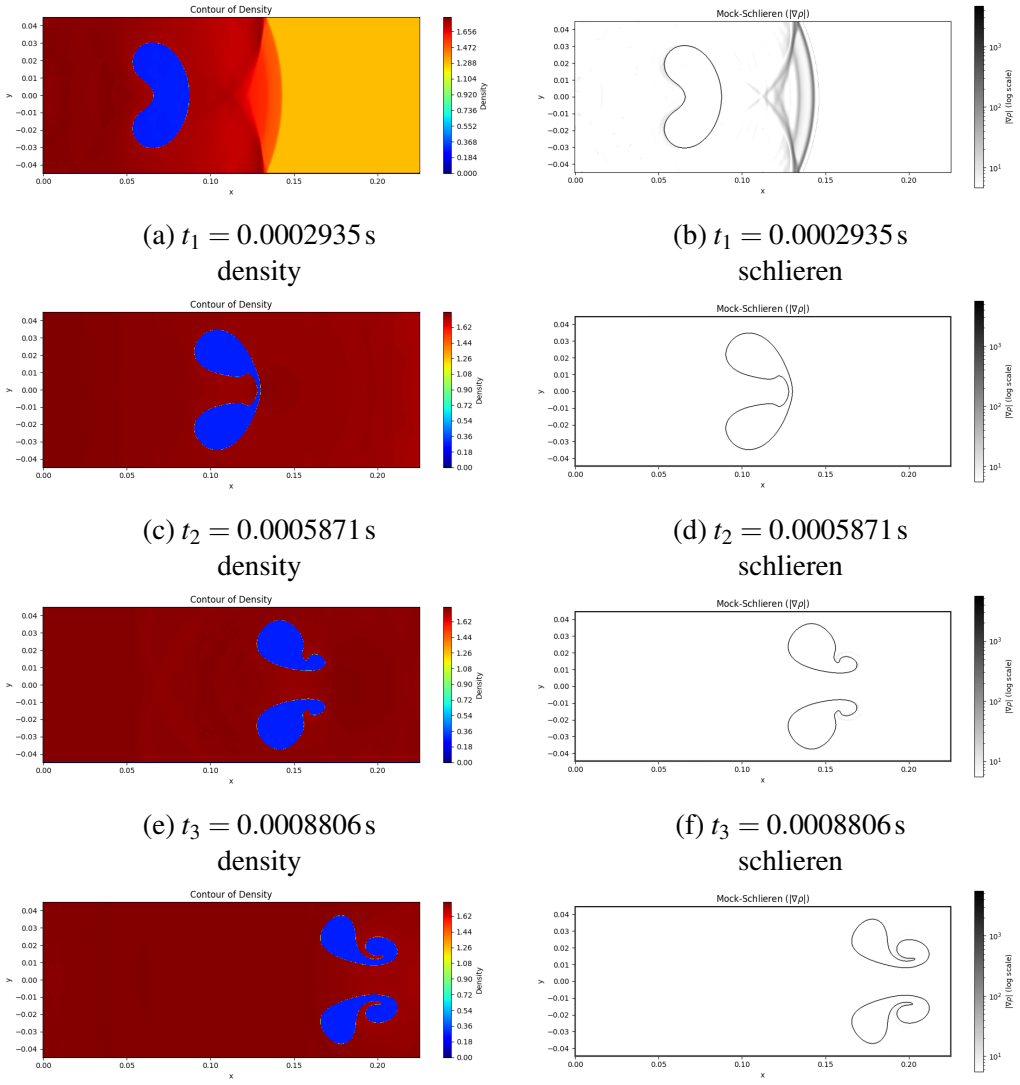
Table 4.6 Initial conditions and numerical setup for the shock–bubble interaction test

Mesh resolution = 1000×394**, CFL = 0.8, Final time** $t = 0.0011741\,\mathrm{s}$
$\gamma_{\mathrm{air}} = 1.4$**,** $\gamma_{\mathrm{He}} = 1.67$**;** $x \in [0, 0.225]\,\mathrm{m}$**,** $y \in [-0.0445, 0.0445]\,\mathrm{m}$

Region	$\rho\,[\mathrm{kg\,m^{-3}}]$	$u\,[\mathrm{m\,s^{-1}}]$	$v\,[\mathrm{m\,s^{-1}}]$	$p\,[\mathrm{Pa}]$
Post-shock air ($x < 0.005\,\mathrm{m}$)	1.776	110.63	0	1.59×10^5
Pre-shock air ($x > 0.005\,\mathrm{m}$, outside bubble)	1.290	0	0	1.01×10^5
Helium bubble ($r \leq 0.025\,\mathrm{m}$, centre $(0.035, 0.0445)\,\mathrm{m}$)	0.214	0	0	1.01×10^5

Figure 4.7, 4.8, 4.9 shows the simulated bubble deformation under a shock using three different GFM: Riemann GFM, Real GFM and Original GFM.

Riemann GFM (Figure 4.7) preserves a smooth, well-defined interface at every time step, which produces a very clean bubble shapes. Real GFM (Figure 4.8) yields slightly asymmetric bubble shape at late time steps. By contrast, Original GFM (Figure 4.9) develops small ripples and sharp corners at later times, and its Mock-Schlieren images are noticeably blurrier, which reflects a greater susceptibility to numerical instabilities in the shock-bubble interaction.



(a) $t_1 = 0.0002935\,\mathrm{s}$ density

(b) $t_1 = 0.0002935\,\mathrm{s}$ schlieren

(c) $t_2 = 0.0005871\,\mathrm{s}$ density

(d) $t_2 = 0.0005871\,\mathrm{s}$ schlieren

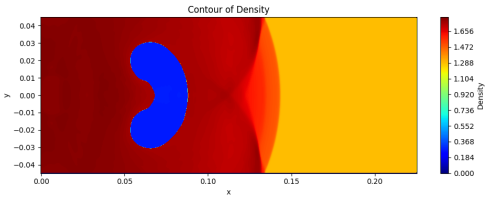
(e) $t_3 = 0.0008806\,\mathrm{s}$ density

(f) $t_3 = 0.0008806\,\mathrm{s}$ schlieren

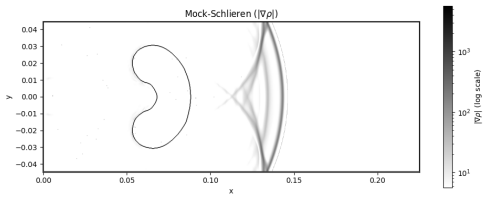
(g) $t_4 = 0.0011741\,\mathrm{s}$ density

(h) $t_4 = 0.0011741\,\mathrm{s}$ schlieren

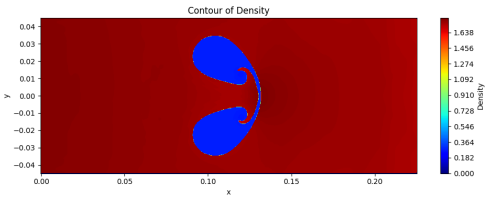
Fig. 4.7 Evolution of the two-phase interface under a planar shock computed with the Riemann GFM on a 1000×394 grid (CFL = 0.8). Shown are density contours (left column) and corresponding shock–schlieren images (right column) at four successive time instants.



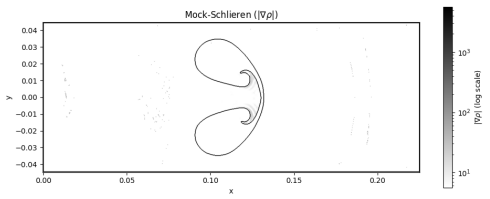
(a) $t_1 = 0.0002935\,\mathrm{s}$
density



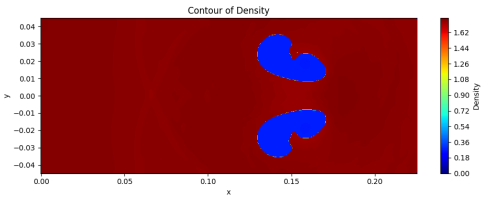
(b) $t_1 = 0.0002935\,\mathrm{s}$
schlieren



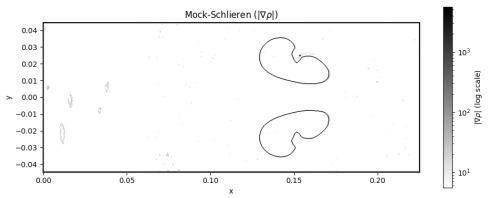
(c) $t_2 = 0.0005871\,\mathrm{s}$
density



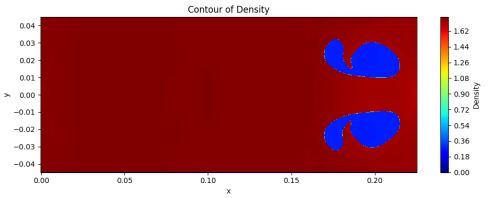
(d) $t_2 = 0.0005871\,\mathrm{s}$
schlieren



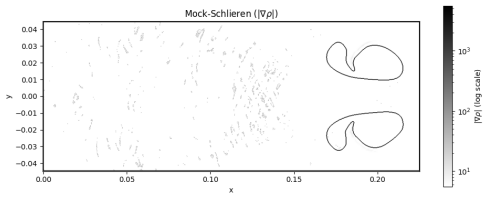
(e) $t_3 = 0.0008806\,\mathrm{s}$
density



(f) $t_3 = 0.0008806\,\mathrm{s}$
schlieren

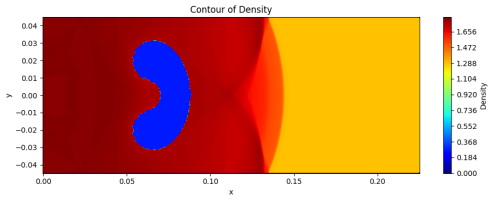


(g) $t_4 = 0.0011741\,\mathrm{s}$
density

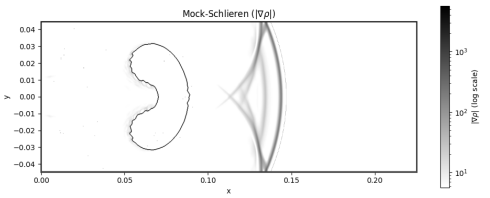


(h) $t_4 = 0.0011741\,\mathrm{s}$
schlieren

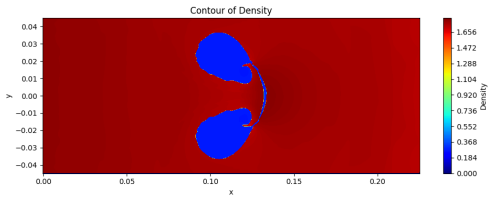
Fig. 4.8 Evolution of the two-phase interface under a planar shock computed with the Real GFM on a 1000×394 grid (CFL = 0.8). Shown are density contours (left column) and corresponding shock–schlieren images (right column) at four successive time instants.



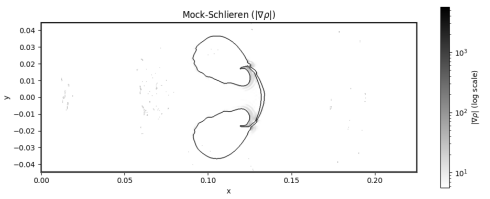
(a) $t_1 = 0.0002935\,\mathrm{s}$ density



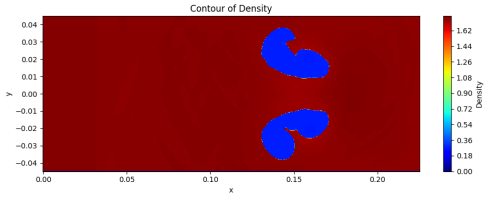
(b) $t_1 = 0.0002935\,\mathrm{s}$ schlieren



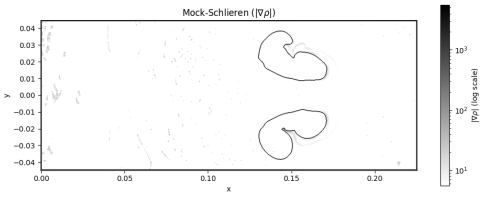
(c) $t_2 = 0.0005871\,\mathrm{s}$ density



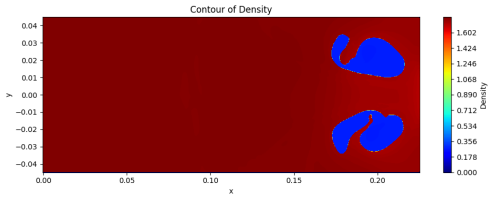
(d) $t_2 = 0.0005871\,\mathrm{s}$ schlieren



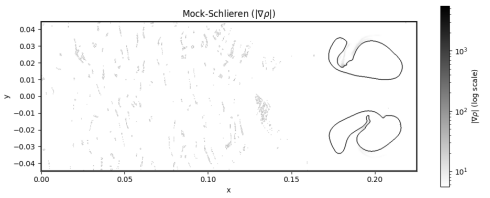
(e) $t_3 = 0.0008806\,\mathrm{s}$ density



(f) $t_3 = 0.0008806\,\mathrm{s}$ schlieren



(g) $t_4 = 0.0011741\,\mathrm{s}$ density



(h) $t_4 = 0.0011741\,\mathrm{s}$ schlieren

Fig. 4.9 Evolution of the two-phase interface under a planar shock computed with the Original GFM on a 1000×394 grid (CFL = 0.8). Shown are density contours (left column) and corresponding shock–schlieren images (right column) at four successive time instants.

Chapter 5

Midterm Conclusion

The previous sections are built on a finite-volume framework in which we have implemented three GFMs: Original GFM, Riemann GFM, and Real GFM. We carried out a series of 1D tests to assess their accuracy and stability.

Firstly, in the Sod shock-tube test (tested along both x and y axes), all three methods accurately capture the left rarefaction, the contact discontinuity, and the right-moving shock in both position and strength. In the Karni Quirk test, each method correctly locates the material interface. However, in the water-air shock tube test, the Original GFM becomes numerically unstable and crashes after only a few time steps, which demonstrates its limitations for strong shock–material-interface interactions. By contrust, both the Riemann GFM and Real GFM remain stable, with the Riemann GFM preserving the sharpest interface profile and the Real GFM exhibiting moderate, physically justified diffusion at fine scales.

In 2D cylindrical-explosion test, all three methods accurately reproduce the radial wave-fronts in close agreement with the reference solution. The two-dimensional shock–bubble interaction test, the Original GFM produces the most pronounced asymmetry and spurious ripples in the bubble deformation, the Real GFM yields a slightly blurred interface, and the Riemann GFM consistently captures the cleanest, most symmetric bubble shapes.

In conclusion, the Riemann GFM achieves the best balance between accuracy and stability, the Real GFM offers a compromise of fidelity and smoothness, and the Original GFM shows clear deficiencies under strong shock–interface conditions. Details of the parallel implementation and optimization are provided in Appendix A. In the next project, we will implement the Modified GFM, the Practical GFM and the Interface GFM, and carry out a suite of 2D test cases to compare their performance and robustness. Future work and final conclusions will be outlined in the end of the project.

Chapter 6

Introduction to Report 2

In the previous part of this project, we explored three foundational Ghost Fluid Methods (GFMs): the Original GFM, the Real GFM, and the Real GFM. We test them in a serious of benchmark tests in one and two dimensions. Each method exhibited distinct behaviours in capturing interface dynamics under varing flow conditions, particularly in terms of stability, sharpness, and accuracy across weak to strong shock regimes. These initial findings provided a critical reference point for understanding the strengths and limitations of early GFM designs.

The second report builds directly on that foundation by implementing and evaluating three more advanced GFMs: the Modified GFM, the Practical GFM, and the Interface GFM. These newer methods aim to address the numerical challenges previously observed, especially the difficulty in preserving sharp interfaces under strong discontinuities and minimizing non-physical oscillations near material boundaries.

Our focus in this report is twofold: Firstly, to assess whether these improved methods provide more reliable and accurate interface treatments in complex multiphase flows. Secondly, to compare their performance against the original three methods, using similar setups to ensure consistency and comparability.

Whereas Report 1 concentrated on foundational mechanisms and limitations, this report emphasizes practical improvements, simplifications in implementation, and overall robustness across a broader range of physical regimes. The methods are tested using the same second-order finite volume solver, HLLC Riemann scheme, and level-set method, which ensures that observed differences arise purely from the GFM design choices.

Chapter 7

1D Validation Tests

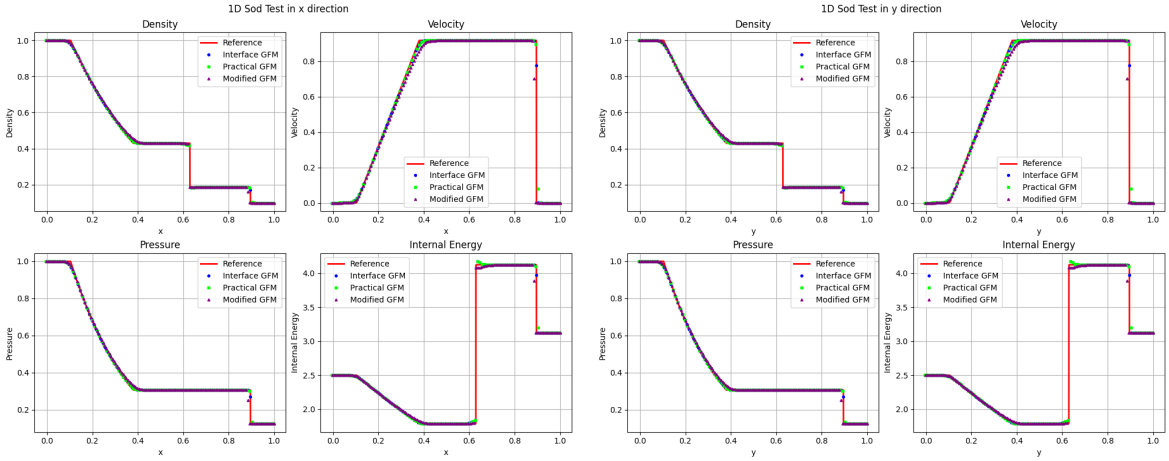
This Chapter is dedicated to validating the performance of the three newly proposed GFMs: the Interface GFM, Practical GFM, and Modified GFM. We will use a seires of classical 1D benchmark tests, such as shock tube test and material interface tests to assess the accuracy, stability and robustness of each method.

7.1 Single-material Sod Shock Tube Test

In this section we perform the classical shock tube test to verify the validity of the three GFMs. The initial condition is the same as listed in Table 4.1.

Figure 7.1 demonstrates the results of the classical Sod shock tube test conducted using the three newly proposed GFMs: Interface GFM, Practical GFM, and Modified GFM. The numerical results are compared against the reference solution for each physical quantity: density, velocity, pressure and specific internal energy.

As shown in the figure, all three GFMs yield solutions that closely align with the reference solution. This confirms that all three methods are capable of accurately capturing single-material flow features and validating their correctness in this benchmark test.



(a) Single-material Sod Shock Tube in the x–direction (200×1 grid).

(b) Single-material Sod Shock Tube in the y–direction (1×200 grid).

Fig. 7.1 One-dimensional Sod shock-tube solutions at t = 0.25 computed with the Interface GFM, Practical GFM and Modified GFM. Panels show the spatial profiles of (a) density ρ, (b) velocity u, (c) pressure p and (d) specific internal energy e. A CFL number of 0.8 was used.

7.2 The Multi-material Karni and Quirk's Test

In this section, we perform again the same Karni and Quirk's Test which conducted in Section 4.2. The initial conditions are listed in Table 4.2.

Figure 7.2 presents a comparison of the numerical results with Interface GFM, Practical GFM and Modified GFM against the exact Riemann solution for Karni and Quirk's test described in [19]. The final time is $t = 0.3$, mesh resolution is 400×1, $CFL = 0.8$.

As shown in the figure, all three GFMs reproduce the correct features, these results verify that the Interface, Practical and Modified GFMs all provide second-order accurate, oscillation-free solutions for this mixed material Riemann problem.

The reference curves shown in Figure 7.2 are generated using an exact Riemann solver, serving as a benchmark to validate the three tested methods.

The three methods aligns with the reference solution well which indicates their robustness and effectiveness in handling 1D multi-fluid problems.

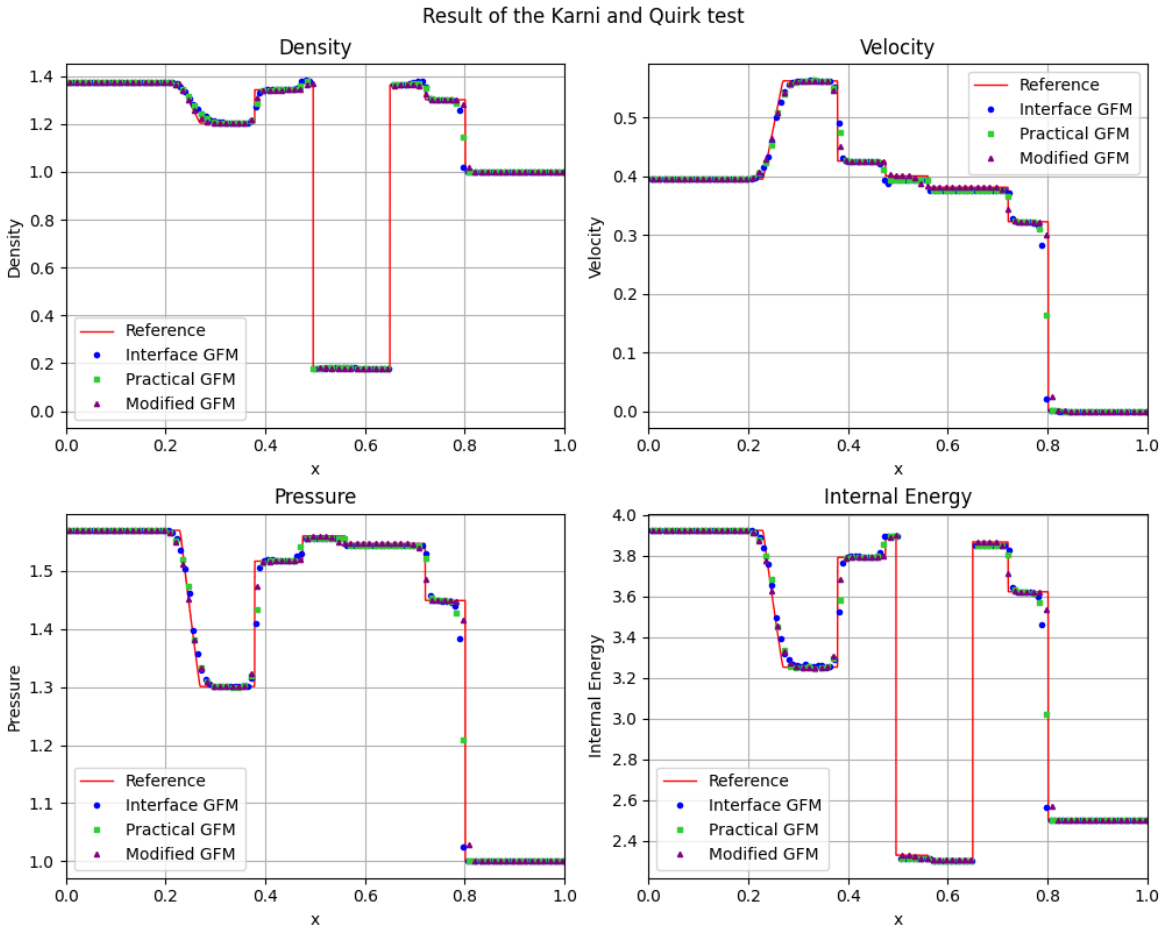


Fig. 7.2 Karni–Quirk test at $t = 0.3\,\mathrm{s}$ (400 cells, CFL $= 0.8$). Red step-line = reference; blue dots = Interface GFM, green square = Practical GFM, purple triangle = Modified GFM.

7.3 The Water Air Shock Tube Test

In this section we repeat the test described in Section 4.3, which aims to test if the three new GFMs can successfully handle the interaction between water and air.

The initial condition is listed in Table 4.3. In this test, we use the final time $t = 0.000229$, Mesh resolution 400×1, and $CFl = 0.8$.

Figure 7.3 shows the numerical results of the Water-Air Shock Tube Test using Interface GFM (blue), Practical GFM (green), and Modified GFM(purple). All three methods successful reproduce the correct shock structures in density, velocity, pressure, and total energy. Compared to the previous GFM methods discussed earlier in the report(i.e. the Original GFM, Real GFM, and Riemann GFM shown in Figure 4.3), these new methods exhibit improved agreement with the exact reference solution (red).

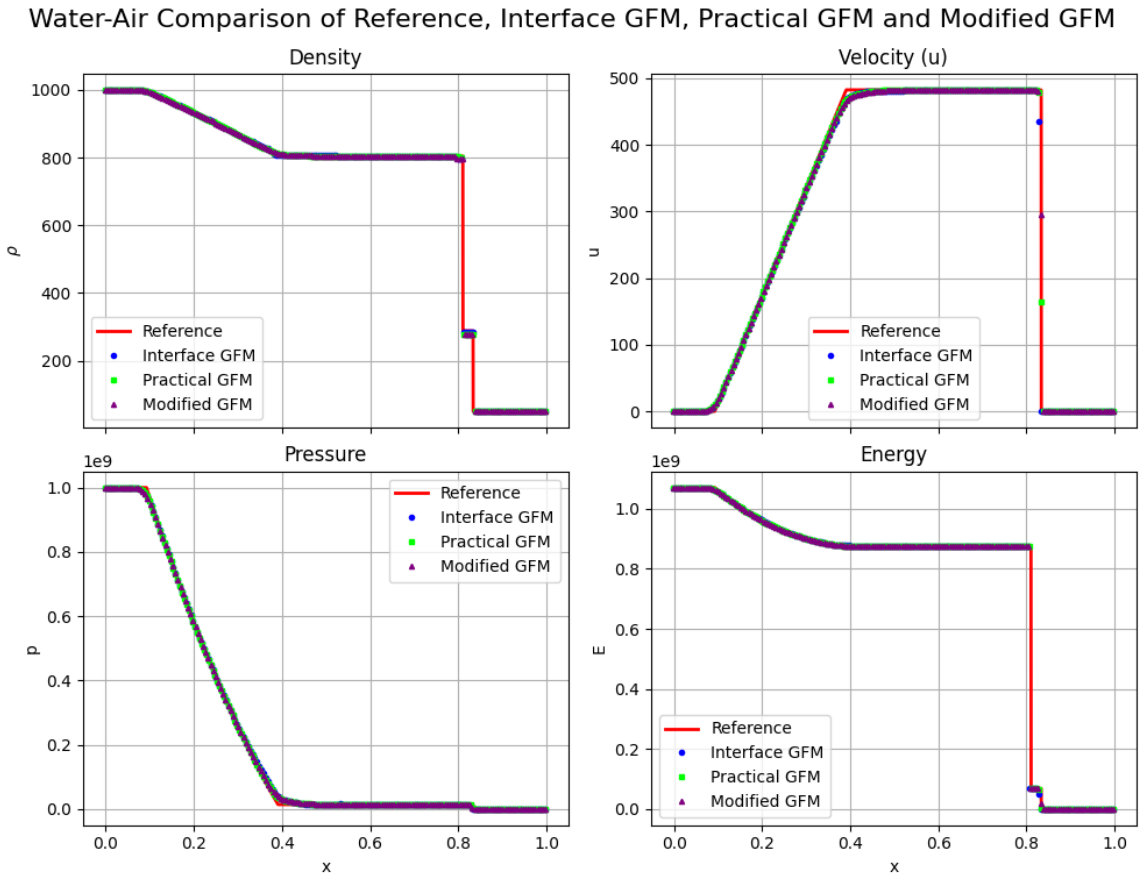


Fig. 7.3 Comparison of numerical and reference solutions for the 1D water–air shock tube test at final time $t = 0.000229$. Shown are the profiles of density ρ, velocity u, pressure p, and total energy E. The numerical results were obtained using the Interface GFM (blue), the Practical GFM (green), the Modified GFM (purple) and are compared against the exact reference solution (red). The simulation was performed on a uniform grid with 400 cells and a CFL number of 0.8.

This multi-material shock tube test demonstrates that the Interface GFM, Practical GFM, and Modified GFM are capable of accurately handling materials governed by different equations of states, such as water and air. For this test, the stiffened gas equation of state was used for water, and the ideal gas equation of state was used for air. The EoS of this two materials are described in Section 2.5. Unlike the Original GFM, Real GFM, and Riemann GFM, which exhibit greater numerical diffusion or overheating errors near material interfaces, these new methods not only capture all physical features correctly but also show significant improvement in numerical stability and consistency with the reference solution.

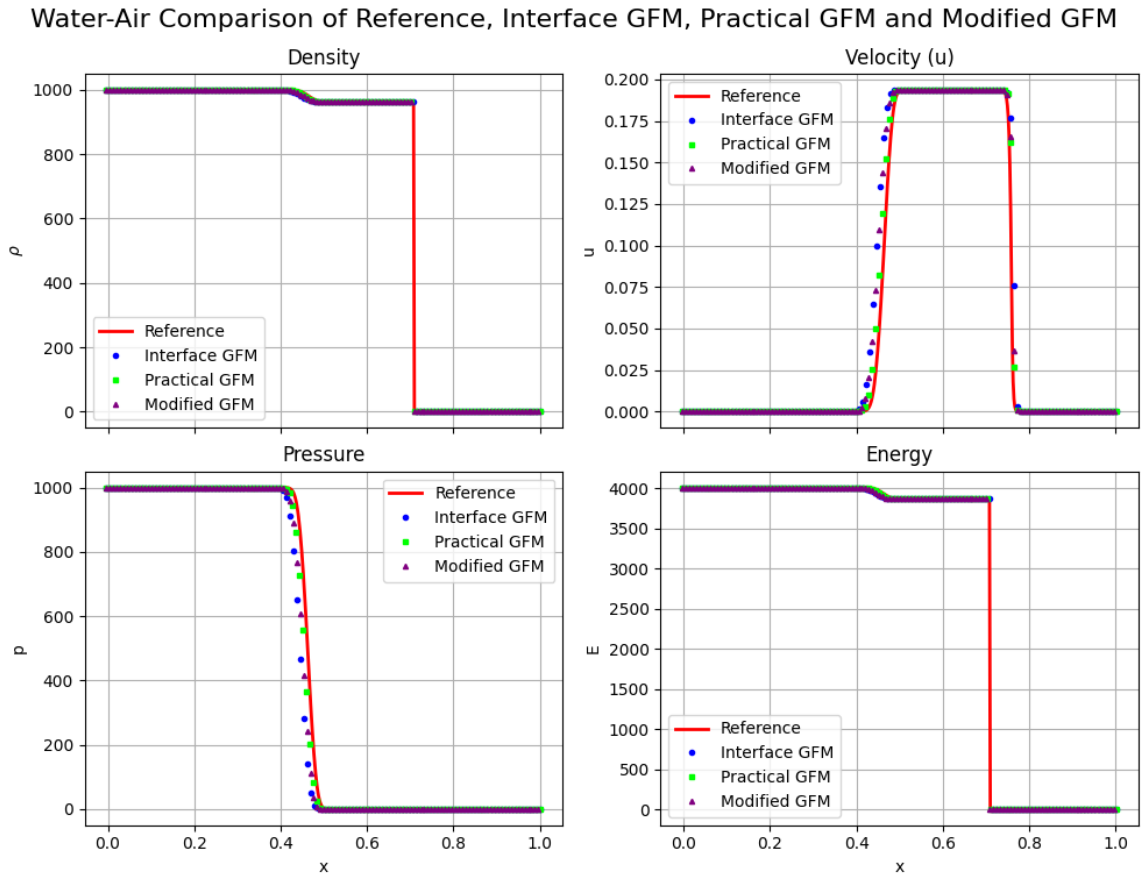


Fig. 7.4 1D underwater shock on a gas–water interface with CFL=0.9, run for 150 time steps, mesh resolution: 200 cells.

7.4 Underwater shock impacting on a gas-water interface

In this section we repeat the same test described in Section 4.4 to see if the three GFMs can still work under this extreme condition. The initial conditions ae the same as listed in Table 4.4.

Figure 7.4 compares the results of three improved GFMs against the reference solution in the context of a 1D underwater shock impacting a gas water interface. All three methods closely match the reference across all variables, which proves their abiity to capture the correct wave structures and interface behaviour even under strong discontinuities.

This result highlights the robustness of the improved GFMs, none of them break down during the simulation, unlike the Original GFM which fails within just a few time steps under this extreme condition. In addition, the improved GFMs exhibit not only enhanced stability

but also significantly higher accuracy compared to the original versions (see Figure 4.4), which makes them well-suited for multi-material shock problems.

Chapter 8

2D Validation Tests

In this chapter, we extend our validation of the improved GFMs (Interface GFM, Practical GFM, and Modified GFM) to 2D problems. While the previous chapters have demonstrated that these methods are stable and accurate in 1D scenarios, it is essential to veify that they retain their robustness and accuracy in more complex 2D settings. Through a series of 2D test cases, we aim to confirm that the improved GFMs can handle multidimensional problems. These tests serve as an important step toward demonstrating the methods' applicability to realistic multiphase flow problems.

8.1 The Cylindrical Explosion Test

In this section, we first perform a 2D validation test using a cylindrical explosion test in a single material. The initial setup is listed in Table 4.5. An accurate method should be able to maintain a circular shock front and avoid unphysical oscillations or artifacts. We compare the results of the three improved GFMs against the reference solution to assess their accuracy and robustness in this 2D radial scenario.

Figure 8.1 and Figure 8.2 demonstrate the performance of the three improved GFMs (Interface GFM, Practical GFM, and Modified GFM) in the 2D cylindrical explosion test with a single material.

From the lineout plots in Figure 8.1, all three methods show excellent agrement with reference solution across all physical variables. This confirms their accuracy in capturing radial shock propagation without introducing significant numerical artifacts.

Furthermore, the surface plots in Figure 8.2 visualize the radial symmetry of the solution at $t = 0.25$, and, all three methods produce consistent and physically realistic results.

These observations indicate that the new GFMs are capable of handling 2D single-material problems robustly and accurately, and also mantain both the symmetry and stability throughout the simulation.

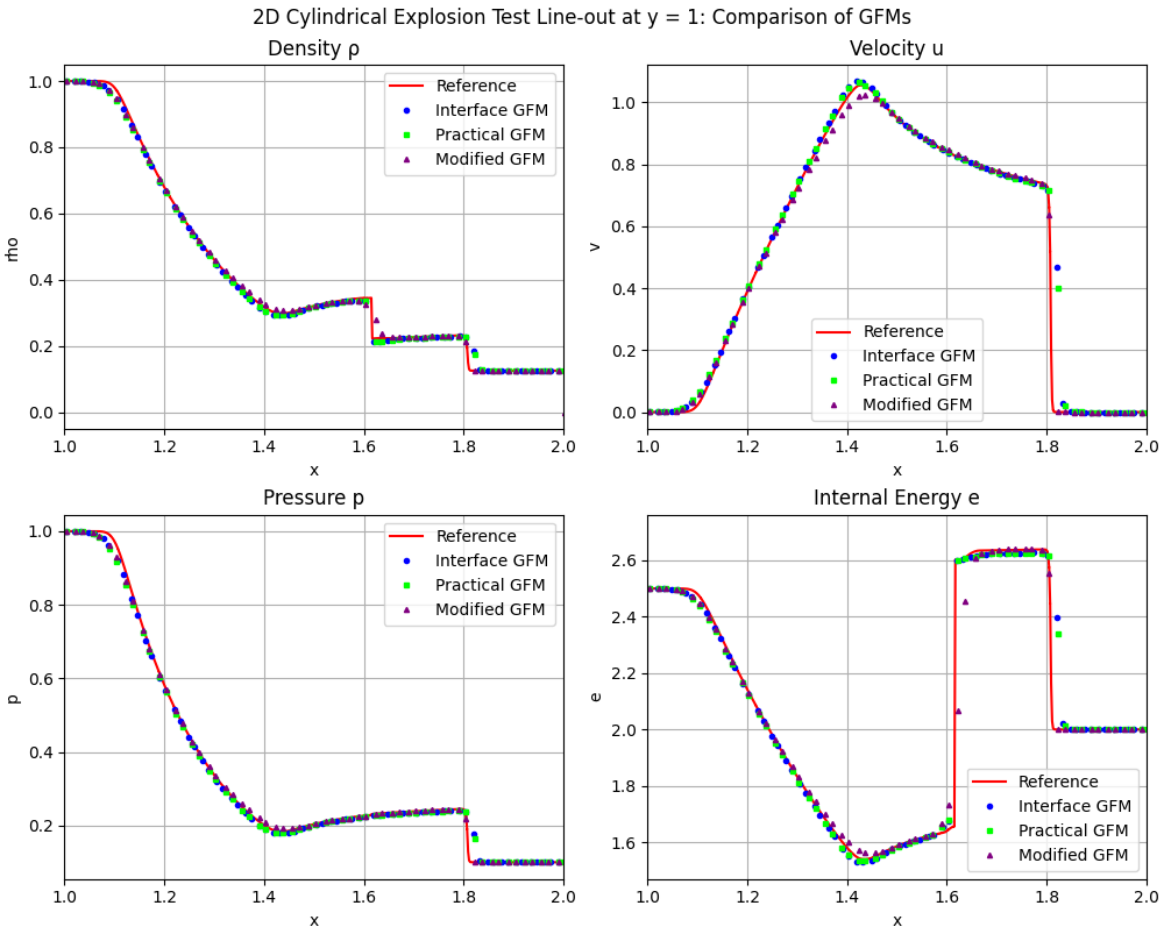
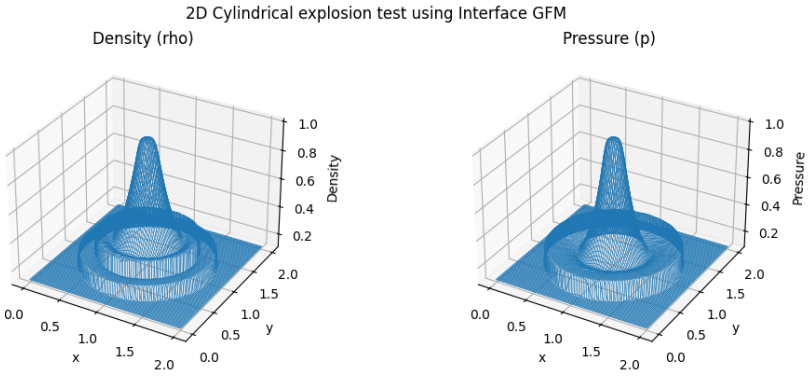
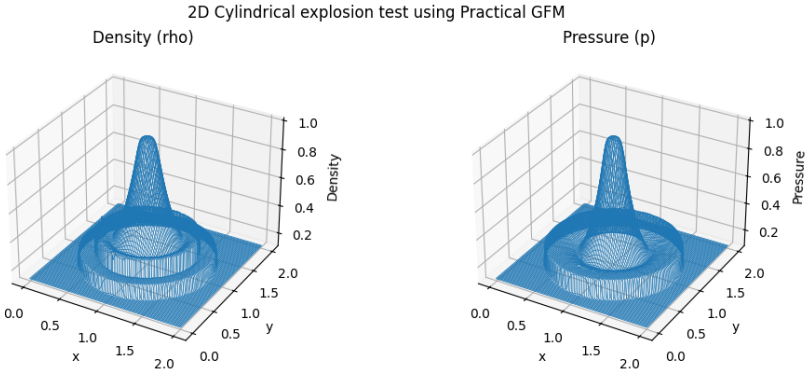


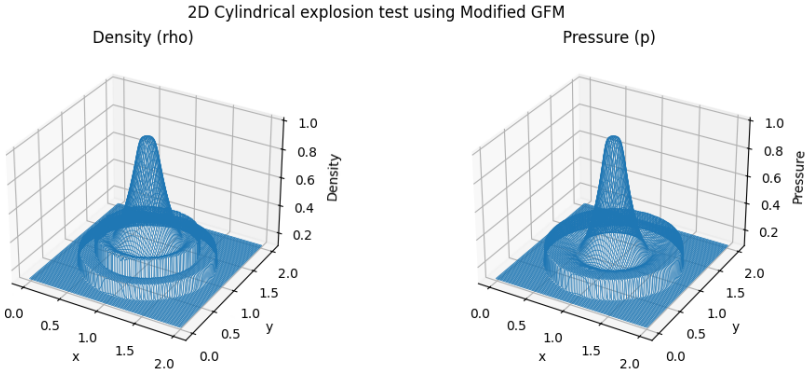
Fig. 8.1 Comparison along $y = 1$ at $t = 0.25\,\mathrm{s}$ for the 2D cylindrical-explosion test (mesh 300×300, CFL=0.8). Panels show density ρ, velocity u, pressure p, and internal energy e.



(a) Interface GFM



(b) Real GFM

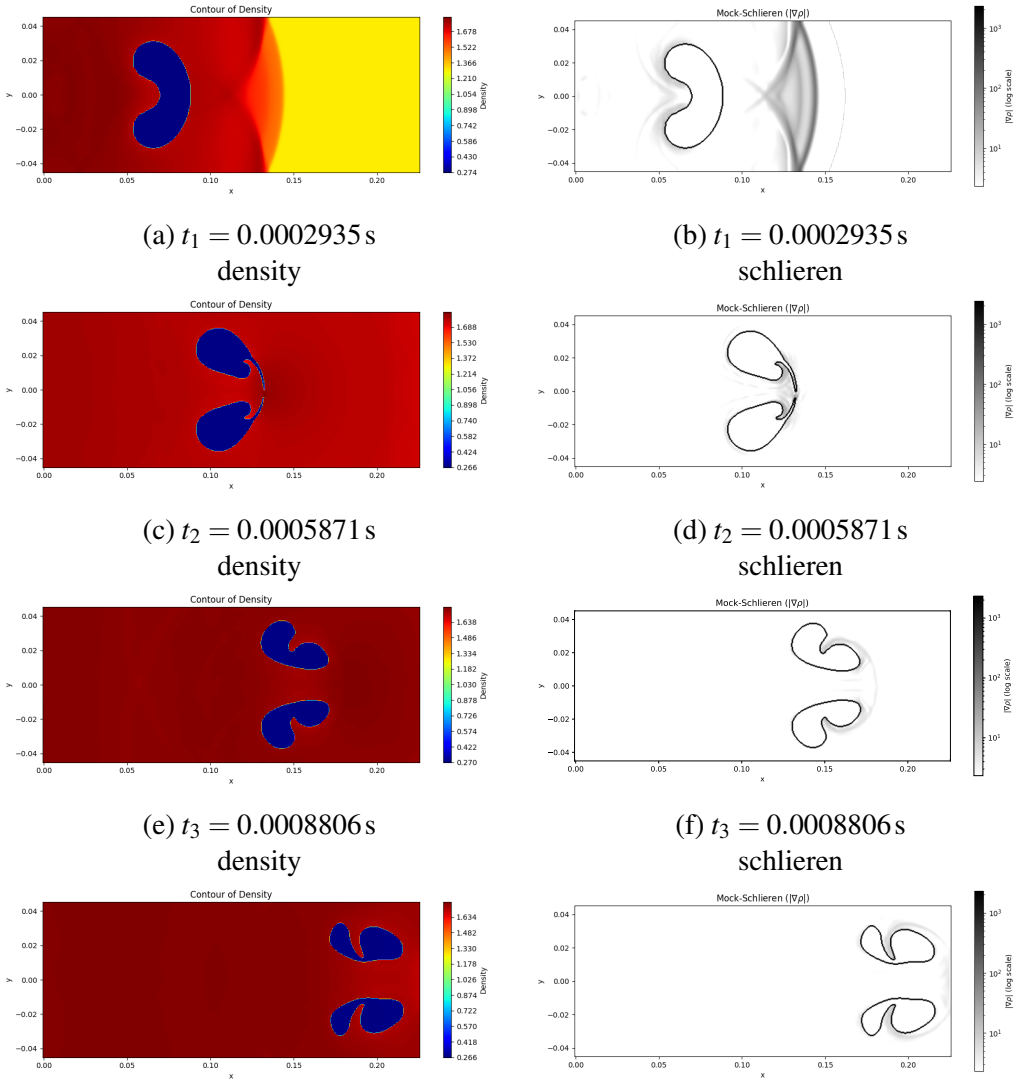


(c) Modified GFM

Fig. 8.2 Surface plots of density (left) and pressure (right) at $t = 0.25\,\mathrm{s}$ for the 2D cylindrical-explosion test on a 300×300 grid (CFL=0.8). Shown are results from (a) the Interface GFM, (b) the Practical GFM, and (c) the Modified GFM.

8.2 The 2D Shock Bubble Test

In this section, we perform the 2D shock bubble test, which serves as a validation case involving two different materials.



(a) $t_1 = 0.0002935\,\mathrm{s}$ density

(b) $t_1 = 0.0002935\,\mathrm{s}$ schlieren

(c) $t_2 = 0.0005871\,\mathrm{s}$ density

(d) $t_2 = 0.0005871\,\mathrm{s}$ schlieren

(e) $t_3 = 0.0008806\,\mathrm{s}$ density

(f) $t_3 = 0.0008806\,\mathrm{s}$ schlieren

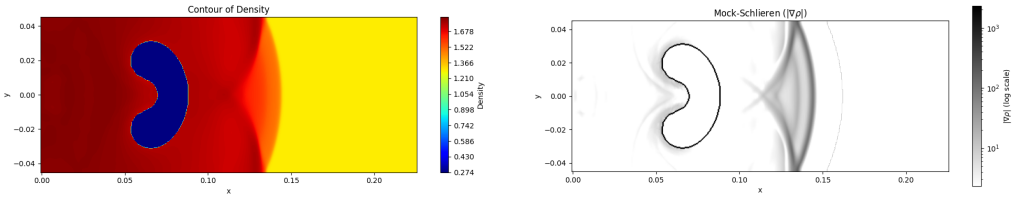
(g) $t_4 = 0.0011741\,\mathrm{s}$ density

(h) $t_4 = 0.0011741\,\mathrm{s}$ schlieren

Fig. 8.3 Evolution of the two-phase interface under a planar shock computed with the Interface GFM on a 1000×394 grid (CFL = 0.8). Shown are density contours (left column) and corresponding shock–schlieren images (right column) at four successive time instants.

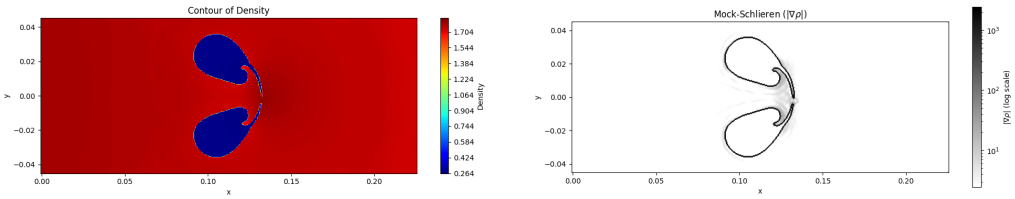
The goal is to assess whether the improved GFMs can robustly handle complex material interfaces in 2D under strong shock interactions. A planar shock wave propagates through one medium and interacts with a bubble of another material, and generate rich interface dynamics. This test is more challenging than the same material explosion case due to the presence of large density and compressibility contrasts across the interface.

The initial conditions for this test are provided in Table 4.6.



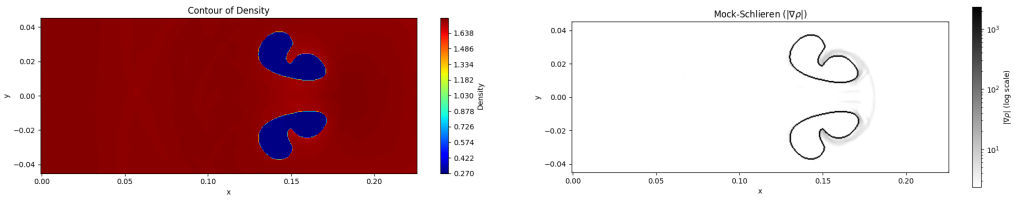
(a) $t_1 = 0.0002935\,\mathrm{s}$ density

(b) $t_1 = 0.0002935\,\mathrm{s}$ schlieren



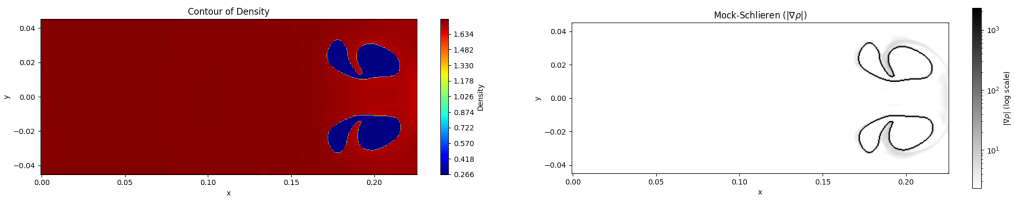
(c) $t_2 = 0.0005871\,\mathrm{s}$ density

(d) $t_2 = 0.0005871\,\mathrm{s}$ schlieren



(e) $t_3 = 0.0008806\,\mathrm{s}$ density

(f) $t_3 = 0.0008806\,\mathrm{s}$ schlieren

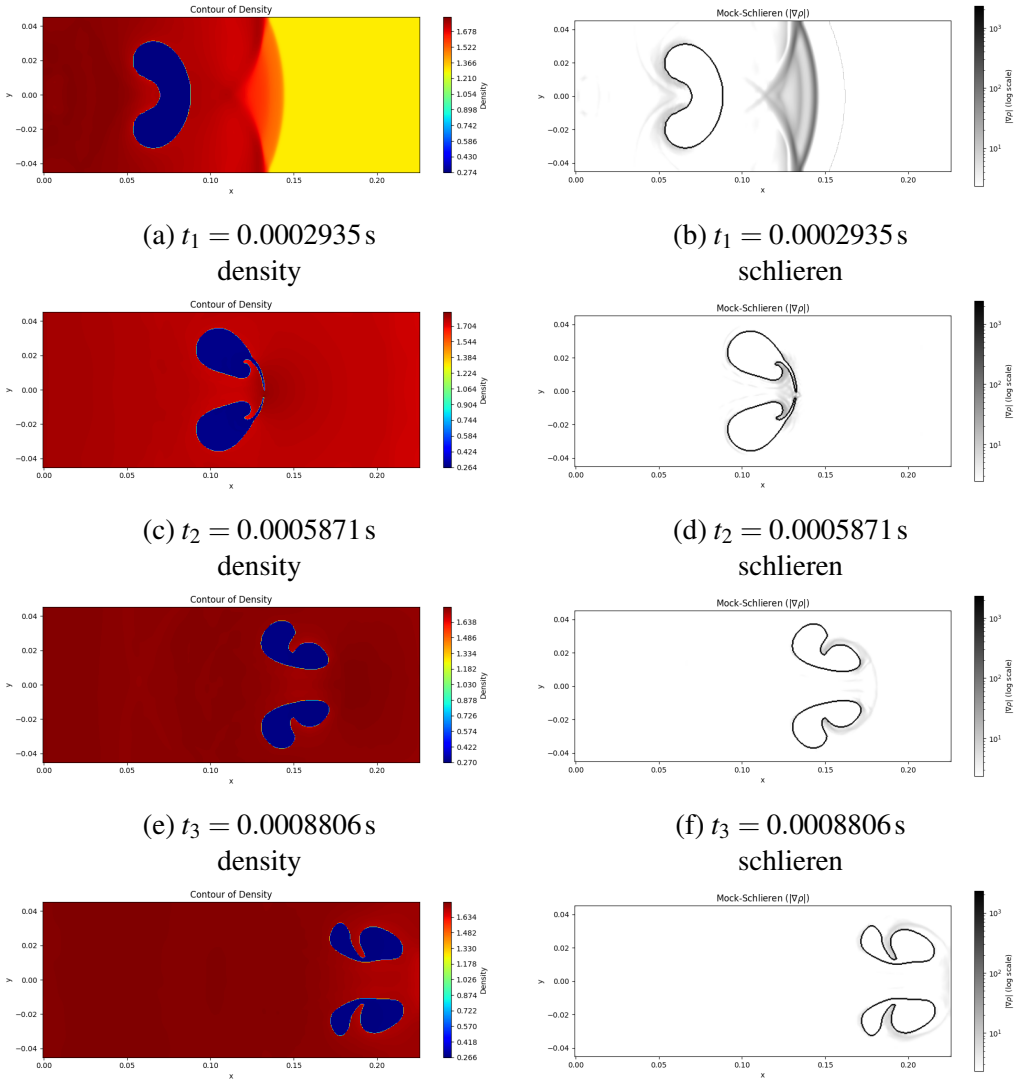


(g) $t_4 = 0.0011741\,\mathrm{s}$ density

(h) $t_4 = 0.0011741\,\mathrm{s}$ schlieren

Fig. 8.4 Evolution of the two-phase interface under a planar shock computed with the Practical GFM on a 1000×394 grid (CFL = 0.8). Shown are density contours (left column) and corresponding shock–schlieren images (right column) at four successive time instants.

Figure 8.3, 8.4, 8.5 show that the Interface GFM, Practical GFM, and Modified GFM reproduce essentially similar interface shapes at all sampled times. This consistency across all three improved methods indicates that they can robustly handle two-material interactions in 2D under moderate shock conditions. The absence of spurious oscillations and the preservation of symmetry in the interface evolution further confirm that each method maintains stability and accuracy throughout the simulation. Although the underlying ghost-fluid con-



(a) $t_1 = 0.0002935\,\mathrm{s}$ density

(b) $t_1 = 0.0002935\,\mathrm{s}$ schlieren

(c) $t_2 = 0.0005871\,\mathrm{s}$ density

(d) $t_2 = 0.0005871\,\mathrm{s}$ schlieren

(e) $t_3 = 0.0008806\,\mathrm{s}$ density

(f) $t_3 = 0.0008806\,\mathrm{s}$ schlieren

(g) $t_4 = 0.0011741\,\mathrm{s}$ density

(h) $t_4 = 0.0011741\,\mathrm{s}$ schlieren

Fig. 8.5 Evolution of the two-phase interface under a planar shock computed with the Modified GFM on a 1000×394 grid (CFL = 0.8). Shown are density contours (left column) and corresponding shock–schlieren images (right column) at four successive time instants.

structions differ slightly, their numerical impact remains negligible in this well-resolved case.

Table 9.1 Initial conditions for Test 2 (Double Rarefaction Case). Mesh resolution: 300×1, CFL = 0.9, $t_{\text{end}} = 0.15$.

Region	**Density** (ρ)	**Velocity** (u)	**Pressure** (p)
Left ($x < 0.5$)	1.0	-2.0	0.4
Right ($x \geq 0.5$)	1.0	2.0	0.4

Chapter 9

Supplementary Tests for GFM Comparison

This chapter presents a set of test problems aimed at supplementing the primary benchmark cases for GFM evaluation. The result in this chapter serve as an extended tookit to deepen the understanding of different GFM performances.

9.1 Toro's Test 2

In this section we perform the test described in the second test case in Toro's book [21]. The initial conditions are listed in Table 9.1.

The solution of Test 2 features two symmetric rarefaction waves moving away from the centre, with only a trivial contact discontinuity in between. The region between the rarefactions, known as the star region, approaches vacuum conditions. This makes Test 2 an effective benchmark for evaluating the ability of numerical schemes to handle extremely low-density flows.

All six GFM variants: Original, Real, Riemann, Interface, Practical, and Modified successfully reproduce the overall wave structure of Toro Test 2. In all cases, the rarefaction waves and linear velocity gradient are correctly captured, and the pressure and density profiles

closely follow the expected "U" shaped solution. This confirms that all schemes have the essential ability to resolve the large-scale features of symmetric rarefaction flows.

However, clear differences emerge when examining the internal energy near the center of the domain, where the solution approaches near-vacuum. The Original GFM exhibits pronounced overshoots, while Modified GFM also display noticeable oscillations. In contrast, the real GFM, Riemann GFM, and interface GFM and practical GFM demonstrate much better behaviour in the same region.

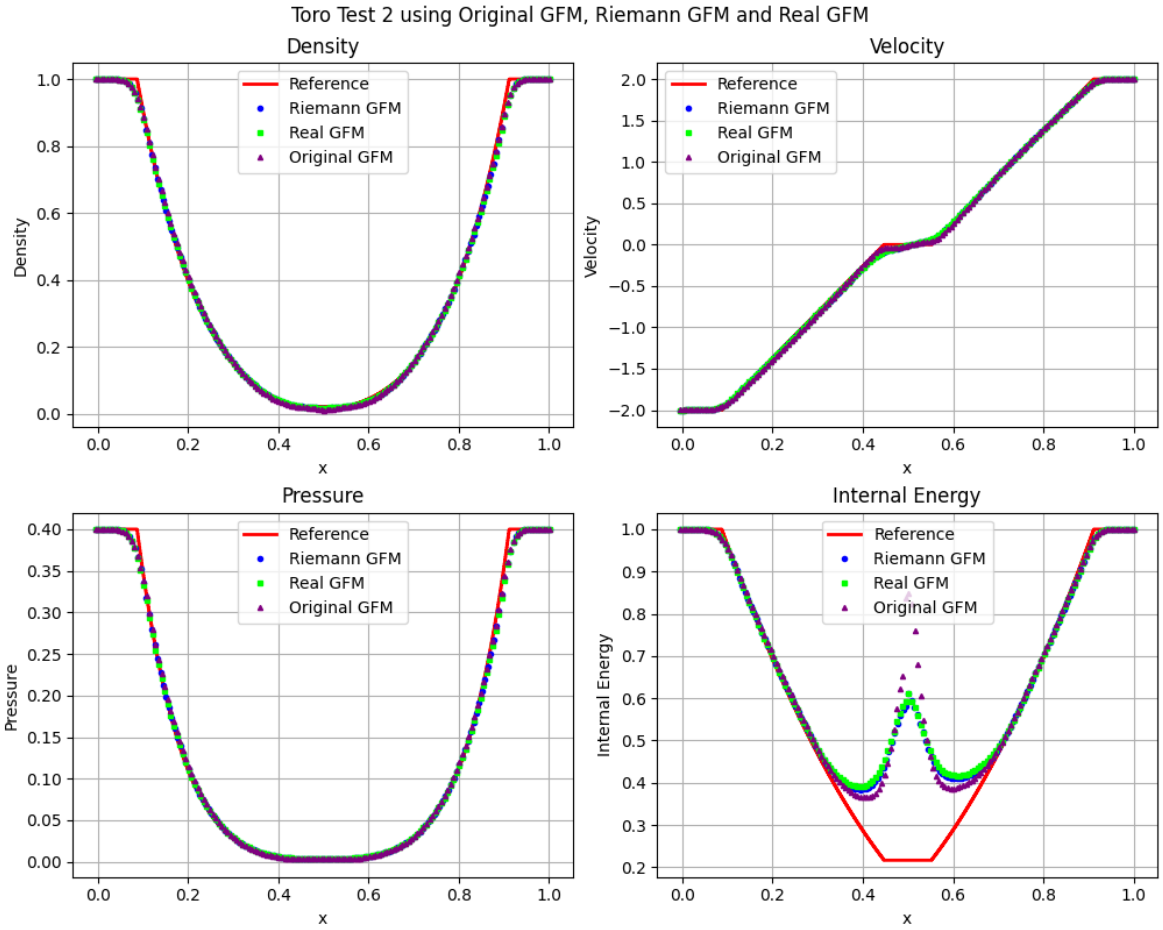


Fig. 9.1 Comparison of numerical results for Toro Test 2 using Original GFM, Real GFM, and Riemann GFM. The four panels show the profiles of density, velocity, pressure, and internal energy at $t = 0.15$ with mesh resolution 300×1 and CFL $= 0.9$. The reference solution (red line) corresponds to the exact Riemann solution.

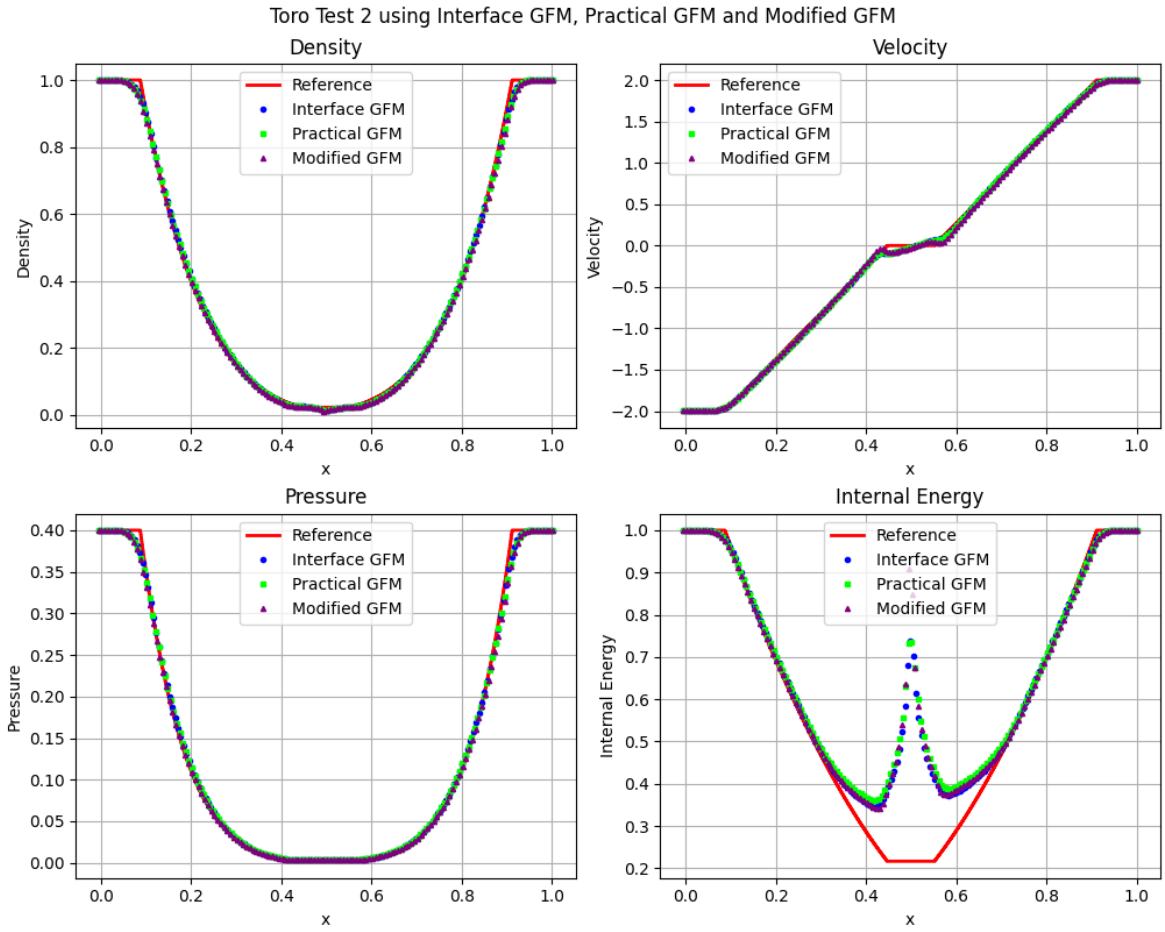


Fig. 9.2 Comparison of numerical results for Toro Test 2 using Interface GFM, Practical GFM, and Modified GFM. The four panels show the profiles of density, velocity, pressure, and internal energy at $t = 0.15$ with mesh resolution 300×1 and CFL $= 0.9$. The reference solution (red line) corresponds to the exact Riemann solution.

9.2 Toro's Test 5

In this section we perform the test case described in Toro's book [21]. Test 5 is not only used to assess the robustness of numerical methods under extreme pressure ratios, but is primarily intended to evaluate their ability to accurately capture slowly moving contact discontinuities. The exact solution consists of a left-going rarefaction wave, a right-going shock wave, and a stationary contact discontinuity, making this case particularly sensitive to numerical diffusion and interface treatment schemes.

The initial conditions of this test is listed in Table 9.2.

Table 9.2 Initial conditions and numerical setup for Test 5

Side	ρ	u	p	Numerical setup
Left ($x < 0.8$)	1.0	-19.59745	1000.0	$\Delta x = \frac{1}{300}$ CFL = 0.9 $t_{\text{end}} = 0.012$
Right ($x \geq 0.8$)	1.0	-19.59745	0.01	
Computational domain: $x \in [0,1]$, $y \in [0,1]$; grid size 300×1				

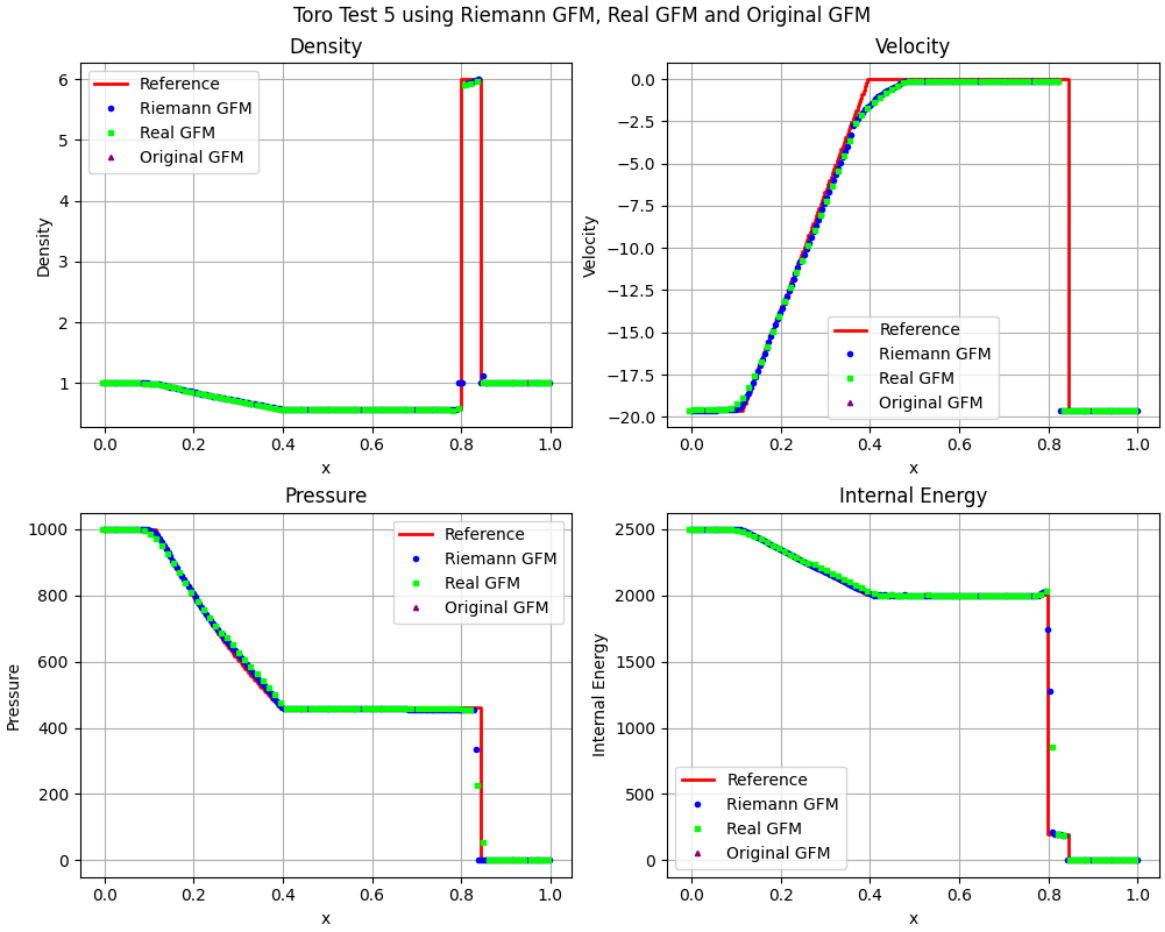


Fig. 9.3 Comparison of numerical results for Toro Test 5 using Original GFM, Real GFM, and Riemann GFM. The four panels show the profiles of density, velocity, pressure, and internal energy at $t = 0.012$ with mesh resolution 300×1 and CFL $= 0.9$. The reference solution (red line) corresponds to the exact Riemann solution.

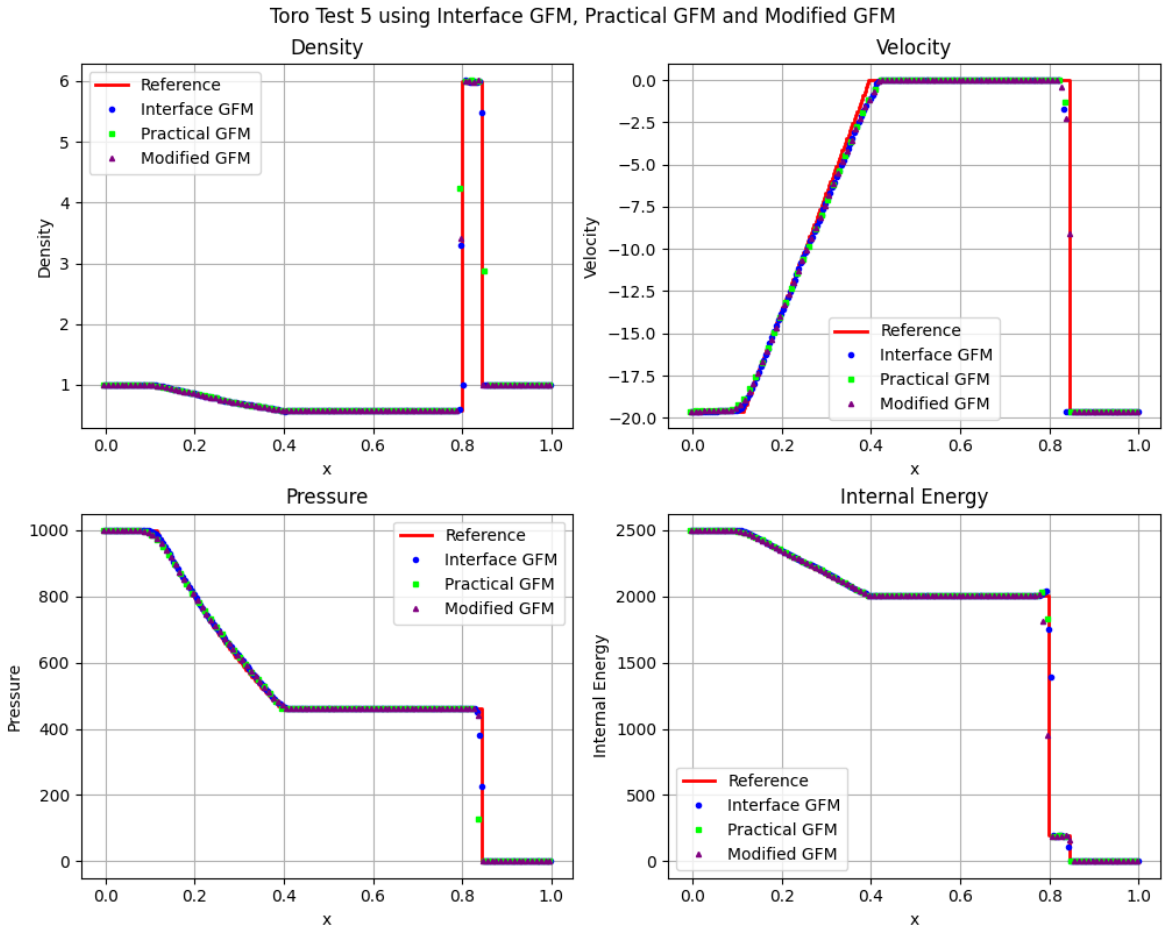


Fig. 9.4 Comparison of numerical results for Toro Test 5 using Interface GFM, Practical GFM, and Modified GFM. The four panels show the profiles of density, velocity, pressure, and internal energy at $t = 0.012$ with mesh resolution 300×1 and CFL $= 0.9$. The reference solution (red line) corresponds to the exact Riemann solution.

In Figure 9.3 and Figure 9.4, we compare the numerical performance of six GFMs on Toro's test 5, which presents a strong shock and vacuum formation in 1D setting with the same material on bith sides of the interface.

As shown in Figure 9.3, the original GFM cannot survive in this extreme shock scenario. In contrast, both Real GFM and Riemann GFM yield correct solutions.

Figure 9.4 presents results form three improved GFMs (Interface GFM, Practical GFM, Modified GFM), which all show more stable and accurate behaviour even under such challenging flow conditions. These methods successfully capture the shock, rarefaction and vacuum without introducing significant numerical oscillations.

Overall, apart from the Original GFM, all five improved GFMs demonstrate the capability to handle strong shock problems in single material configurations, which confirms their robustness in this benchmark test.

Method	Core Idea
Original GFM	Copy-and-mix (no Riemann).
Riemann GFM	Two point interpolation + 1-D Riemann (HLLC).
Real GFM	Nearest-face 1D Riemann.
Modified GFM	Solve $p_I, v_{n,I}$ via characteristic relations.
Practical GFM	Estimate only u_I; mirror rest.
Interface GFM	Full two-fluid Riemann at the interface.

Table 10.1 Algorithmic kernels of the six GFMs.

Chapter 10

Comparative Analysis and Discussion

We have implemented and validated six GFMs (Original GFM, Riemann GFM, Real GFM, Modified GFM, Practical GFM, Interface GFM). Each scheme introduces targeted improvements, yet their relative performance, especially with respect to stability under strong shock and the preservation of 2D interfaces has not been systematically compared. Accordingly, this chapter presents a comprehensive evaluation of the six methods across the key criteria.

To ground the subsequent comparison, section10.1 offers a concise overview of the governing ideas behind each GFM.

10.1 Overview of GFMs

Table 10.1 summarizes the core algorithmic ideas behind the six GFMs cond=sidered in this work. Each method constructs ghost states at fluid interfaces using a different strategy. The spectrum ranges from the simplest Original GFM, which constructs ghost states by copying the pressure and normal velocity from the opposite material (and extrapolating density and tangential velocity from the same side) without solving a Riemann problem, to

more physically consistent schemes such as Interface GFM, which solves a full two-fluid Riemann problem at the interface.

Several methods (e.g. Riemann GFM and Real GFM) use 1D Riemann solvers at the interface, while others (e.g. Modified GFM and Practical GFM) aim to modify the Original GFM to deal with the strong shock situation.

This overview highlights that while all six GFMs share the same underlying philosophy, which is constructing ghost states to enforce interfacial jump conditions, their algorithmic realizations differ significantly in complexity, physical accuracy, and extensibility to multidimensional problems. These differences will be evaluated in the following sections.

10.2 Evaluation Strategy: Criteria and Benchmark Suite

Before presenting quantitative results, we first introduce what will be measured and where the measurements are taken. Table 10.2 lists five practical evaluation dimensions together with the concrete proxy we will extract from every run. Table 10.3 completes these criteria by listing the eight benchmark problems drwan from previous chapters.

Dimension	**What we actually look at**
Accuracy near interface	Interface resolution quality
Stability under strong shocks	Crash / no-crash flag and max oscillation
Ease of implementation	Core lines of code + need for special data structures
2-D robustness	Symmetry error (cylindrical blast) + presence of spurious waves
Computational cost	Number of 1-D Riemann solves per time step($0 \rightarrow$ many)

Table 10.2 Five practical evaluation dimensions used in the comparative study.

10.3 Results and Cross-Test Comparison

The Shock Tube test (Figure 4.1, Figure 7.1) primarily stresses accuracy in capturing shock and rarefaction positions and overall fidelity. All six schemes capture the shock and rarefaction at the correct locations, Riemann GFM and Real GFM follow virtyally identical path, while Modified GFM and Original GFM form a very similar path. Overall, Interface GFM and Practical GFM track the exact solution most closely, delivering the highest fidelity among the six methods.

In the Karni-Quirk test (Figure 4.2 and Figure 7.2), Real GFM and Riemann GFM exhibit noticeable oscillations in the internal energy profile near $x = 0.5$ and $x = 0.7$. By contrast,

Test	Dim.	Scenario & Purpose	Main Challenge
4.1 and 7.1	1D	Shock-tube (Sod)	Interface sharpness
4.2 and 7.2	1D	Karni–Quirk	Multiple interaction with interfaces with different acoustic impedance.
4.3 and 7.3	1D	Water–air shock-tube	High density ratio, prevent overshoot
4.4 and 7.4	1D	Under-water shock on gas–water interface	Strong shock–interface interaction
4.5 and 8.1	2D	Cylindrical explosion	Cylindrical symmetry, spurious waves
4.6 and 8.2	2D	Shock-bubble interaction	Multidimensional deformation
9.1	1D	Toro Test 2	Vacuum stability and double rarefaction interaction
9.2	1D	Toro Test 5	Very narrow discontinuities pattern

Table 10.3 Benchmark suite for GFM comparison.

Original GFM, Interface GFM, Practical GFM and Modified GFM follow the reference curve much more closely, providing the most accurate results for this mixed-EOS, low-March test case.

In the Water-air shock tube test (Figure 4.3 and Figure 7.3), which stresses strong shocks and multi-material interaction, stability under extreme discontinuities and accurate interface treatment are both critical. The Original GFM breaks down early, failing to complete the simulation, which demonstrates it lacks the necessary robustness for this harsh shock regime. Of the remaining schemes, Interface GFM, Practical GFM, and Modified GFM not only remain stable but also track the reference solution most closely, delivering the best combination of fidelity and shock-handling precision. By contrast, Riemann GFM and Real GFM stay intact but show noticeably larger deviations from the exact solution, suggesting their interface treatments are less effective here, potentially due to weaker coupling or oscillatory artifacts induced by the strong shock. Overall, the enhanced interface strategies in Interface, Practical, and Modified GFM allow them to cope with the situation more reliably and accurately, whereas Original GFM fails and the standard Riemann-based approaches trade off some accuracy.

In the Underwater shock on a gas-water interface test (Figure 4.4, Figure 7.4), this extreme scenario again exposes the limited robustness of Original GFM, which fails to complete the simulation, while all other schemes remain stable. Among the successful methods, Interface GFM, Practical GFM, and Modified GFM achieve significantly higher accuracy than Real GFM and Riemann GFM, exhibiting closer agreement with the reference solution. The advantage of Interface, Practical, and Modified GFM stems from their more effective interface treatment and coupling, which better resolve the complex shock transmission and

reflection dynamics across the gas–water boundary, whereas the standard Riemann-based approaches show larger discrepancies.

In Toro Test 2 (Figure 9.1, Figure 9.2), This case features interacting rarefaction waves that create a vacuum region, so it stresses both stability in low-density zones and the ability to preserve sharp, non-diffused structures. All six GFMs complete the simulation, but differences appear in smoothness and fidelity. Interface GFM and Practical GFM best reproduce the exact solution, with minimal diffusion and sharp rarefaction profiles. Real GFM and Riemann GFM remain stable but introduce slight smoothing, indicating a mild loss of resolution. Modified GFM has small oscillations slight over and undershoots in the delicate vacuum region. Original GFM stays stable but departs from the reference more than any other method. For this double rarefaction scenario, Interface and Practical GFM achieve the best balance between robustness and sharp feature preservation.

Toro Test 5, characterized by a left-going rarefaction wave, a right-going strong shock wave, and a stationary contact discontinuity with an extreme pressure ratio, poses a significant challenge to numerical stability (Figures 9.3 and 9.4). Among the six GFMs, Interface GFM and Practical GFM demonstrate the best performance, with density and pressure distributions closely matching the exact Riemann solution (red line), minimal diffusion, and stable velocity and internal energy profiles, particularly excelling in capturing the slowly moving contact discontinuity. Riemann GFM and Real GFM follow with good accuracy in density and pressure, slight smoothing in velocity and internal energy at the shock region ($x > 0.8$), and robust stability. Modified GFM shows comparable density and pressure alignment with moderate diffusion, minor oscillations in velocity, and a stable internal energy profile, indicating decent but slightly inferior performance. In contrast, Original GFM fails to simulate the test effectively, with flat and deviating profiles across all variables, suggesting a lack of stability and inability to handle the test's physical characteristics.

The 2D cylindrical explosion test, evaluated along $y = 1$ at $t = 0.25$ s (Figures 4.5, 4.6, and 8.1, 8.2), presents a challenging scenario involving a strong shock wave and a contact discontinuity, testing the numerical stability and accuracy of the GFMs. All six GFMs perform well in completing the simulation, which demonstrats robust stability across the test conditions. Interface GFM, Practical GFM and Modified GFM exhibit the best performance, with density and pressure profiles closely aligning with the reference solution (red line), minimal diffusion, and stable velocity and internal energy distributions, particularly excelling in capturing the shock and contact discontinuity features. Riemann GFM, Real GFM and Original GFM demonstrate comparable performance to Modified GFM, with density and pressure closely matching the reference, slight smoothing in velocity and internal energy at the shock, and robust stability.

The shock-bubble interaction test, evaluated at 4 time steps (Figures 4.7-4.9, and 8.3-8.5), presents a complex scenario involving a shock wave interacting with a cylindrical bubble, challenging the numerical stability and accuracy of the GFMs in a 2D background. All six GFMs successfully complete the simulation, demonstrating robust stability under the test conditions. Interface GFM and Practical GFM exhibit the best performance, and clear preservation of the bubble interface and shock front, particularly excelling in capturing the Richtmyer-Meshkov instability. Modified GFM follows with good accuracy in density and pressure, moderate diffusion, and stable interface tracking, though it shows slight blurring near the shock-bubble boundary. Riemann GFM and Real GFM perform comparably to Modified GFM, with density and pressure fields aligning well with the reference, minor smoothing at the interface, and stable shock propagation.

However, both Original GFM and Real GFM exhibit noticeable asymmetry in the deformation of the bubble. This asymmetry is not purely physical, but can be traced back to the numerical components used in these implementations. Specifically, the Minbee slope limiter, employed in these methods, is known to be non-symmetric and may introduce directional bias in wave propagation. Moreover, the HLLC Riemann solver used in their flux computation calculates wave speeds in an asymmetric manner, especially under strong shock conditions. Together, these two components can significantly contribute to asymmetry in otherwise symmetric initial conditions.

Original GFM, while stable, displays the weakest performance, with noticeable diffusion and distorted bubble shapes, which indicates limitations in handling the test's multi-material interface dynamics, as evident from the contour plots and cross-sectional data. In contrast, Interface GFM, Practical GFM, and Modified GFM do not rely on the HLLC Riemann solver near the interface, and instead use more symmetric ghost state constructions, which likely explains their more symmetric results.

The Original GFM is the easiest to implement because it avoids any Riemann solves entirely. Its copy-and-mix logic is local and algebraic, making the data flow straightforward and the code minimal. In contrast, both Riemann GFM and Real GFM require solving a 1D Riemann problem at or near the interface (with Riemann GFM using two-point interpolation plus a full Riemann solver and Real GFM using a nearest-face Riemann), so they introduce additional solver modules. The Modified, Practical, and Interface GFM are more involved: Modified GFM uses characteristic relations to estimate interface pressure and normal velocity, Practical GFM streamlines that by reducing to a single degree of freedom (reflecting normal velocity and mirroring other quantities), and Interface GFM fully resolves the two-fluid interaction via coupled interface Riemann-type conditions. These latter methods require interface localization, nonlinear solves (e.g. Newton iterations), and tighter coupling between

real and ghost states, making their implementation significantly more complex than the basic Riemann based approaches.

Below we summarize those qualitative observations in a compact form. Table 10.4 lists the main and weaknesses of each GFof each GFM and providesvides practical guidance on when to deploy or combine them. It makes explicit the trade-offs already hinted at above: Original GFM is the simplest but breaks down in extreme interactions; Riemann GFM and Real GFM offer moderate coupling with some loss of sharpness. Modified and Practical GFM improve robustness and accuracy at the cost of added algorithmic complexity. Interface GFM delivers the highest fidelity where critical interface resolution is needed, albeit with the heaviest implementation burden. This summary serves as a lens for interpreting the per-test behaviors and motivates hybrid or fallback strategies in complex flow scenarios.

10.4 Final Conclusion

This project has systematically implemented and compared six GFMs across a suite of challenging multi-material benchmark problems. The comparative study revealed a clear performance hierarchy: methods that incorporate richer physical coupling and characteristic information (Modified, Practical, and Interface GFM) consistently outperform the classical approaches in demanding scenarios involving strong shocks, mixed equations of state, vacuum/double rarefaction interactions, and complex interface dynamics. While the Original GFM remains attractive for rapid prototyping or low-intensity interactions due to its minimal implementation cost, it lacks robustness and fidelity under extreme conditions. Riemann and Real GFM occupy a middle ground but sacrifice sharp-feature resolution through smoothing or oscillations in sensitive flows.

Based on these observations, we offer targeted recommendations: deploy Interface or Modified GFM where high-fidelity interface resolution and shock robustness are paramount, use Practical GFM as an efficient and stable compromise in moderately complex scenarios, and reserve Original GFM for prototyping or weakly interacting regions. A dynamic hybrid strategy: switching or blending GFM variants according to local flow characteristics such as pressure gradients, Mach number, or interface curvature—holds promise for balancing accuracy and efficiency in spatially heterogeneous simulations.

Looking ahead, we plan to extend these methods to three-dimensional grids and conduct systematic comparisons in gas–liquid–solid three-phase and other multi-material configurations to better understand interface coupling and stability in coexisting phases. We will also integrate more general non-ideal equations of state, including phase-change and temperature-dependent effects, to evaluate method robustness and accuracy in complex material behaviors;

develop ghost-state update and reconstruction strategies that maintain conservation and numerical stability under interface topological changes such as merging and breakup, explore adaptive switching mechanisms driven by local physical or error indicators to optimize the trade-off between computational cost and simulation fidelity, and accelerate high-overhead schemes (particularly Interface GFM) through parallel Newton solvers and optimized data-coupling workflows to enhance scalability in large scale parallel environments. These efforts will not only validate and enrich the current comparative results but also equip multi-material interface flow simulations with more powerful and flexible tools for engineering and scientific applications.

Table 10.4 Summary of observed advantages, weaknesses, and practical recommendations for the six GFMs.

Method	Strengths	Weaknesses	Suggested use / trade-off
Original GFM	Easiest to implement (no Riemann solves); simple local algebraic logic; stable in moderate regimes.	Breaks down or loses accuracy under strong shocks / extreme interface dynamics; largest deviation in demanding cases.	Lightweight baseline or initial guess; avoid in high-fidelity or harsh multi-material interactions.
Riemann GFM	Uses localized 1-D Riemann solve with two-point interpolation; reasonably stable.	Introduces mild smoothing / oscillatory artifacts in sensitive regimes; moderate implementation complexity.	Use when moderate coupling is acceptable and full interface coupling is not needed; decent baseline with some resolution loss.
Real GFM	Nearest-face Riemann solver gives better local consistency than pure copying.	Similar to Riemann GFM, may smooth sharp features; requires solver module.	Good for cases needing localized refinement without full interface coupling; moderate trade-off between complexity and accuracy.
Modified GFM	Robust and accurate under strong shocks due to characteristic-based interface estimation.	Higher implementation complexity; can exhibit small oscillations in delicate scenarios (e.g.vacuum / double rarefaction).	Use in regions with strong discontinuities where fidelity matters, accepting a bit more complexity.
Practical GFM	Simplifies Modified GFM to a single degree of freedom; easier implementation than full interface solvers; avoids some oscillations.	Slightly more diffusive than the most accurate schemes.	Good compromise when implementation simplicity and robustness are prioritized over minimal diffusion.
Interface GFM	Highest fidelity; fully coupled two-fluid resolution at the interface; excels in preserving sharp features.	Most complex to implement (nonlinear solves, interface localization, tight coupling).	Preferred in critical interface/strong interaction zones; use when accuracy outweighs implementation cost.

References

[1] Dhruv Pandit. Multi-phase flow simulation using computational fluid dynamics. *International Journal of Machine Systems and Manufacturing Technology*, 01(02):28–34, 2024.

[2] Zhipeng Yao, Caixia Chen, Teng Wang, Zihong Xia, Luhaibo Zhao, Zhiyong Tang, and Shuai Yan. Effects of gas distributor on hydrodynamics in gasüliquid bubble column by visual experiments and cfd simulations. *Chemical Engineering Journal*, 504:158476, 2025.

[3] Jing Gong, Qi Kang, Haihao Wu, Xiaoping Li, Bohui Shi, and Shangfei Song. Application and prospects of multi-phase pipeline simulation technology in empowering the intelligent oil and gas fields. *Journal of Pipeline Science and Engineering*, 3(3):100127, 2023.

[4] Francesco Lucci, Edo Frederix, and Arkadiusz K. Kuczaj. Aerosolved: Computational fluid dynamics modeling of multispecies aerosol flows with sectional and moment methods. *Journal of Aerosol Science*, 156:105794, 2021.

[5] Ajit Yoganathan and Fotis Sotiropoulos. Using computational fluid dynamics to examine the hemodynamics of artificial heart valves. *US Cardiology*, pages 1–5, 2004. Wallace H. Coulter Department of Biomedical Engineering, Georgia Institute of Technology.

[6] Vivek Kumar Srivastav, Anuj Jain, and Akshoy Ranjan Paul. Computational studies of aerosolized drug deposition in human respiratory tract. In *Proceedings of the 16th Asian Congress of Fluid Mechanics*, Lecture Notes in Mechanical Engineering, pages 11–23. Springer, 2021.

[7] Jianjian Xin, Fulong Shi, Shi Fan, and Qiu Jin. A sharp interface multiphase flow model for two-dimensional water impact of a symmetric and asymmetric wedge. *Applied Ocean Research*, 119:102988, 2022.

[8] Murray Cutforth, Philip T. Barton, and Nikos Nikiforakis. An efficient moment-of-fluid interface tracking method. *Computers Fluids*, 224:104964, 2021.

[9] Holger Marschall, Stephan Boden, Christoph Lehrenfeld, Carlos J. Falconi D., Uwe Hampel, Arnold Reusken, Martin Wrner, and Dieter Bothe. Validation of interface capturing and tracking techniques with different surface tension treatments against a taylor bubble benchmark problem. *Computers Fluids*, 102:336–352, 2014.

[10] Victor Boniou, Thomas Schmitt, and Aymeric Vi. Comparison of interface capturing methods for the simulation of two-phase flow in a unified low-mach framework. *International Journal of Multiphase Flow*, 149:103957, 2022.

[11] X.Y. Hu and B.C. Khoo. An interface interaction method for compressible multifluids. *Journal of Computational Physics*, 198(1):35–64, 2004.

[12] Mark Sussman, Peter Smereka, and Stanley Osher. A level set approach for computing solutions to incompressible two-phase flow. *Journal of Computational Physics*, 114(1):146–159, 1994.

[13] James Glimm, D Marchesin, and O McBryan. A numerical method for two phase flow with an unstable interface. *Journal of Computational Physics*, 39(1):179–200, 1981.

[14] Ronald P Fedkiw, Tariq Aslam, Barry Merriman, and Stanley Osher. A non-oscillatory eulerian approach to interfaces in multimaterial flows (the ghost fluid method). *Journal of Computational Physics*, 152(2):457–492, 1999.

[15] T.G. Liu, B.C. Khoo, and K.S. Yeo. Ghost fluid method for strong shock impacting on material interface. *Journal of Computational Physics*, 190(2):651–681, 2003.

[16] X. Y. Hu and B. C. Khoo. A real ghost fluid method for the simulation of multimedium compressible flows. *SIAM Journal on Scientific Computing*, 26(5):1595–1620, 2004.

[17] Liang Xu, Chengliang Feng, and Tiegang Liu. Practical techniques in ghost fluid method for compressible multi-medium flows. *Communications in Computational Physics*, 20(3):619–659, 2016.

[18] Hongkai Zhao. A fast sweeping method for eikonal equations. *Mathematics of computation*, 74(250):603–627, 2005.

[19] Shao-Ping Wang, Mark H Anderson, Jason G Oakley, Michael L Corradini, and Riccardo Bonazza. A thermodynamically consistent and fully conservative treatment of contact discontinuities for compressible multicomponent flows. *Journal of Computational Physics*, 195(2):528–559, 2004.

[20] Richard Saurel and Rémi Abgrall. A multiphase godunov method for compressible multifluid and multiphase flows. *Journal of Computational Physics*, 150(2):425–467, 1999.

[21] Eleuterio F Toro. *Riemann solvers and numerical methods for fluid dynamics: a practical introduction*. Springer Science & Business Media, 2013.

Appendix A

OpenMP Parallelization

In each GFM implementation all grid-scale operations including interface cells identification, normal vector computation are parallelized via OpenMP. Specifically, within the main time-stepping loop we wrap the core double loop over 2D indices `(i,j)` in a static-scheduled parallel region:

Listing A.1 OpenMP parallel core in GFM

```
#include <omp.h>
int main() {
  omp_set_num_threads(16);
  initializephiGrid();
  #pragma omp parallel for
  for (int i = ghost_cells; i < nxCells + ghost_cells; i++) {
    for (int j = ghost_cells; j < nyCells + ghost_cells; j++) {
     indentify interface(...);
    }
  }
  compute_normal_vector();

  double t = 0.0, dt;
  int iter = 0;
  do {
    gfm(...);
    constant extrap(...);
    boundary conditions(...);
    update dt(...);
    t += dt;
    iter++;
```

```
22      advectPhi(...);
23
24      #pragma omp parallel for schedule(static)
25      for (int i = ghost_cells; i < nxCells + ghost_cells; i++) {
26        for (int j = ghost_cells; j < nyCells + ghost_cells; j++)
             {
27          identify interface(...);
28        }
29      }
30
31      reinitialisation(...);
32      compute_normal_vector(...);
33      evolve x;
34      evolve y;//using operator splitting
35    } while (t < T_end);
36      store data(...);
37    return 0;
38  }
```